Largan Precision

Core weakness outweighs progress in AI optics

Equities

Taiwan

Electric Components & Equipment

12-month rating	**Sell**
12m price target	**NT$5,000.00**
Price (08 Oct 2026)	**NT$6,660**

RIC: 3008.TW **BBG:** 3008 TT

Trading data and key metrics

52-wk range	NT$7,810.00-2,025.00
Market cap.	NT$875b/US$27.4b
Shares o/s	131m (ORD)
Free float	70%
Avg. daily volume ('000)	2,379
Avg. daily value (m)	NT$13,134.2
Common s/h equity (12/26E)	NT$199b
P/BV (12/26E)	4.4x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From	To	% ch	Cons.
12/26E	188.21	160.04	-15	184.91
12/27E	214.00	180.00	-16	204.02
12/28E	250.66	220.00	-12	238.78

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CPO progress is more tangible but substantial success is already priced in

We maintain Sell and our NT$5,000 target price as erosion in the core smartphone business outweighs CPO progress. The Q326 call confirmed that Largan's fiber array has passed customer qualification although its customer qualified another vendor's prism micro-lens assembly (PMLA) that Largan would now buy in. If Largan qualifies the full FAU plus carrier assembly for 2H27, content rises but with more pass-through cost and less margin upside. We roll our valuation to 2028E, the first full year of CPO production, though 25% below the NT$6,660 close on October 8.

Full FAU assembly would lift content, but with more pass-through

We believe Largan's FA customer is one of two module suppliers on NVIDIA's 2H27 program, which we estimate could reach 40% share. The program is still subject to successful yield ramp and could move to metalens in 2028, which would reopen supplier qualification. Assembling the full FAU could lift Largan's ASP to US$125-150 leveraging its precision alignment and automation but the PMLA, V-groove and carrier parts would be pass through that cuts into margin leverage. We expand our FAU TAM analysis (Figure 9) with FAU market in 2028/30 expanding to 12.3mn/35.6mn units, with Largan base case set at 25% of the TAM.

Core smartphone business stays soft into 2027

Q326 sales of NT$15.7bn, +15% QoQ/-11% YoY missedUBSe/Street +36%/+33% on weaker flagship smartphone demand. GMs fell 7.2ppts QoQ to 42.2%, lowest in 13 years on renewable energy costs and low variable-aperture yields, dropping OpM to 17 year lows at 30.3% and EPS to NT$39.6, below UBS/street NT$51.50/NT$49.69. We cut Q4 to NT$16.7bn +7% QoQ, 11% below consensus as its customer trimmed orders after a pricier iPhone 18 launch and possible lag to Genius which is picking up periscope share and outgrew in Q3 (+1% YoY vs. -11% YoY). We cut 2027 from NT$77bn to NT $67bn, before a recovery to NT$79bn in 2028 with the next periscope upgrade. We see GM recovering to 47.5%/48.9% in 2027-28 as yields improve and energy costs pass through in 2028 pricing.

Maintain Sell and our NT$5,000 target

We cut 2026E/27E EPS by 14%/16% to NT$160/NT$180 and introduce 2028E EPS of NT$220 for the core smartphone business. We value the lens business at NT$3,300 (15x 2028E ex-CPO EPS of NT$220, the middle of Largan's 13–18x historical range) and CPO option at NT$1,700 or US$7bn (49x our base-case 2028E CPO EPS of NT$35, near the top of optical peers' 14–54x). Our base case gives Largan 25% of a 12.3mn-unit, US $1.85bn FAU market at US$150 per unit and a 50% gross margin. That credits it with yields and margins it still needs to prove in a quarter when yields and energy costs cut 7.2ppts from its lens margin. At ~NT$6,660, the shares imply ~NT$69 of 2028E CPO EPS, about twice our base case.

Highlights (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	48,842	59,458	61,148	61,550	67,438	79,464	78,546	81,569
EBIT (UBS)	17,807	24,033	23,558	21,139	24,150	30,110	29,487	30,648
Net earnings (UBS)	18,047	22,480	23,613	20,953	23,567	28,805	28,397	29,473
EPS (UBS, diluted) (NT$)	135.22	168.43	176.92	160.04	180.00	220.00	216.89	225.11
DPS (net) (NT$)	85.50	67.50	97.50	80.00	81.97	90.34	110.41	108.85
Net (debt) / cash	111,655	117,781	122,309	109,131	120,267	127,842	144,348	160,697

Profitability/valuation	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
EBIT (UBS) margin %	36.5	40.4	38.5	34.3	35.8	37.9	37.5	37.6
ROIC (EBIT) %	39.3	46.2	40.9	30.3	30.1	35.1	33.1	35.3
EV/EBITDA (UBS core) x	7.7	7.1	6.0	24.7	21.7	18.1	17.7	16.6
P/E (UBS, diluted) x	16.2	15.1	13.3	41.6	37.0	30.3	30.7	29.6
Equity FCF (UBS) yield %	2.0	4.3	3.0	2.1	2.5	2.2	3.6	3.5
Dividend yield (net) %	3.9	2.7	4.1	1.2	1.2	1.4	1.7	1.6

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 6,660.00 on 12-Oct-2026 06:29:19 CST

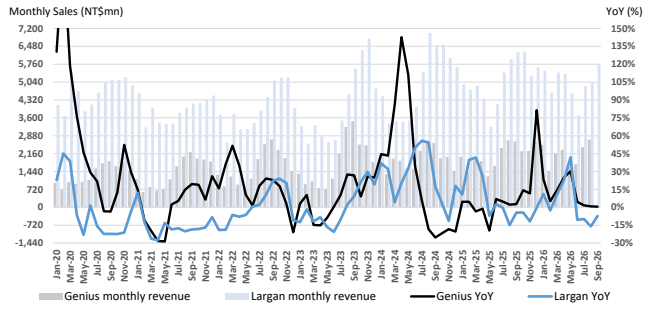
This report has been prepared by UBS Securities Pte. Ltd., Taipei Branch. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 13.**

Core smartphone lens seeing weakness with declining margins

Q326 sales was reported light at NT$15.7bn, +15% QoQ missing UBSe/Street +36%/ +33% amid weaker flagship smartphone demand. GMs fell 720bp to 42.2%, lowest in 13 years on renewable energy costs and low variable-aperture yields, dropping OpM to 17 year lows at 30.3% and EPS to NT$39.6, below UBS/street NT$51.50/NT$49.69. Renewable-energy costs are structural under Largan's long-term power commitments and rise with Apple's share of the mix; with 2027 pricing set, pass-through is more likely in 2028.

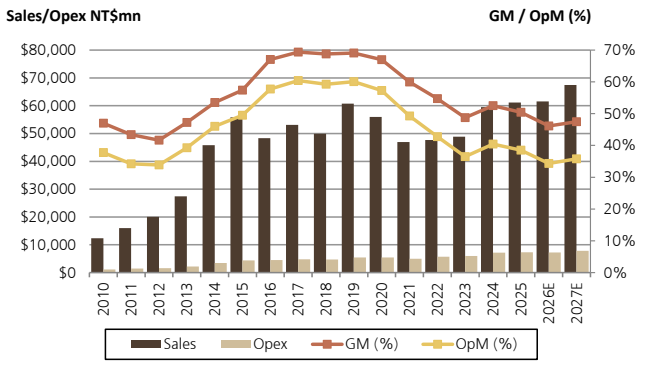
We cut Q4 to NT$16.7bn +7% QoQ, 11% below consensus as its customer trimmed orders after a pricier iPhone 18 launch and possible lag to Genius which is picking up periscope share and outgrew in Q3 (flat vs. -11% YoY). October revenue should be flat MoM and November lower as the key customer has shortened its forecast after cutting flagship orders in September. The spring-2027 model starts shipping in October but will not offset the flagship decline.

Figure 1: Genius outpacing Largan in recent months



Source: Company data

Figure 2: Largan margins dropping in recent years



Source: Company data, UBS estimates

We cut 2026E/27E revenue 9%/13% to NT$61.6bn/NT$67.4bn and EPS 14%/16% to NT$163/NT$180, and introduce 2028E revenue of NT$79.5bn, +18% YoY for the new smartphone lens upgrade, taking EPS to NT$220. Our estimates exclude CPO, which we value separately under base, bull and bear scenarios. Including our base case, 2008E EPS would be NT$255, 6% above the NT$240 consensus which likely includes some CPO. We see GM recovering to 47.5%/48.9% in 2027-28 as yields improve and energy costs pass-through in 2028 pricing. With a still a range of outcomes on 2028 CPO, we keep it in our valuation as a scenario for our upside/downside/base cases.

Figure 3: UBS vs. VA Consensus Q126A-Q426E and 2025A-28E

	Q126	Q226	Q326E			Q426E			2025	2026E			2027E			2028E		
(NT$m)	Actual	Actual	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA	Actual	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA
Revenue	**15,544**	**13,665**	**15,651**	**18,566**	**18,147**	**16,690**	**20,094**	**16,173**	**61,148**	**61,550**	**67,869**	**61,033**	**67,438**	**77,189**	**68,359**	**79,464**	**83,044**	**115,261**
- QoQ / YoY	-9.7%	-12.1%	14.5%	35.9%	32.8%	6.6%	28.4%	3.3%	2.8%	0.7%	11.0%	-0.2%	9.6%	13.7%	12.0%	17.8%	7.6%	68.6%
Gross profit	**7,680**	**6,753**	**6,607**	**9,242**	**8,968**	**7,346**	**10,044**	**6,980**	**30,837**	**28,386**	**33,718**	**28,019**	**32,008**	**38,405**	**30,319**	**38,823**	**42,004**	**53,520**
- Gross margin	49.4%	49.4%	42.2%	49.8%	49.4%	44.0%	50.0%	43.2%	50.4%	46.1%	49.7%	45.9%	47.5%	49.8%	44.4%	48.9%	50.6%	46.4%
Operating profit	**5,812**	**5,148**	**4,735**	**7,144**	**6,819**	**5,443**	**7,833**	**5,105**	**23,558**	**21,139**	**25,937**	**20,800**	**24,150**	**29,796**	**21,482**	**30,110**	**34,945**	**39,917**
- OP margin	37.4%	37.7%	30.3%	38.5%	37.6%	32.6%	39.0%	31.6%	38.5%	34.3%	38.2%	34.1%	35.8%	38.6%	31.4%	37.9%	42.1%	34.6%
Non-op profit	**1,476**	**951**	**1,357**	**1,027**	**1,071**	**1,064**	**1,064**	**1,477**	**2,342**	**4,847**	**4,518**	**5,261**	**4,467**	**4,467**	**4,734**	**4,595**	**4,595**	**4,937**
Pre-tax profit	**7,288**	**6,099**	**6,092**	**8,171**	**7,889**	**6,507**	**8,897**	**6,582**	**25,900**	**25,986**	**30,455**	**26,061**	**28,618**	**34,263**	**26,543**	**34,705**	**39,540**	**44,854**
Tax	**-1,112**	**-1,373**	**-853**	**-1,385**	**-1,250**	**-1,041**	**-1,499**	**-1,306**	**-4,340**	**-4,379**	**-5,369**	**-4,763**	**-5,050**	**-6,244**	**-5,150**	**-5,900**	**-6,722**	**-8,572**
- Tax rate	15.3%	22.5%	14.0%	17.0%	15.8%	16.0%	16.9%	19.8%	16.8%	16.9%	17.6%	18.3%	17.6%	18.2%	19.4%	17%	17%	19%
Net profit	**6,123**	**4,670**	**5,182**	**6,736**	**6,614**	**5,409**	**7,348**	**5,275**	**21,275**	**21,385**	**24,877**	**21,298**	**23,567**	**28,019**	**21,414**	**28,805**	**32,818**	**36,282**
EPS (NT$)	**46.63**	**35.70**	**39.62**	**51.50**	**49.69**	**41.36**	**56.18**	**40.04**	**159.41**	**163.33**	**190.00**	**161.25**	**180.00**	**214.00**	**161.64**	**220.00**	**250.66**	**275.28**

Source: Company data, Visible Alpha, UBS estimates. Note: EPS based on reported and may differ with cover page diluted EPS.

CPO fiber array qualified, but rising pass-through as it upgrades to FAU production

Largan's fiber array (FA) has passed qualification with a volume customer, and management expects mass-production capacity by mid-2027 in a newly completed fab dedicated to FA/FAU, using in-house equipment. The company's next step is to produce the full detachable FAU mounted on a carrier, a board with three alignment beams that sits on the optical engine's receptor. Largan can build FAUs now and says the customer

would prefer it to supply the full assembly, but qualification is pending with the customer preferring FAU and carrier sampled separately.

Even as its potential to supply the full FAU improve, Largan will buy in most of the content. The customer has qualified another PMLA vendor and prefers semiconductor-made optics; V-grooves and carrier beams and boards are outsourced and the customer does not want Largan to make MT ferrules. Management described its role as closer to a contractor and said margins hinge on yields, as the FA carries a high ASP but also a large bill of materials. Supplying the complete FAU could raise billed content, but a larger purchased bill of materials also dilutes the relationship between revenue and value added. Our full-FAU ASP assumption requires Largan to capture that invoice scope.

Next year's product is grating-coupled and works for both CPO and NPO while edge-coupled designs are being explored with the customer. The current product does not go directly to end customers, but management expects to work directly with end customers on next-generation designs (likely 2028) as it progresses toward multi-row FAs.

To ramp production, Largan has bought several land parcels and will build new fabs from scratch, with construction finishing around 2030. It may outsource some capacity in the interim and said it cannot yet be sure capacity will meet demand.

Annual FAU TAM reaches US$1.85bn in 2028 and US$4.5bn in 2030 in our base case

We update our FAU (fiber array unit) TAM factoring CPO/NPO penetration into AI networking switches, NVIDIA multi-rack systems, scale-up on compute and switch boards, and non-NVIDIA accelerators. We assume one passive FAU assembly connecting a fiber array to a photonic engine with one unit per optical engine.

We assume one FAU per optical engine and 32 per switch ASIC in our base case, matching NVIDIA's 32x3.2T engines on its 102.4T Spectrum-X package. Broadcom's 102.4Tb/s Davisson uses 16 engines at 6.4Tb/s, so our bear case assumes and even mix of these two designs (24 per ASIC). We note that higher switch bandwidth does not automatically increase FAU count if engine bandwidth rises at the same rate. Our 2030 assumption maintains the 32 FAUs per switch based on bandwidth rising to 204.8Tb/s ASICs using 6.4Tb/s optical engines.

Figure 4: UBS scenario based on 32 FAUs per switch although switches to date carry a range of specifications

	TH5 Bailly	TH6 Davisson	Quantum-X Q3450	Spectrum-X SN6810	Spectrum-X SN6800
Year of Introduction	**2023**	**H225**	**H225**	**H226**	**H226**
Switch ASIC	Tomahawk 5	Tomahawk 6	Quantum-3	Spectrum 6	Spectrum 6
Bandwidth per Switch box	51.2T	102.4T	115.2T	102.4T	409.6T
Switch ASICs per box	1	1	4	1	4
OE bandwidth (Tb/s each way)	6.4T	6.4T	1.6T	3.2T	3.2T
Optical engines per ASIC	8	16	18	32	32
FAU per Engine	1	1	1	1	1
FAU per Switch	**8**	**16**	**72**	**32**	**128**

Source: Company Data, UBS

Our base case for the FAU market sums up 4 opportunities for CPO/NPO attach:

1. Scale-out and scale-across switches

We estimate annual switch ASIC shipments x CPO + NPO penetration x optical engines per ASIC x FAUs per engine. In 2028, 25% CPO/NPO penetration or 1.0m 102.4T switch ASICs at 32 FAUs each produces 8.0m FAUs. In 2030, 55% of 1.1mn 204.8T switch ASICs at 32 FAUs drives a market size of 19.4mn FAUs.

Figure 5: Scale-out/scale-across could drive 8.0/19.4mn FAUs in 2028/2030

	2028			2030			Assumption
Scale out/Scale across	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
Switch ASIC shipments (k units)	800	1,000	1,050	880	1,100	1,155	
NPO penetration	3%	5%	7%	8%	10%	12%	Optics on the board beside the ASIC
CPO penetration	10%	20%	25%	35%	45%	55%	Optics inside the switch package.
NPO/CPO switch ASICs (k units)	**104**	**250**	**336**	**378**	**605**	**774**	
Bandwidth per Switch (Tbps)	102.4	102.4	102.4	204.8	204.8	204.8	
Optical engine (Tb)	3.2	3.2	3.2	6.4	6.4	6.4	
FAUs per switch ASIC	24	32	32	24	32	32	NVIDIA: 32 OE per ASIC, Broadcom 16
Scale out/Scale across # of FAUs (mn)	**2.5**	**8.0**	**10.8**	**9.1**	**19.4**	**24.8**	

Source: UBS

2. NVIDIA's multi-rack connections for Rubin Ultra and Feynman

NVIDIA at its 2026 GTC San Jose proposed a multi-rack option to network together 8 Oberon NVL72 racks for the Rubin Ultra generation and 8 Kyber NVL144 racks. Feedback at the time showed possible transceiver or NPO connections for Rubin Ultra and CPO connections for Feynman. Our 2028 base case uses 110,000 racks, 15% multi-rack adoption and 75% CPO/NPO penetration, giving approximately 12,400 connected racks.

For 2030, our rack count falls to 35,000 racks as we assume Kyber would have 4x the number of GPUs, equivalent to 140k Oberon equivalents. We assume the 35,000 racks at 55% multi-rack adoption x 80% CPO/NPO attach gives 15,400 connected racks. We assume about 216 and 576 inter-rack FAUs per rack, respectively, driving a market opportunity of 2.7mn FAUs in 2028 and 8.9mn in 2030.

Figure 6: NVIDIA rack connected pods could drive 2.7mn/8.9mn FAUs in 2028/2030

	2028			2030			Assumption
NVIDIA Pod Connected GPUs	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
NV platform	Oberon / Rubin Ultra			Kyber / Feynman			
NV racks shipments (k)	95	110	125	30	35.0	40	UBS rack assumptions
8-rack pods penetration (%)	10%	15%	20%	40%	55%	70%	
NPO/CPO penetration within pods (%)	50%	75%	80%	65%	80%	95%	Pluggable FAUs excluded from this model
NPO/CPO connected racks (k)	**4.8**	**12.4**	**20.0**	**7.8**	**15.4**	**26.6**	
GPUs per rack	72	72	72	144	144	144	
Switches per rack	72	72	72	144	144	144	72 ASICs in Oberon, 144 per Kyber
Optical engine (Tb) bandwidth	3.2	3.2	3.2	6.4	6.4	6.4	
FAUs attached per switch	2.5	3.0	4.0	2.5	4.0	4.5	3.2T OE in 2028; 6.4T in 2030 (each way)
NVLink bandwidth per NVIDIA GPU	14.4	14.4	14.4	28.8	28.8	28.8	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
FAUs per connected racks	**180**	**216**	**288**	**360**	**576**	**648**	
NV scale up # of FAUs (mn)	**0.9**	**2.7**	**5.8**	**2.8**	**8.9**	**17.2**	

Source: UBS. Switch count increases to connect to other racks

3. Initial CPO/NPO optical connections on compute and switch boards by 2030

We include by 2030 CPO native optical connections on compute and switch boards in 5% of racks each adding 1,152 FAUs across the GPUs and switches or 2.0mn FAUs by 2030.

Figure 7: NVIDIA compute board optics could drive 2.0mn FAUs in 2030

	2028			2030			Assumption
NVIDIA Compute Board Optics	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
NV racks shipments (k)	95	110	125	30	35.0	40	UBS rack assumptions
Optical GPU penetration	0%	0%	0%	3%	5%	15%	CPU-native GPUs
Compute tray FAUs #	0	0	0	144	144	144	36 trays x 16
GPUs per rack	72	72	72	144	144	144	
Optical engine (Tb)	3.2	3.2	3.2	6.4	6.4	6.4	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
NVLink bandwidth per NVIDIA GPU	14.4	14.4	14.4	28.8	28.8	28.8	
FAUs per connected racks	0	0	0	720	1,152	1,296	
NV optical GPU # of FAUs (mn)	**0.0**	**0.0**	**0.0**	**0.6**	**2.0**	**7.8**	

Source: UBS.

4. Non NVIDIA scale-up opportunity

Non-NVIDIA scale-up uses accelerator rack shipments x optical attach rate x FAUs per rack, producing about 0.5m units in 2028 and 3.7m in 2030.

Figure 8: ASIC market could drive additional compute board optics could drive 3.7mn FAUs in 2030

	2028			2030			Assumption
Non-NV scale up	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
ASIC rack shipments (k)	43.6	48.5	50.9	76.6	85.1	89.3	*Trainium 4/5/6, MI500, MI600, Jalapeno
Addressable NPO/CPO attach rate (%)	0%	5%	10%	5%	15%	25%	
NPO/CPO connected racks (k)	**0.0**	**2.4**	**5.1**	**3.8**	**12.8**	**22.3**	
Accelerators per rack	72	72	72	72	72	72	Average accelerators per rack
FAUs per accelerator	2.5	3.0	4.0	2.5	4.0	4.5	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
Scale-up per non-NVIDIA accelerator (Tb)	14.4	14.4	14.4	28.8	28.8	28.8	
FAUs per connected racks	180	216	288	180	288	324	
Non-NV # of FAUs (mn)	**0.0**	**0.5**	**1.5**	**0.7**	**3.7**	**7.2**	

Source: UBS

Overall market opportunity: 12.3mn/35.6mn generates a US$1.85bn/US $4.5bn TAM in 2028/2030

We sum four pools: scale-out/scale-across switches, NVIDIA multi-rack pods, CPO native NVIDIA GPUs and non-NVIDIA scale-up. We add 10%/5% for FAUs lost at downstream attach and test and for field spares, driving a market size of in our base case of 12.3mn FAUs in 2028 and 35.6mn in 2030. At US$150/US$125 per FAU, that is a US$1.85bn market in 2028 and US$4.5bn in 2030 (bear/bull US$0.5bn/US$3.5bn in 2028; US $1.4bn/US$9.0bn in 2030). The expansion to 2030 comes from wider optical adoption and greater content in scale-up systems, despite an assumed FAU ASP decline from US $150 to US$125.

Figure 9: Summary of FAU drivers: 12.3mn/35.6mn base case in 2028-2030

	2028			2030		
Summary of FAU drivers	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**
Scale out/Scale across # of FAUs (mn)	2.5	8.0	10.8	9.1	19.4	24.8
NV scale up # of FAUs (mn)	0.9	2.7	5.8	2.8	8.9	17.2
NV optical GPU # of FAUs (mn)	0.0	0.0	0.0	0.6	2.0	7.8
Non-NV # of FAUs (mn)	0.0	0.5	1.5	0.7	3.7	7.2
Downstream yield loss and spares	10%	10%	10%	5%	5%	5%
FAUs (mn units)	**3.7**	**12.3**	**19.8**	**13.9**	**35.6**	**59.9**
FAU ASP	$125	$150	$175	$100	$125	$150
Total FAU TAM (US$bn)	**$0.5**	**$1.8**	**$3.5**	**$1.4**	**$4.5**	**$9.0**

Source: UBS

Scenario analysis on Largan's CPO opportunity

Our base case gives Largan the benefit of the doubt for successfully ramping up a full FAU production and capturing 25% of the overall FAU TAM, yielding NT$35 of 2028E EPS, about 16% of our 2028E lens EPS. A 25% share means 3.1mn FAUs in its first full year of production: about 29% of NVIDIA-linked units and roughly three-quarters of the allocation our checks suggest its module partner holds. We book these at the full US $150 FAU price and 50% gross margin, allowing for an early-supply premium even though the optics, V-grooves and carrier parts are bought in. That gives 2028E CPO sales of US$462mn, rising to US$1.1bn and NT$84 of EPS in 2030 on a larger market.

Our bear case (15% share, 45% GM) gives NT$5 of CPO EPS in 2028E and NT$14 in 2030E; our bull case (35% share, 55% GM) gives NT$103 and NT$267. Each 5ppt of share moves 2028E CPO EPS by ~NT$6.9 (~NT$340/share at 49x), and each 5ppt of gross margin by ~NT$4.6 (~NT$225/share). If gross margin were 35% rather than 50%, holding operating expenses at 12% of sales, EPS would fall to approximately NT$21. A next-generation shift in lens or coupling design could also change content and alignment requirements including a potential metalens design which could invite in different suppliers.

Our 2030 base case implies ~8.9mn Largan FAUs a year (21mn in the upside case), though with risks around competition in a crowded field, yields and technology. A shift to semiconductor-made optics or metalens would loosen alignment tolerances and erode the precision premium behind Largan's pitch. The upside lies in direct work with cloud providers on multi-row FAs from 2028, where Largan's multi-layer alignment is most differentiated.

Figure 10: Largan downside/base/upside cases for 2028 and 2030

Largan CPO	2028			2030		
Scenario analysis	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**
FAUs (mn units)	3.7	12.3	19.8	13.9	35.6	59.9
FAU ASP	125	150	175	100	125	150
Total FAU TAM (US$mn)	**$461**	**$1,847**	**$3,461**	**$1,389**	**$4,452**	**$8,980**
Largan Sales (US$mn)	**$69.1**	**$461.9**	**$1,211.3**	**$208.3**	**$1,113.1**	**$3,142.8**
Largan share of the TAM (%)	15%	25%	35%	15%	25%	35%
GM %	45%	50%	55%	45%	50%	55%
Pretax %	33%	38%	43%	33%	38%	43%
Pretax income (US$mn)	**$22.8**	**$175.5**	**$520.8**	**$68.7**	**$423.0**	**$1,351.4**
Tax	18%	18%	18%	18%	18%	18%
Net income (US$mn)	**$18.7**	**$143.9**	**$427.1**	**$56.4**	**$346.8**	**$1,108.2**
Net income (NT$mn)	**$589**	**$4,533**	**$13,453**	**$1,776**	**$10,925**	**$34,907**
EPS (NT$)	**$5**	**$35**	**$103**	**$14**	**$84**	**$267**
Shares	130.8	130.8	130.8	130.8	130.8	130.8

Source: UBS

Valuation and investment implications

We roll our valuation forward to 2028E and value Largan on a sum of the parts. We value the lens business at NT$3,300 (15x 2028E EPS, the middle of Largan's 13–18x historical range) and CPO at NT$1,700 (49x our base-case 2028E CPO EPS of NT$35, near the top of optical peers' 14–54x). Our base case gives Largan 25% of a 12.3mn-unit, US$1.85bn FAU market at US$150 per unit and a 50% gross margin. The 49x multiple also assigns a substantial value to a business whose complete assembly still requires qualification and whose margins still need to demonstrate can reach similar levels to its mature smartphone business. At ~NT$6,660 recent share price level, the shares imply ~NT$69 of 2028E CPO EPS, about twice our base case.

Upside risks include faster full-FAU qualification, firm volume commitments from additional customers, stronger lens demand and better yields. Near-term catalysts are the customer's capacity forecast and its visit to the new fab in the coming month. A large forecast or faster FAU plus carrier qualification would be upside risks to our sell.

Figure 11: UBS SOTP valuation for Largan

(NT$)		UBSe			Prior UBS		Upside Case			Downside Case		
	2025	**2026**	**2027**	**2028**	**2026**	**2027**	**2026**	**2027**	**2028**	**2026**	**2027**	**2028**
Lens Business ex-CPO												
Sales	$61,148	$61,550	$67,438	$79,464	$67,869	$77,189	$62,166	$70,135	$82,642	$59,704	$64,066	$75,491
YoY Growth	2.8%	0.7%	9.6%	17.8%	11.0%	13.7%	1.7%	12.8%	17.8%	-2.4%	7.3%	17.8%
Gross Profit	**$30,837**	**$28,386**	**$32,008**	**$38,823**	**$33,718**	**$38,405**	**$28,981**	**$34,410**	**$41,699**	**$26,997**	**$29,446**	**$35,750**
GM%	**50.4%**	**46.1%**	**47.5%**	**48.9%**	**49.7%**	**49.8%**	**46.6%**	**49.1%**	**50.5%**	**45.2%**	**46.0%**	**47.4%**
Opex	$7,279	$7,247	$7,857	$8,714	$7,781	$8,609	$7,283	$8,015	$8,888	$7,138	$7,661	$8,496
Op Profit	**$23,558**	**$21,139**	**$24,150**	**$30,110**	**25,937**	**29,796**	**21,697**	**$26,396**	**$32,811**	**19,858**	**$21,785**	**$27,254**
OpM%	**38.5%**	**34.3%**	**35.8%**	**37.9%**	**38.2%**	**38.6%**	**34.9%**	**37.6%**	**39.7%**	**33.3%**	**34.0%**	**36.1%**
Non-Op	$2,342	$4,847	$4,467	$4,595	$4,518	$4,467	$4,847	$4,467	$4,595	$4,847	$4,467	$4,595
Pretax	$25,900	$25,986	$28,618	$34,705	$30,455	$34,263	$26,545	$30,863	$37,405	$24,706	$26,253	$31,849
Tax	($4,340)	($4,379)	($5,050)	($5,900)	($5,369)	($6,244)	($4,473)	($5,447)	($6,601)	($4,163)	($4,633)	($5,621)
Net income	**$21,275**	**$21,385**	**$23,567**	**$28,805**	**$24,877**	**$28,019**	**$21,849**	**$25,416**	**$30,804**	**$20,321**	**$21,620**	**$26,228**
EPS	**$159.41**	**$163.33**	**$180.00**	**$220.00**	**$190.00**	**$214.00**	**$166.88**	**$194.12**	**$235.28**	**$155.20**	**$165.13**	**$200.32**
Shares	133	131	131	131	131	131	131	131	131	131	131	131
EPS vs. base case							**-94.1%**	**7.8%**	**6.9%**	**-5.0%**	**-8.3%**	**-8.9%**
Current Price		**$6,660**										
P/E				15		15			16			14
Implied Valuation				**$3,300**		**$3,270**			**$3,670**			**$2,810**
CPO Scenario												
EPS				**$35**		**$35**			**$103**			**$5**
P/E				49		50			52			47
Implied Valuation				**$1,700**		**$1,730**			**$5,300**			**$210**
Total Largan												
Implied Share Price				**$5,000**		**$5,000**			**$8,970**			**$3,020**
Upside/Downside				**-25%**					**35%**			**-55%**

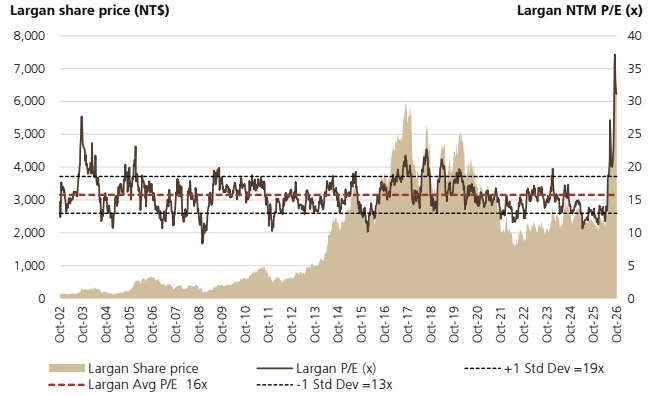
Source: Company data, UBS estimates

Figure 12: Optical component peers average 27x 2028 with a broad range of 12-75x 2028 P/E

Optical Company	Ticker	Market cap (US$m)	Price 11-Oct (LC$)	Price target (LC$)	P/E target (x)	Rating	EPS (LC$) 2025	EPS (LC$) 2026E	EPS (LC$) 2027E	EPS (LC$) 2028E	Earnings growth (%) 2026E	Earnings growth (%) 2027E	Earnings growth (%) 2028E	PE (x) 2026E	PE (x) 2027E	PE (x) 2028E	ROE (%) 2026E	ROE (%) 2027E	ROE (%) 2028E	Historical P/E (x) -1 Std Dev	Historical P/E (x) Avg.	Historical P/E (x) +1 Std Dev
Lens																						
Largan	3008.TW	27,378	6,660.0	5,000	23.4	Sell	176.92	188.21	214.00	250.66	4.4	13.7	17.1	35.4	31.1	26.6	12.7	13.7	14.8	13.1	15.9	18.8
Genius	3406.TW	3,741	1,060.0	1,250	16.9	Buy	36.28	38.72	41.47	47.75	6.7	7.1	15.1	27.4	25.6	22.2	15.8	15.6	16.4	1.1	2.3	3.5
Average											**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**
CPO Supply Chain																						
Lumentum	LITE.O	93,344	1,103.4	960	40.4	Neutral	2.12	10.52	23.78	35.81	434.4	160.4	51.7	52.0	56.0	37.9	27.1	36.1	37.9	8.5	19.4	30.3
Corning	GLW.N	137,095	156.7	201	42.1	Buy	2.57	3.34	4.78	7.12	30.6	43.5	47.3	47.7	33.4	22.4	22.9	31.8	41.1	-168.2	32.9	234.0
Coherent	COHR.N	123,129	312.6	-	-	-	4.49	7.42	11.74	16.82	86.0	64.0	42.0	42.1	26.6	18.6	13.5	19.7	23.8	14.1	19.6	25.1
Senko	9069.T	416,040	2,368.0	-	-	-	119.98	140.89	159.70	171.34	19.5	12.8	7.4	16.8	14.8	13.8	10.4	11.3	11.6	9.2	11.6	14.0
TFC	300394.SZ	256,388	235.0	-	-	-	2.02	3.32	5.36	6.88	65.5	60.7	28.5	70.8	43.9	34.2	47.0	47.4	42.1	21.2	33.9	46.5
FOCI	3363.TWO	75,798	667.0	-	-	-	0.85	0.22	13.89	43.01	(66.3)	6684.0	210.2	3,102.3	48.0	15.5	-1.6	39.2	74.6	51.6	115.2	178.8
EZconn	6442.TW	146,044	1,785.0	-	-	-	20.95	62.01	130.00	NULL	206.9	109.6	n.a.	28.8	13.7	n.a.	56.1	63.9	NULL	12.9	23.7	34.5
Himax	HIMX.N	2,320	13.3	-	-	-	0.32	0.52	1.01	1.96	62.3	88.2	93.5	25.5	13.2	6.8	10.5	14.5	26.1	0.6	21.5	42.3
Browave	3163.TWO	65,204	689.0	-	-	-	4.36	9.33	21.17	NULL	n.a.	n.a.	n.a.	73.8	32.5	n.a.	22.2	38.6	NULL	10.1	32.2	54.2
VisEra	6789.TW	153,102	481.0	-	-	-	3.60	6.13	7.45	8.95	70.9	21.5	20.2	78.4	64.6	53.7	10.7	13.3	12.2	31.3	50.4	69.6
Average											**101.1**	**805.0**	**62.6**	**353.8**	**34.7**	**25.4**	**21.9**	**31.6**	**33.7**	**-0.9**	**36.0**	**72.9**
Optical Average											**76.4**	**606.4**	**50.7**	**325.1**	**35.5**	**27.2**	**19.9**	**27.5**	**28.6**	**1.2**	**30.5**	**59.8**

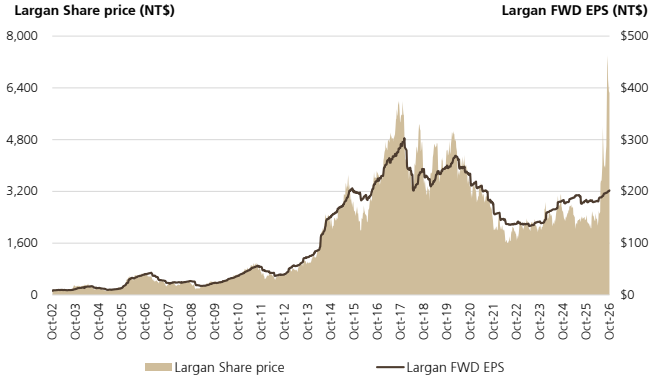
Source: UBSe, Eikon

Figure 13: Share price rally has re-rated Largan well above its historical range



Source: LSEG Workspace

Figure 14: Largan's share price has run well ahead of forward EPS as investors price in CPO



Source: LSEG Workspace

Figure 15: UBS P&L for Largan - mild growth at lower margins in recent years

Ticker		3008.TW	Current Price (NT$)			$6,660
Shares outstanding (mn)		130.8	Mkt cap (NT$ mn)			$871,115
Year	**2022**	**2023**	**2024**	**2025**	**2026E**	**2027E**
Revenues (NT$ mn)	**$47,675**	**$48,842**	**$59,458**	**$61,148**	**$61,550**	**$67,438**
YoY Growth (%)		**2.4%**	**21.7%**	**2.8%**	**0.7%**	**9.6%**
Gross profit (NT$mn)	$26,083	$23,794	$31,209	$30,837	$28,386	$32,008
GM (%)	**54.7%**	**48.7%**	**52.5%**	**50.4%**	**46.1%**	**47.5%**
Operating profit (NT$mn)	$20,384	$17,807	$24,033	$23,558	$21,139	$24,150
OPM (%)	**42.8%**	**36.5%**	**40.4%**	**38.5%**	**34.3%**	**35.8%**
Net profit (NT$ mn)	$22,625	$17,902	$25,915	$21,275	$21,385	$23,567
EPS (NT$)	**$169.52**	**$134.13**	**$194.17**	**$159.41**	**$163.33**	**$180.00**
EPS growth (%)		-21%	45%	-18%	2%	10%
P/E (x)	**39.3**	**49.7**	**34.3**	**41.8**	**40.8**	**37.0**
Dividend yield (%)	2%	1.1%	1.2%	1.3%	2.0%	1.2%
P/B (x)	**5.7**	**5.4**	**4.8**	**4.7**	**4.4**	**4.1**
ROE (%)	15%	11%	14%	11%	11%	12%

Quarters	**Q325**	**Q425**	**Q126**	**Q226**	**Q326E**	**Q426E**
Revenues (NT$ mn)	**$17,677**	**$17,219**	**$15,544**	**$13,665**	**$15,651**	**$16,690**
QoQ Growth (%)		**-2.6%**	**-9.7%**	**-12.1%**	**14.5%**	**6.6%**
Gross profit (NT$mn)	$8,352	$8,260	$7,680	$6,753	$6,607	$7,346
GM (%)	**47.2%**	**48.0%**	**49.4%**	**49.4%**	**42.2%**	**44.0%**
Operating profit (NT$mn)	$6,265	$6,329	$5,812	$5,148	$4,735	$5,443
OPM (%)	**35.4%**	**36.8%**	**37.4%**	**37.7%**	**30.3%**	**32.6%**
Net profit (NT$ mn)	$7,080	$6,720	$6,123	$4,670	$5,182	$5,409
EPS (NT$)	**$53.05**	**$50.35**	**$45.88**	**$34.99**	**$39.58**	**$41.32**

Source: Company data, UBS estimates. Note: 2026E EPS based on reported.

Figure 16: UBS Cash Flow Summary - staying positive to invest in CPO

Annual (NT$mn)	2021	2022	2023	2024	2025	2026E	2027E	21-25 Avg
Revenue	$46,962	$47,675	$48,842	$59,458	$61,148	$61,550	$67,438	$52,817
Capital spending	$5,850	$8,256	$8,080	$11,126	$12,338	$11,721	$11,135	$9,130
Capex/revenue (%)	**12%**	**17%**	**17%**	**19%**	**20%**	**19%**	**17%**	**17%**
Dep and amort	$4,650	$5,031	$5,299	$6,038	$7,511	$9,221	$10,460	$5,706
Depr/revenue (%)	**10%**	**11%**	**11%**	**10%**	**12%**	**15%**	**16%**	**11%**
Operating cash flow	$20,878	$44,207	$18,199	$31,579	$26,080	$20,614	$33,003	$28,188
Free cash flow	**$15,028**	**$35,951**	**$10,118**	**$20,453**	**$13,743**	**$8,893**	**$21,868**	**$19,058**

FCF and Dividends	2021	2022	2023	2024	2025	2026E	2027E	21-25 Avg
FCF / share (NT$)	$112	$269	$76	$153	$103	$68	$167	$143
FCF / EV Yield (%)	**2%**	**5%**	**1%**	**3%**	**2%**	**1%**	**3%**	**3%**
FCF Yield (%)	**2%**	**4%**	**1%**	**2%**	**2%**	**1%**	**3%**	**2%**
Dividend/share (NT$)	$91.51	$109.66	$72.50	$81.00	$85.50	$131.95	$81.89	$88.03
Payout rate (%)	**50%**	**79%**	**43%**	**60%**	**44%**	**83%**	**50%**	**55%**
Dividend amount (NT$)	$12,274	$14,636	$9,676	$10,811	$11,412	$17,276	$10,722	$11,762
Pre-Dividend close (NT$)	$3,040.00	$1,990.00	$2,310.00	$3,120.00	$2,260.00	$6,660.00	$6,660.00	
Dividend Yield (%)	**3.0%**	**5.5%**	**3.1%**	**2.6%**	**3.8%**	**2.0%**	**1.2%**	**3.6%**

Source: Company data, UBS estimates

Largan Precision (3008.TW)

Income Statement (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**48,842**	**59,458**	**61,148**	**61,550**	*0.7*	**67,438**	*9.6*	**79,464**	**78,546**	**81,569**
Gross profit	23,794	31,209	30,837	28,386	*-7.9*	32,008	*12.8*	38,823	38,398	39,903
EBITDA (UBS)	**23,106**	**30,070**	**31,069**	**30,360**	*-2.3*	**34,610**	*14.0*	**41,028**	**41,108**	**42,938**
Depreciation & amortisation	(5,299)	(6,038)	(7,511)	(9,221)	*-22.8*	(10,460)	*-13.4*	(10,918)	(11,621)	(12,290)
EBIT (UBS)	**17,807**	**24,033**	**23,558**	**21,139**	*-10.3*	**24,150**	*14.2*	**30,110**	**29,487**	**30,648**
Associates & investment income	212	0	3	12	*352.1*	11	*-7.1*	11	11	11
Other non-operating income	294	303	419	458	*9.5*	200	*-56.4*	200	200	200
Net interest	3,934	4,404	4,259	3,946	*-7.3*	4,256	*7.9*	4,384	4,516	4,651
Exceptionals (incl goodwill)	(145)	3,435	(2,338)	432	*-*	0	*-*	0	0	0
Pre-tax profit	**22,102**	**32,174**	**25,900**	**25,986**	*0.3*	**28,618**	*10.1*	**34,705**	**34,213**	**35,510**
Tax	(4,200)	(5,963)	(4,340)	(4,379)	*-0.9*	(5,050)	*-15.3*	(5,900)	(5,816)	(6,037)
Profit after tax	**17,902**	**26,211**	**21,560**	**21,607**	*0.2*	**23,567**	*9.1*	**28,805**	**28,397**	**29,473**
Preference dividends	0	0	0	0	*-*	0	*-*	0	0	0
Minorities	0	(296)	(285)	(222)	*22.1*	0	*-*	0	0	0
Extraordinary items	0	0	0	0	*-*	0	*-*	0	0	0
Net earnings (local GAAP)	**17,902**	**25,915**	**21,275**	**21,385**	*0.5*	**23,567**	*10.2*	**28,805**	**28,397**	**29,473**
Net earnings (UBS)	**18,047**	**22,480**	**23,613**	**20,953**	*-11.3*	**23,567**	*12.5*	**28,805**	**28,397**	**29,473**
Tax rate (%)	19.0	18.5	16.8	16.9	*0.6*	17.6	*4.7*	17.0	17.0	17.0

Per Share (NT$)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	135.22	168.43	176.92	160.04	*-9.5*	180.00	*12.5*	220.00	216.89	225.11
EPS (local GAAP, diluted)	134.13	194.17	159.41	163.33	*2.5*	180.00	*10.2*	220.00	216.89	225.11
EPS (UBS, basic)	135.22	168.43	176.92	160.04	*-9.5*	180.00	*12.5*	220.00	216.89	225.11
DPS (net) (NT$)	85.50	67.50	97.50	80.00	*-17.9*	81.97	*2.5*	90.34	110.41	108.85
Cash EPS (UBS, diluted) [1]	174.92	213.67	233.20	230.47	*-1.2*	259.89	*12.8*	303.39	305.65	318.97
Book value per share	1,234.64	1,382.91	1,430.07	1,518.08	*6.2*	1,616.11	*6.5*	1,745.87	1,849.64	1,963.23
Average shares (diluted)	133	133	133	131	*-1.9*	131	*0.0*	131	131	131

Balance Sheet (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	111,655	117,985	122,309	109,131	*-10.8*	120,267	*10.2*	127,842	144,348	160,697
Other current assets	22,666	27,779	31,977	42,218	*32.0*	45,034	*6.7*	56,368	55,712	57,851
Total current assets	**134,321**	**145,764**	**154,285**	**151,350**	*-1.9*	**165,301**	*9.2*	**184,210**	**200,060**	**218,547**
Net tangible fixed assets	41,135	46,936	51,472	54,428	*5.7*	55,103	*1.2*	54,763	53,191	50,448
Net intangible fixed assets	0	0	0	0	*-*	0	*-*	0	0	0
Investments / other assets	19,682	23,827	15,029	23,254	*54.7*	24,367	*4.8*	24,215	24,225	24,236
Total assets	**195,138**	**216,527**	**220,787**	**229,032**	*3.7*	**244,770**	*6.9*	**263,188**	**277,476**	**293,232**
Trade payables & other ST liabilities	29,517	30,375	29,749	30,078	*1.1*	32,982	*9.7*	34,573	35,276	36,158
Short term debt	0	203	0	0	*-*	0	*-*	0	0	0
Total current liabilities	**29,517**	**30,578**	**29,749**	**30,078**	*1.1*	**32,982**	*9.7*	**34,573**	**35,276**	**36,158**
Long term debt	0	0	0	0	*-*	0	*-*	0	0	0
Other long term liabilities	110	561	171	194	*13.7*	194	*0.0*	31	31	31
Preferred shares	0	0	0	0	*-*	0	*-*	0	0	0
Total liabilities (incl pref shares)	**29,627**	**31,139**	**29,919**	**30,272**	*1.2*	**33,176**	*9.6*	**34,604**	**35,307**	**36,189**
Common s/h equity	**165,510**	**185,388**	**190,868**	**198,760**	*4.1*	**211,594**	*6.5*	**228,584**	**242,170**	**257,042**
Minority interests	0	0	0	0	*-*	0	*-*	0	0	0
Total liabilities & equity	**195,138**	**216,527**	**220,787**	**229,032**	*3.7*	**244,770**	*6.9*	**263,188**	**277,476**	**293,232**

Cash Flow (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	17,902	25,915	21,275	21,385	*0.5*	23,567	*10.2*	28,805	28,397	29,473
Depreciation & amortisation	5,299	6,038	7,511	9,221	*22.8*	10,460	*13.4*	10,918	11,621	12,290
Net change in working capital	(8,990)	(6,633)	(7,328)	(1,082)	*85.2*	(6,064)	*NM*	(9,743)	1,359	(1,256)
Other operating	(212)	295	282	210	*-25.5*	5,040	*NM*	(11)	(11)	(11)
Operating cash flow	**13,999**	**25,615**	**21,740**	**29,735**	*36.8*	**33,003**	*11.0*	**29,968**	**41,366**	**40,496**
Tangible capital expenditure	(8,080)	(11,126)	(12,338)	(11,721)	*5.0*	(11,135)	*5.0*	(10,578)	(10,049)	(9,547)
Intangible capital expenditure	0	0	0	0	*-*	0	*-*	0	0	0
Net (acquisitions) & disposals	(2,111)	(13)	(1,200)	962	*-*	0	*-*	0	0	0
Other investing	(959)	(4,927)	4,302	(965)	*-*	0	*-*	0	0	0
Investing cash flow	**(11,151)**	**(16,066)**	**(9,236)**	**(11,724)**	*-26.9*	**(11,135)**	*5.0*	**(10,578)**	**(10,049)**	**(9,547)**
Equity dividends paid	(9,676)	(10,811)	(11,412)	(17,276)	*-51.4*	(10,732)	*37.9*	(11,816)	(14,811)	(14,601)
Share issues / (buybacks)	0	0	0	0	*-*	0	*-*	0	0	0
Other financing	(34)	1,255	(2,937)	(4,133)	*-40.7*	0	*-*	0	0	0
Change in debt & pref shares	(19)	212	(212)	0	*-*	0	*-*	0	0	0
Financing cash flow	**(9,729)**	**(9,344)**	**(14,561)**	**(21,408)**	*-47.0*	**(10,732)**	*49.9*	**(11,816)**	**(14,811)**	**(14,601)**
Cash flow inc/(dec) in cash	**(6,881)**	**205**	**(2,057)**	**(3,398)**	*-65.2*	**11,136**	*-*	**7,575**	**16,506**	**16,349**
FX / non cash items	6,823	6,125	6,380	(9,780)	*-*	0	*-*	0	0	0
Balance sheet inc/(dec) in cash	**(59)**	**6,330**	**4,324**	**(13,177)**	*-*	**11,136**	*-*	**7,575**	**16,506**	**16,349**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

Largan Precision (3008.TW)

Valuation (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	16.4	13.1	14.8	40.8	37.0	30.3	30.7	29.6
P/E (UBS, diluted)	16.2	15.1	13.3	41.6	37.0	30.3	30.7	29.6
P/CEPS	12.6	11.9	10.1	28.9	25.6	22.0	21.8	20.9
Equity FCF (UBS) yield %	2.0	4.3	3.0	2.1	2.5	2.2	3.6	3.5
Dividend yield (net) %	3.9	2.7	4.1	1.2	1.2	1.4	1.7	1.6
P/BV	1.8	1.8	1.6	4.4	4.1	3.8	3.6	3.4
EV/revenues (core)	3.6	3.6	3.1	NM	NM	9.3	9.3	8.7
EV/EBITDA (UBS core)	7.7	7.1	6.0	24.7	21.7	18.1	17.7	16.6
EV/EBIT (core)	10.0	8.9	8.0	35.4	31.1	24.6	24.7	23.2
EV/OpFCF (core)	8.1	7.4	6.3	25.1	22.0	18.0	17.7	16.5
EV/op. invested capital	3.9	4.1	3.3	10.7	9.3	8.6	8.2	8.2

Enterprise value (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	294,465	340,820	315,906	874,578	874,578	874,578	874,578	874,578
Net debt (cash)	(111,675)	(114,718)	(120,045)	(115,720)	(114,699)	(124,054)	(136,095)	(152,522)
Buy out of minorities	0	0	0	0	0	0	0	0
Pension provisions/other	0	0	0	0	0	0	0	0
Total enterprise value	**182,790**	**226,102**	**195,861**	**758,858**	**759,879**	**750,524**	**738,483**	**722,055**
Non core assets	(4,916)	(12,460)	(8,517)	(9,934)	(9,945)	(9,955)	(9,966)	(9,976)
Core enterprise value	**177,874**	**213,642**	**187,344**	**748,924**	**749,934**	**740,568**	**728,517**	**712,079**

Growth (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	2.4	21.7	2.8	0.7	9.6	17.8	(1.2)	3.8
EBITDA (UBS)	(9.1)	30.1	3.3	(2.3)	14.0	18.5	0.2	4.5
EBIT (UBS)	(12.6)	35.0	(2.0)	(10.3)	14.2	24.7	(2.1)	3.9
EPS (UBS, diluted)	3.8	24.6	5.0	(9.5)	12.5	22.2	(1.4)	3.8
Net DPS	21.9	(21.1)	44.4	(17.9)	2.5	10.2	22.2	(1.4)

Margins & Profitability (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	48.7	52.5	50.4	46.1	47.5	48.9	48.9	48.9
EBITDA margin	47.3	50.6	50.8	49.3	51.3	51.6	52.3	52.6
EBIT (UBS) margin	36.5	40.4	38.5	34.3	35.8	37.9	37.5	37.6
Net earnings (UBS) margin	37.0	37.8	38.6	34.0	34.9	36.2	36.2	36.1
ROIC (EBIT)	39.3	46.2	40.9	30.3	30.1	35.1	33.1	35.3
ROIC post tax	31.8	36.6	34.6	25.1	24.8	29.1	27.5	29.3
ROE (UBS)	11.3	12.8	12.6	10.8	11.5	13.1	12.1	11.8

Capital structure & Coverage (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	(4.8)	(3.9)	(3.9)	(3.6)	(3.5)	(3.1)	(3.5)	(3.7)
Net debt / total equity %	(67.5)	(63.5)	(64.1)	(54.9)	(56.8)	(55.9)	(59.6)	(62.5)
Net debt / (net debt + total equity) %	NM	NM	NM	NM	NM	NM	NM	NM
Net debt/EV %	(61.1)	(50.7)	(61.3)	(15.2)	(15.1)	(16.5)	(18.4)	(21.1)
Capex / depreciation %	152.5	184.3	164.3	127.1	106.5	96.9	86.5	77.7
Capex / revenue %	16.5	18.7	20.2	19.0	16.5	13.3	12.8	11.7
EBIT / net interest	-	-	-	-	-	-	-	-
Dividend cover (UBS)	1.6	2.5	1.8	2.0	2.2	2.4	2.0	2.1
Div. payout ratio (UBS) %	63.2	40.1	55.1	50.0	45.5	41.1	50.9	48.4

Revenues by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	48,842	59,458	61,148	61,550	67,438	79,464	78,546	81,569
Total	**48,842**	**59,458**	**61,148**	**61,550**	**67,438**	**79,464**	**78,546**	**81,569**

EBIT (UBS) by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	17,807	24,033	23,558	21,139	24,150	30,110	29,487	30,648
Total	**17,807**	**24,033**	**23,558**	**21,139**	**24,150**	**30,110**	**29,487**	**30,648**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	-24.9%
Forecast dividend yield	1.2%
Forecast stock return	-23.7%
Market return assumption	6.3%
Forecast excess return	-30.0%

Company Description

Founded in 1987, Largan Precision (Largan) is one of the world's leading manufacturers of lenses used in cameras and IT products. It specialises in plastic/hybrid lenses and, to a lesser extent, glass lenses and modules. In 2022, 20MP+ camera lenses were 10-20% of Largan's shipment mix, 10MP lenses were 50-60%, 8MP lenses were 0-10% and other lenses were 30-40%.

Valuation Method and Risk Statement

Our price target is based on SOTP valuation on the CPO opportunity and traditional lens business.
We think Largan's downside risks include: 1) ongoing pricing pressure from smartphone OEM clients and increasingly intense competition among lens suppliers; and 2) a slower-than-expected smartphone demand recovery in China, and 3) CPO commercial success below the market's elevated expectations.
We believe the key upside risk is a faster-than-expected CPO qualification process at Largan. In addition, if the company demonstrates exceptionally efficient and highly accurate automation capabilities, it could support greater market share gains within the industry than we currently anticipate.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Largan Precision

Question	Response
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	3
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	4
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	2
5. What's driving the difference?	We expect downside to consensus in 2026 due to slower smartphone sales through August. We indicate things getting better for the stock in the past 3-6 months on the smartphone traction.
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	2
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	Negative Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

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12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
Largan Precision[28]	3008.TW	Sell	NT$6,660	08 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.
28. UBS holds a long position of 0.5% or more of the listed shares of this company.

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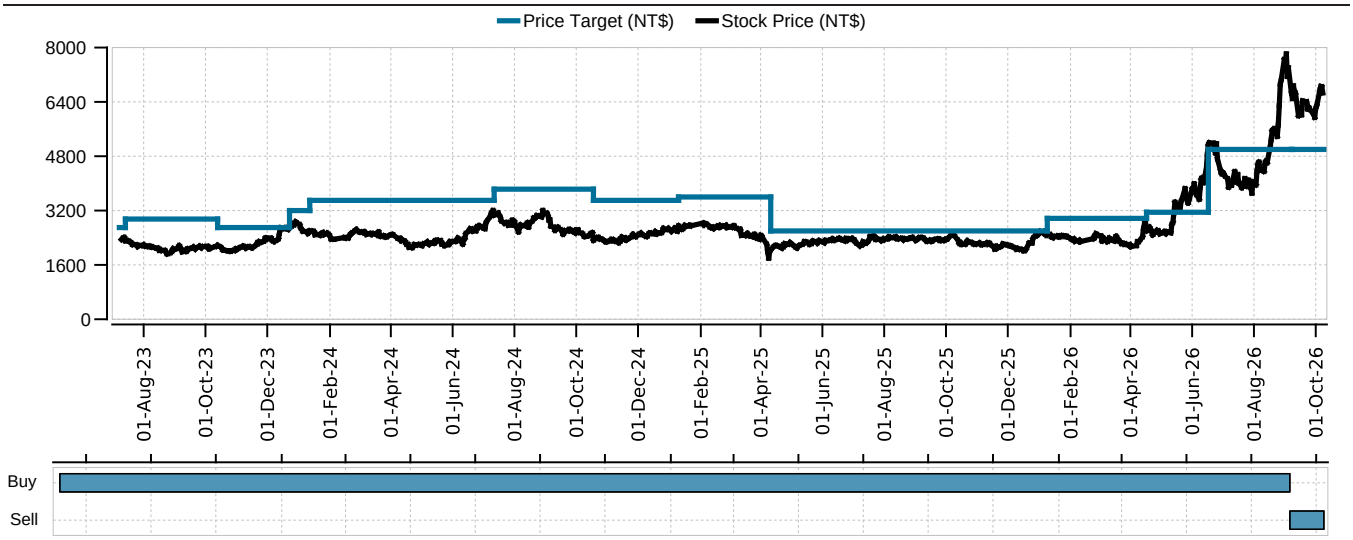
Largan Precision (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-07	2340.00	2700.00	Buy
2023-07-13	2405.00	2950.00	Buy
2023-10-12	2150.00	2700.00	Buy
2023-12-22	2650.00	3200.00	Buy
2024-01-11	2550.00	3500.00	Buy
2024-07-11	3200.00	3830.00	Buy
2024-10-17	2545.00	3500.00	Buy
2025-01-09	2595.00	3600.00	Buy
2025-04-10	1980.00	2600.00	Buy
2026-01-08	2475.00	2970.00	Buy
2026-04-16	2820.00	3150.00	Buy
2026-06-16	4860.00	5000.00	Buy
2026-09-06	7400.00	5000.00	Sell

Source: UBS Global Research; LSEG Eikon as of 08-Oct-2026. All prices as of local market close. Ratings as of date shown.

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Subject: S&P 500 SECTORS CALL: More Green On The Screen
Date: 7 hours ago

S&P 500 SECTORS CALL: More Green On The Screen

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S&P 500 SECTORS CALL: More Green On The Screen

By Ed Yardeni & Toby Hearst • 11 Oct 2026

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YRI S&P 500 SECTORS WEIGHTING RECOMMENDATIONS* (Oct 09, 2026)

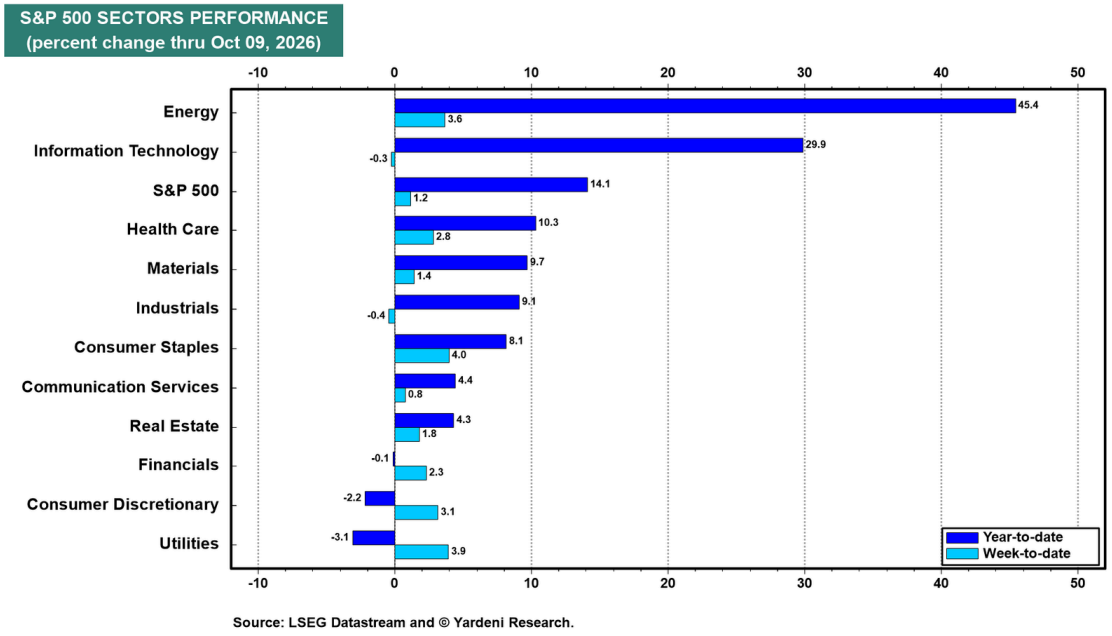
S&P 500 Sectors	Weight Recommendation	Week-to-Date	Month-to-Date	Year-to-Date
S&P 500		1.2	2.1	14.1
Communication Services	MW	0.8	0.4	4.4
Consumer Discretionary	UW	3.1	4.4	-2.2
Consumer Staples	UW	4.0	3.9	8.1
Energy	OW	3.6	5.8	45.4
Financials	OW	2.3	2.5	-0.1
Health Care	OW	2.8	1.5	10.3
Industrials	OW	-0.4	1.4	9.1
Information Technology	MW	-0.3	1.6	29.9
Materials	OW	1.4	2.0	9.7
Real Estate	UW	1.8	1.5	4.3
Utilities	MW↓	3.9	5.0	-3.1

Source: LSEG Datastreamand © Yardeni Research.

* OW = overweight, MW = market weight, and UW = underweight.

Bond yields edged down last week. As a result, the S&P 500 edged up, led by interest-rate-sensitive sectors. The AI-related sectors Industrials (-0.4%) and Information Technology (-0.3%) edged down and in fact were the only down sectors last week. The big interest-rate-sensitive gainers were Consumer Staples (4.0%), Utilities (3.9%), Consumer Discretionary (3.1%), and Financials (2.3%).

The year's laggards were last week's leaders (chart).

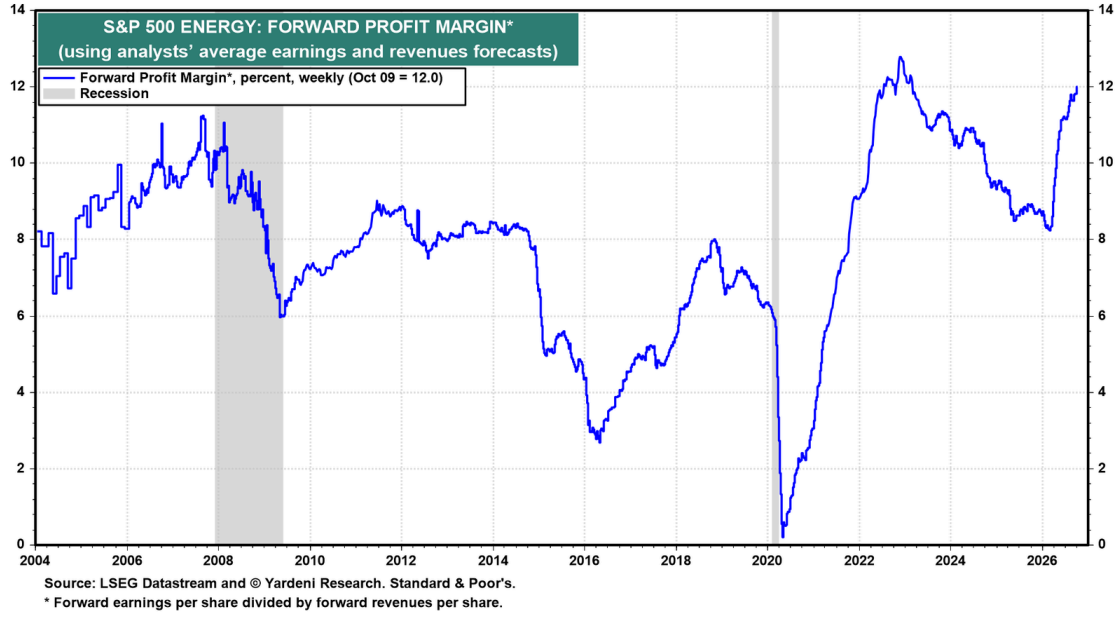


Let's have a look at the Energy, Financials, and Industrials sectors:

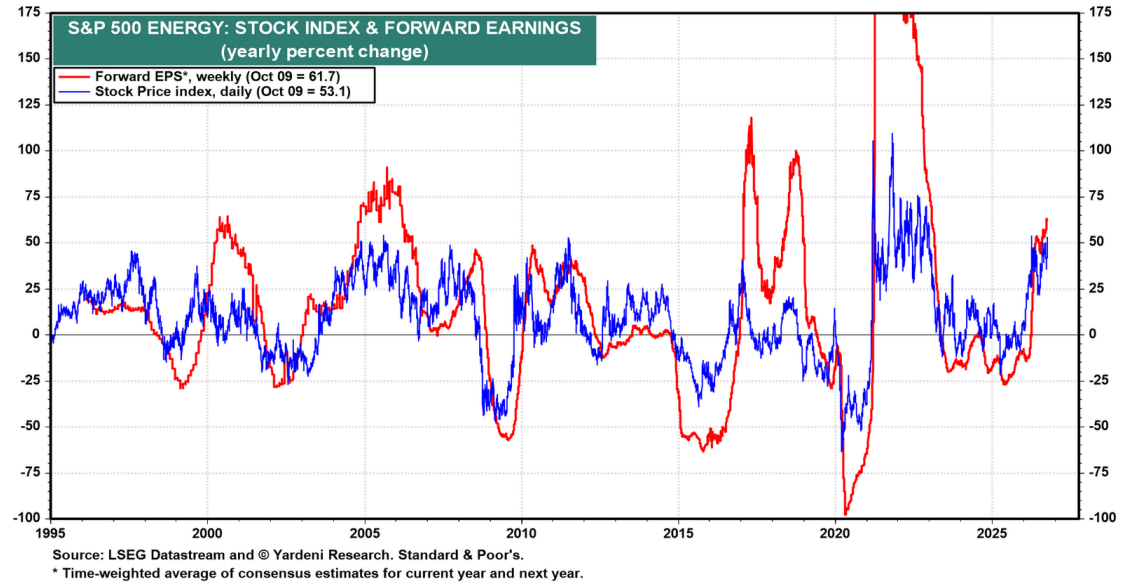
(**1) Energy.** S&P 500 Energy (3.6%) powered ahead on renewed risks to ships passing through the Strait of Hormuz (chart). Over the weekend, the Houthis attacked King Khalid International Airport in Riyadh and declared Saudi airspace a zone of military operations, warning airlines, travelers, and airport workers to steer clear of airports across the kingdom.

Sector momentum turned decisively positive mid-week, bolstered by a late-week surge as energy equities reacted to escalating Middle East conflicts and subsequent spikes in crude oil benchmarks (chart). Brent crude is quiet tonight at $103.65 per barrel. President Donald Trump helped calm the oil market by announcing that an energy ceasefire between Russia and Ukraine took effect immediately, saying on social media that both parties had agreed to the terms.

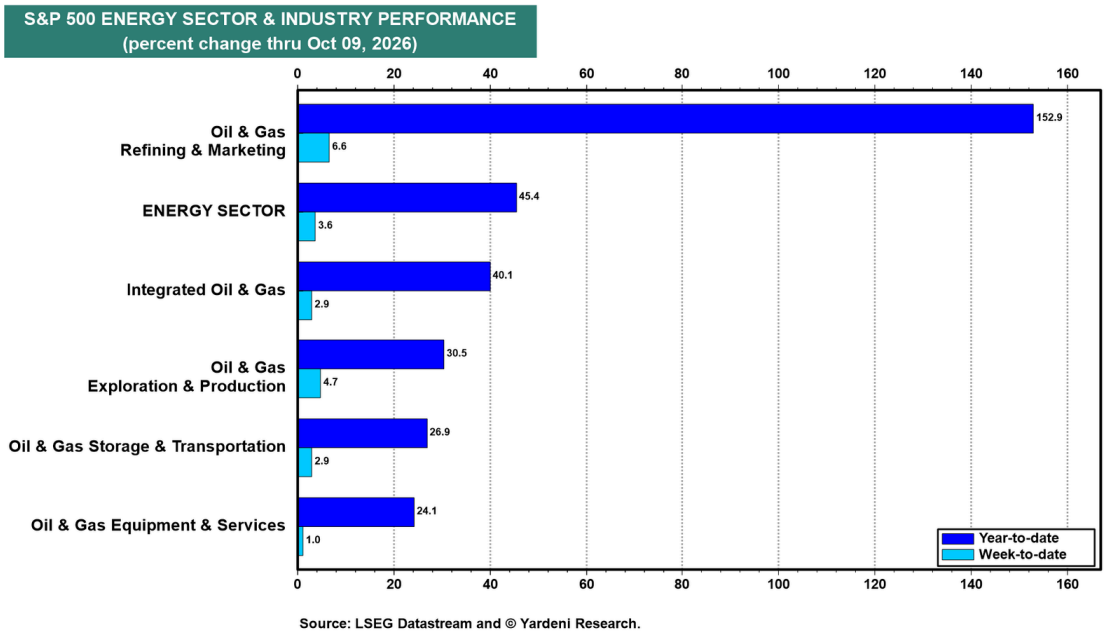
The rebound in the Energy sector's forward earnings this year has been attributable to the jump in the sector's forward profit margin, fueled by the jump in oil prices (chart).



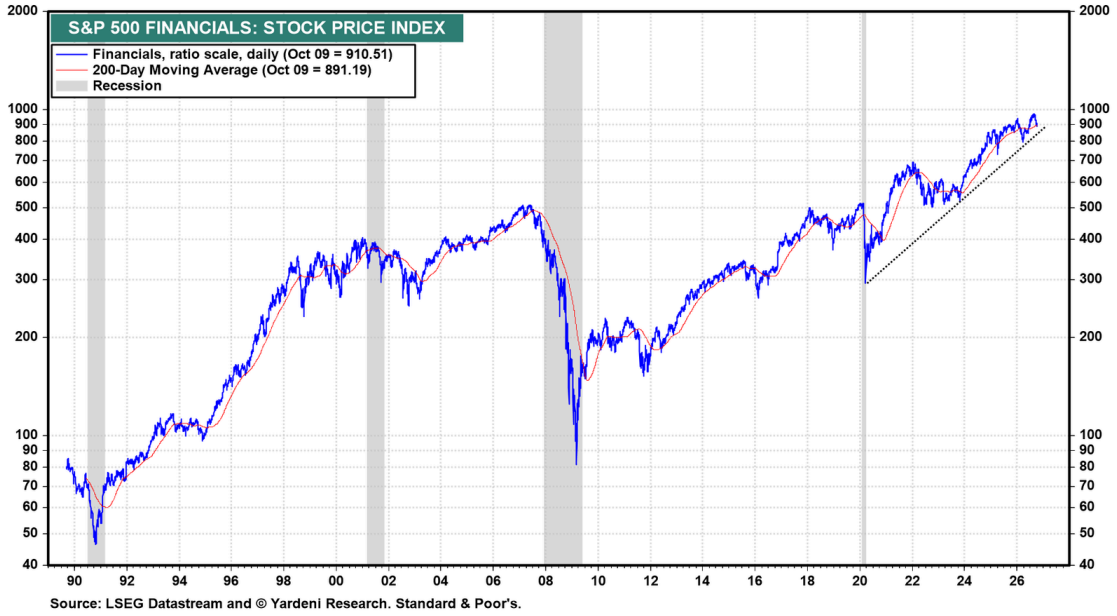
The sector's forward earnings is up 61.7% y/y, and its stock price index is up 53.1% (chart). There should be more upside for the rest of the year, especially if the war in the Middle East heats up after the mid-term elections, as Trump has threatened.



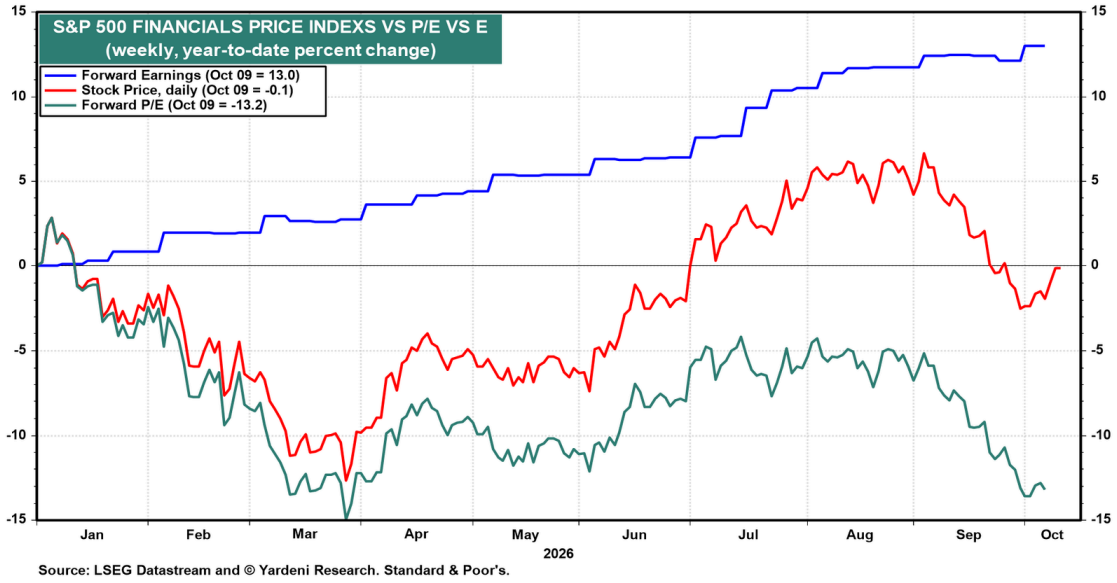
Oil & Gas Refining & Marketing has been the sector's big winner so far this year (chart). A global shortage of refinery capacity might fuel more gains.



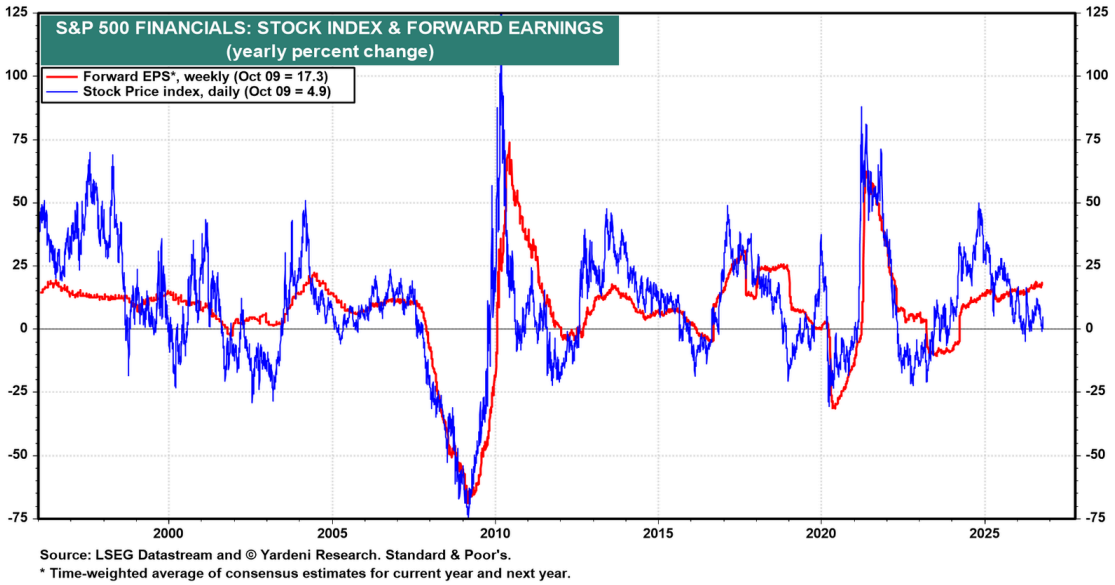
(**2) Financials.** The Q3-2026 earnings reporting season starts this week with the big banks. The sector's stock price index is nearly down at its 200-day moving average (chart). Solid earnings reports from the banks are likely, as we explained last week. If so, then the sector's uptrend should hold.



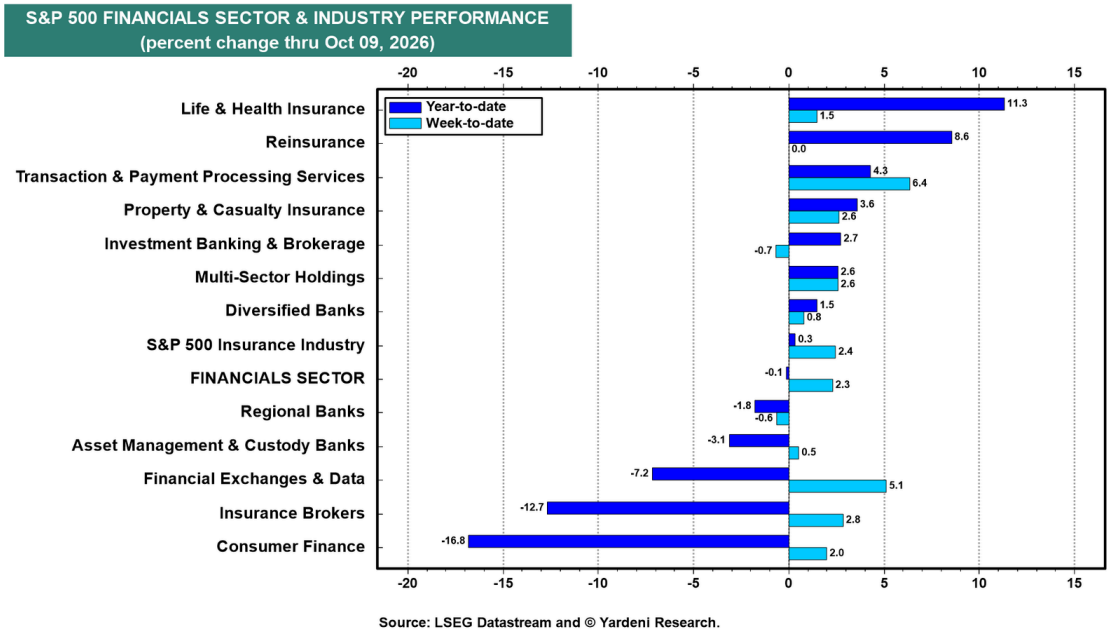
Rising interest rates are probably boosting the net interest margins of the large banks. However, the rapid ascent in bond yields in recent weeks has unnerved investors, as evidenced by the sharp drop in the sector's forward P/E since mid-August (chart). Bond yields may be stabilizing as investors are drawn to higher returns.



The sector's forward earnings is up 17.3% y/y, while its stock price index is up only 4.9% y/y (chart). A solid Q3 earnings season should lift the sector.

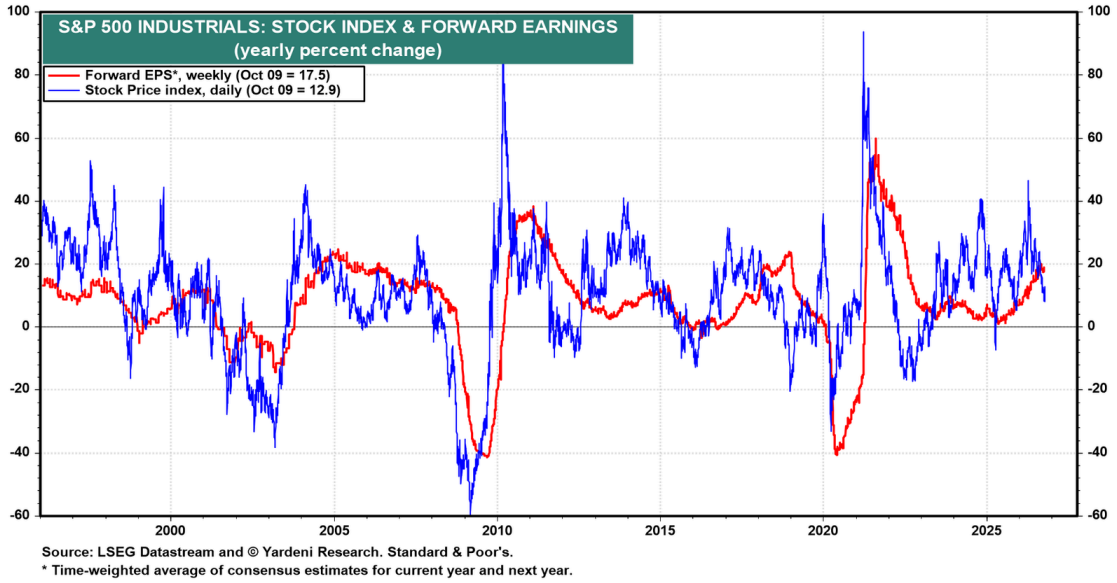


So far this year, insurance industries have performed best in the sector (chart).

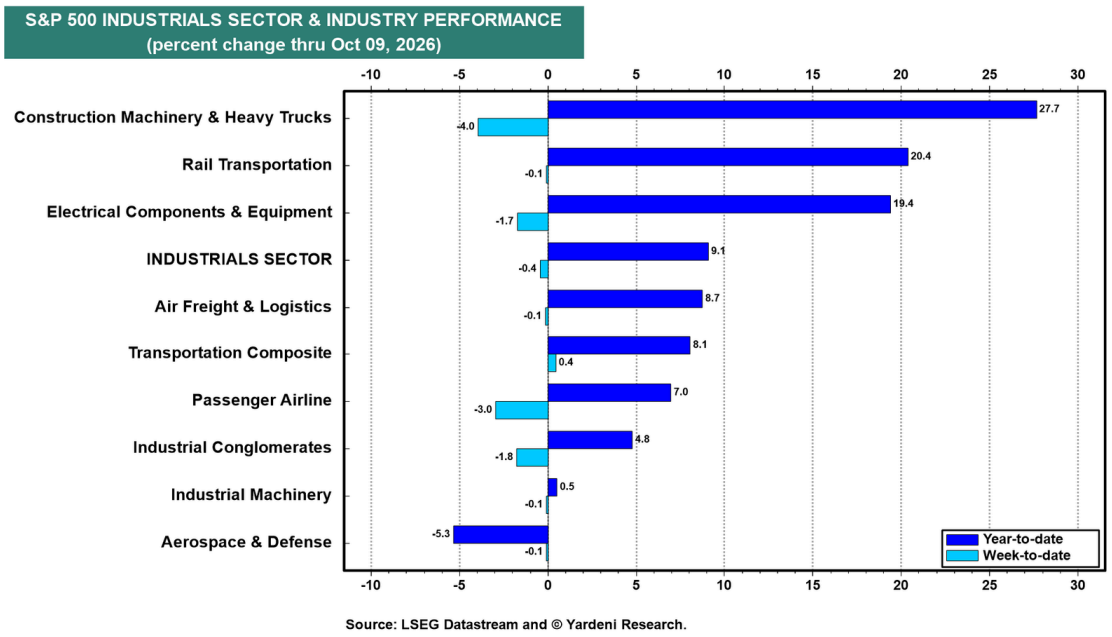


(3) Industrials. Industrials balanced competing forces this week—benefiting from safe-haven rotation and robust structural demand, while facing occasional headwinds from surging energy input costs and persistent high-interest-rate pressures. This S&P 500 sector is also testing support at its 200-day moving average (chart).

This is another sector where forward earnings is rising faster than the sector's stock price index (chart). The price action sometimes leads the growth rate in forward earnings. This time may be different as the AI buildout continues to boost the sector's earnings.



Construction Machinery & Heavy Trucks is the sector's best-performing industry so far this year (chart). Second best has been Rail Transportation. Notwithstanding the wars in the Middle East and between Russia and Ukraine, Aerospace & Defense has been the weakest industry. Military budgets for large munitions and battlefield weapons may have to make room for more spending on AI-guided drones and "Terminators."



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ECONOMIC WEEK AHEAD: October 12-16

YARDENI QUICKTAKES

ECONOMIC WEEK AHEAD: October 12-16

By Ed Yardeni & Elias Griepentrog • 11 Oct 2026

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Photo by Monica Sauro / Unsplash

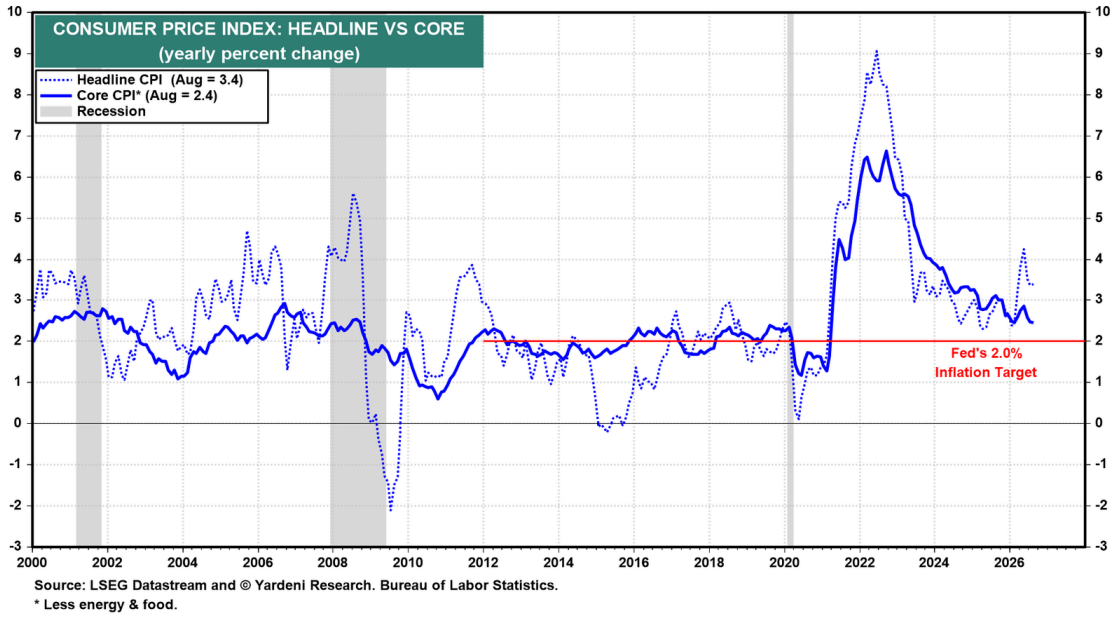
Last week was relatively quiet on the economic data front, but the indicators released pointed to continued economic strength. ADP weekly payroll gains increased for the fifth consecutive week, reaching a three-month high, while jobless claims fell to multi-year lows. The ISM NM-PMI remained solid at 54.9 in September, though its prices paid component jumped to 74.0, the highest since July 2022. Redbook same-store sales rose 8.3% y/y during the week of October 2, signaling continued strength in consumer spending, while the New York Fed's consumer survey showed expected spending growth and one-year-ahead inflation expectations both reaching three-year highs.

The data suggest that the Fed has good reason to be hawkish. Accordingly, September's FOMC meeting minutes showed that most policymakers favored another rate hike by year-end, with several judging monetary policy to be barely restrictive, if at all. We agree that further tightening is warranted. However, recent comments by New York Fed President John Williams, Fed Vice Chair Philip Jefferson, and Governor Christopher Waller suggest flexibility on the timing. The markets assign a 17.2% probability to an October hike and a 71.6% probability of a December hike.

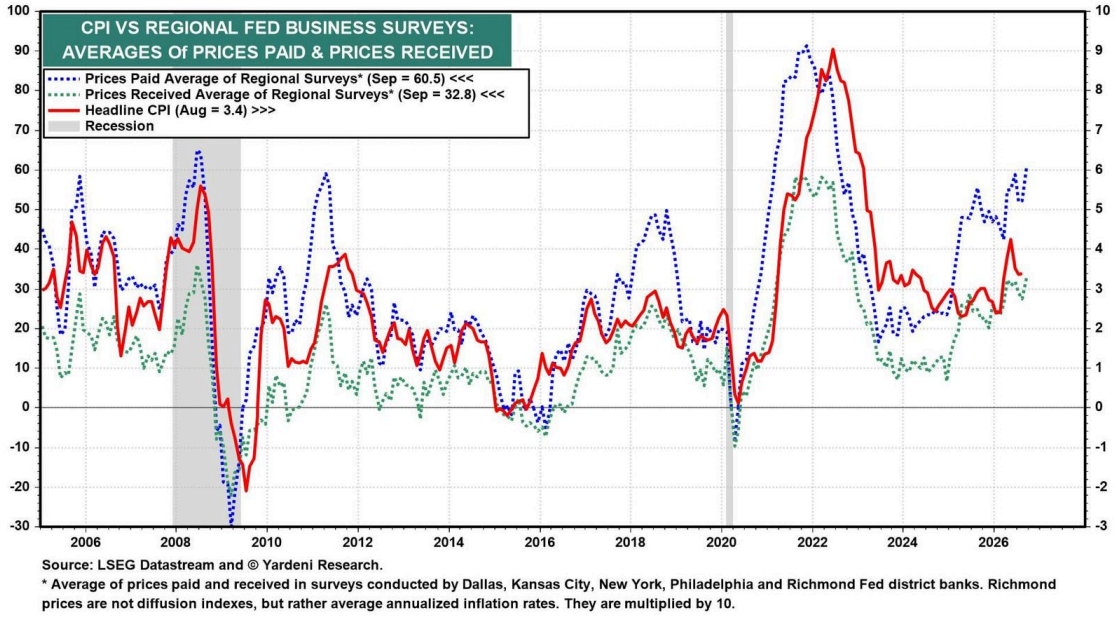
This week's inflation reports could change those odds. A hotter-than-expected CPI reading on Wednesday could prompt the markets to price in an October hike, pressuring the Fed after September's unanimous 25bps increase, which Fed Chair Kevin Warsh described as removing "a dose of accommodation." An in-line or cooler CPI reading would support patience. Warsh speaks Thursday evening at the IMF and World Bank annual meetings in Bangkok.

Here's more:

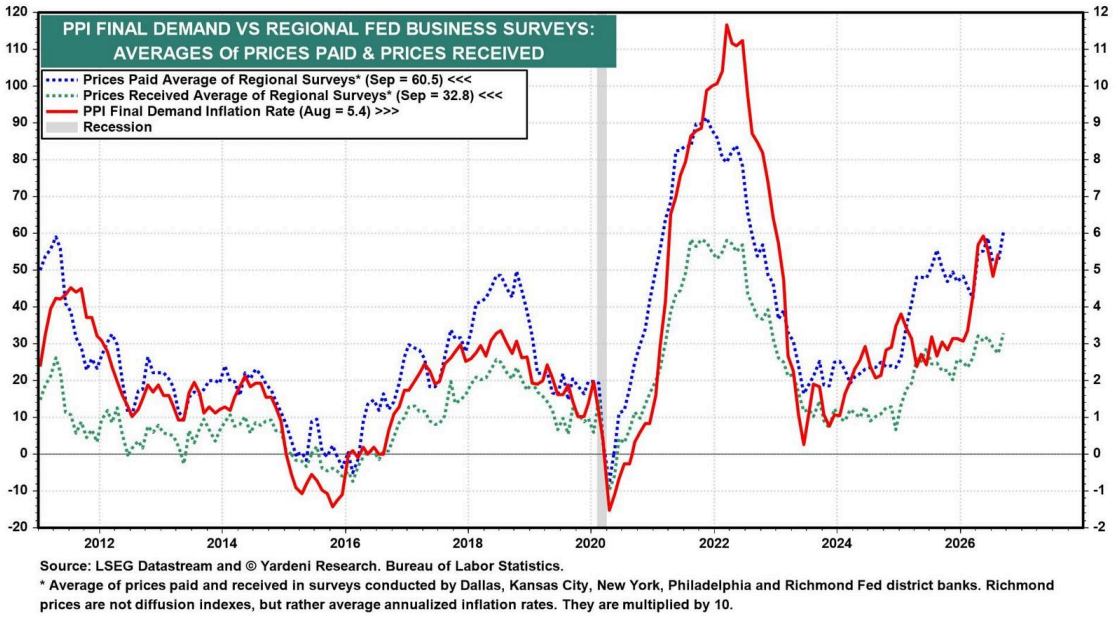
(1) Inflation. The September CPI and PPI reports will be released on Wednesday and Thursday, respectively. The Cleveland Fed's Inflation Nowcast model projects a hot 0.53% m/m increase in headline CPI, while core CPI is expected to rise a more modest 0.2%. That would put headline and core CPI inflation at 3.6% and 2.4% y/y, respectively. September is expected to mark the 67th consecutive month of CPI inflation exceeding 2.0% y/y, underscoring the persistence of inflationary pressures (chart).



Adding to inflation concerns, the averages of the prices-paid and prices-received indexes of five regional Fed banks' business surveys rose to their highest levels since 2022 in September (chart). Both have historically tracked headline CPI inflation closely.

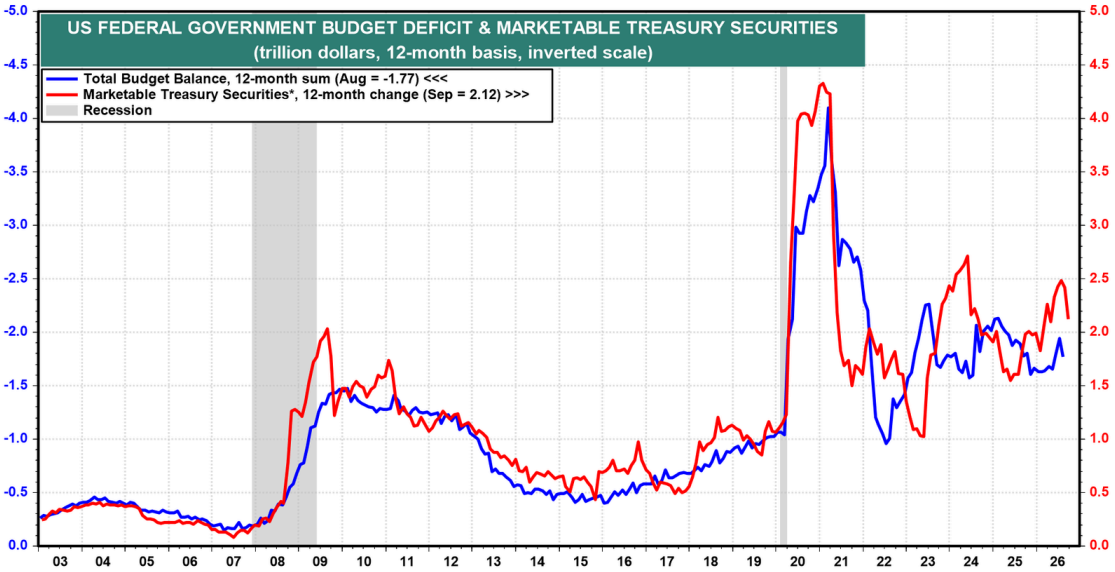


The same indicators have also closely tracked PPI final demand inflation, which was already running hot at 5.4% y/y in August (chart).



Source: LSEG Datastream and © Yardeni Research. Bureau of Labor Statistics.
* Average of prices paid and received in surveys conducted by Dallas, Kansas City, New York, Philadelphia and Richmond Fed district banks. Richmond prices are not diffusion indexes, but rather average annualized inflation rates. They are multiplied by 10.

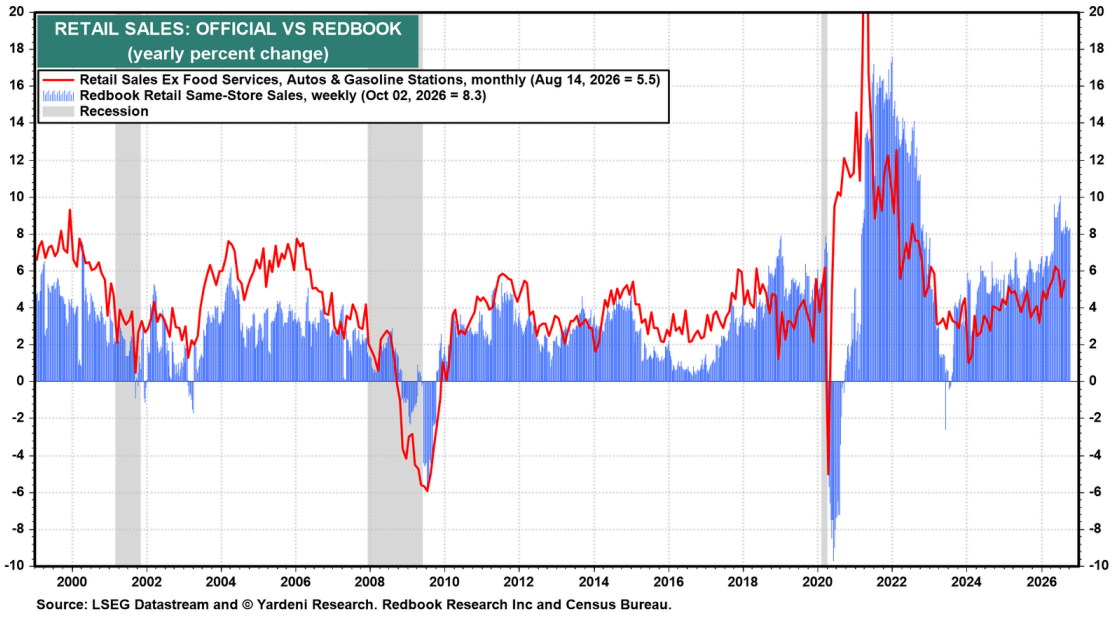
(2) Monthly Treasury statement. The federal budget deficit totaled $1.77 trillion over the 12 months through August, while outstanding marketable Treasury securities increased by $2.12 trillion over the 12 months through September (chart). The recent slowdown in Treasury borrowing suggests that the deficit may have narrowed somewhat. Tuesday's report will reveal the final budget deficit for fiscal 2026, which ended on September 30.



Source: LSEG Datastream and © Yardeni Research. US Treasury.
* Marketable interest-bearing bills, notes, and bonds.

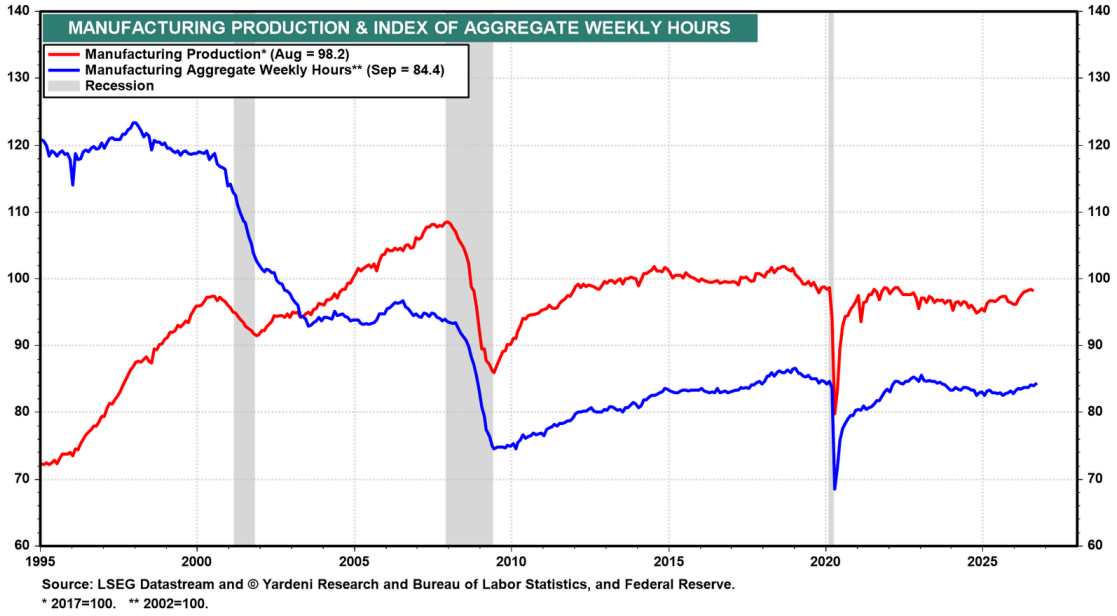
(3) Consumer spending. Thursday's September retail sales report will provide the first official reading on consumer spending for the month. We expect another solid increase, as Redbook same-store sales rose 8.3% y/y in September, well above the 2025 average of 5.8%. Redbook sales

historically have tracked official retail sales closely, suggesting that consumer spending maintained strong momentum through the end of Q3 (chart).

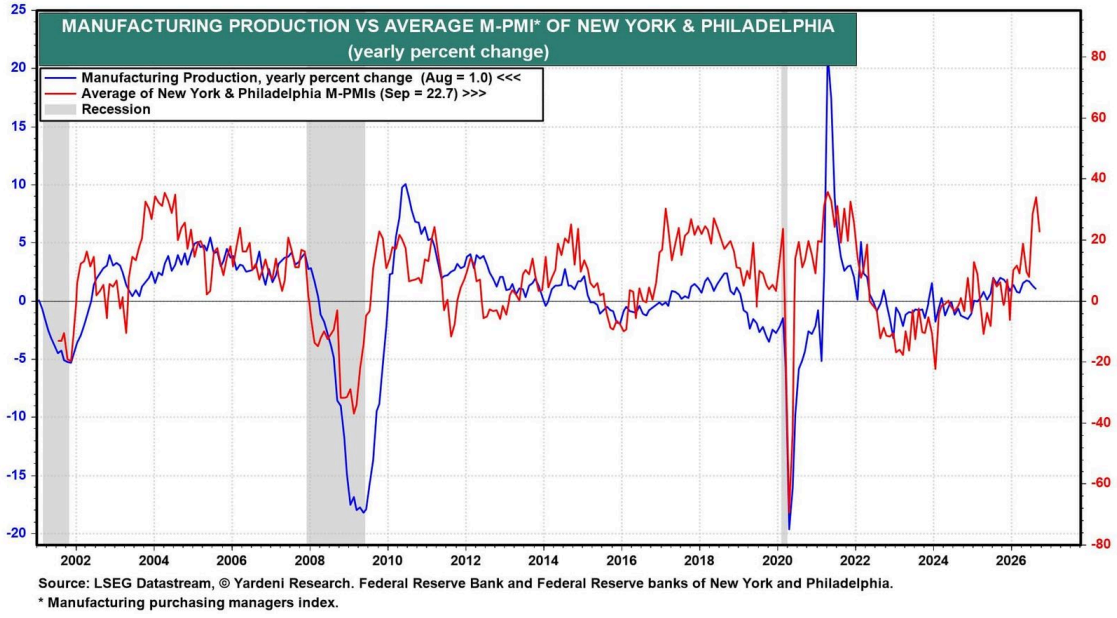


(4) Manufacturing. The week will also provide important insights into the manufacturing sector, with the October New York and Philadelphia Fed banks' business surveys due Thursday, followed by September's industrial production and capacity utilization reports on Friday.

Manufacturing aggregate weekly hours rose to 84.4 in September, the highest since August 2023, continuing their recent upward trend (chart). Immediate expensing under the OBBBA, reshoring, and the AI buildout are helping to reinvigorate manufacturing activity. With skilled labor in short supply, manufacturers appear to be meeting stronger demand partly by increasing the hours worked by existing employees. The rise in aggregate hours worked suggests that manufacturing production expanded at a healthy pace in September.



The New York and Philadelphia Fed manufacturing surveys provide another reason for optimism. Their average manufacturing PMI rose to 22.7 in September, the highest since 2021 (chart). Historically, the average has closely tracked manufacturing production growth, suggesting that factory output could accelerate further in the months ahead.



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Weekly Roundup - October 11th

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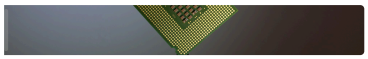

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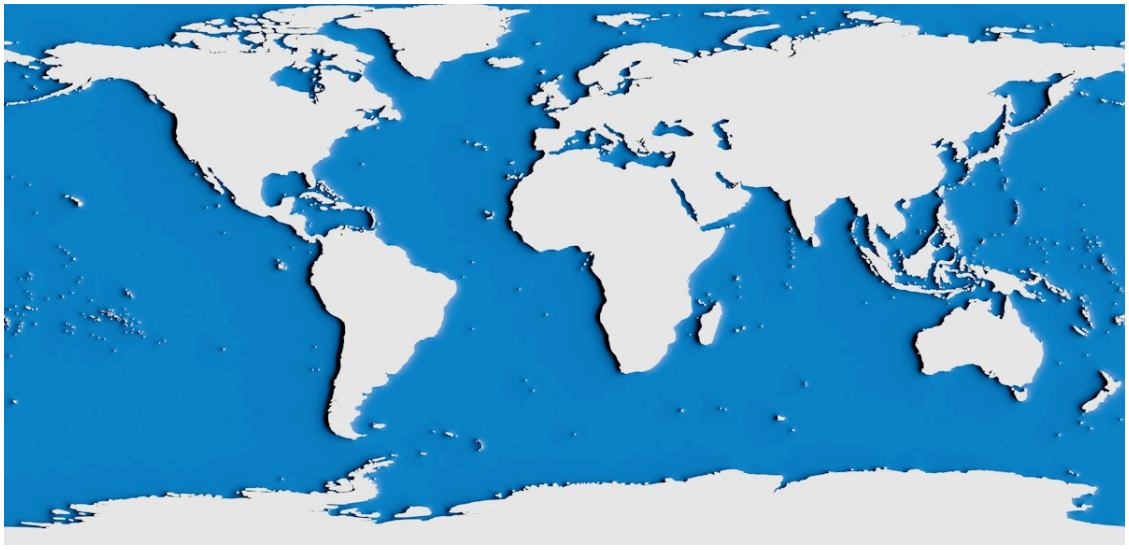
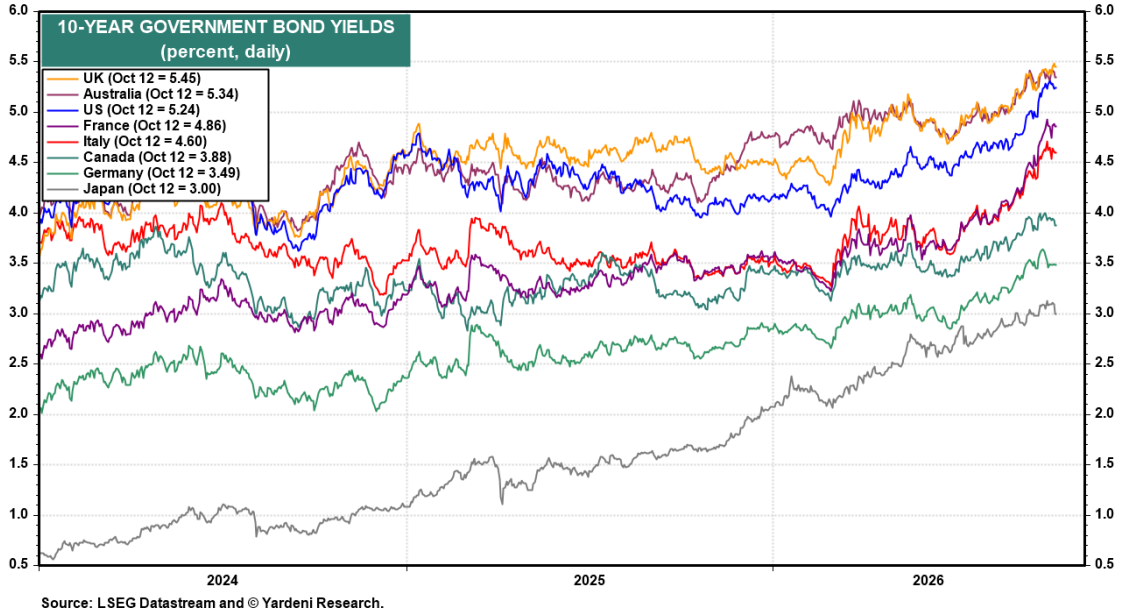


Photo by Insaanu Studio / Unsplash

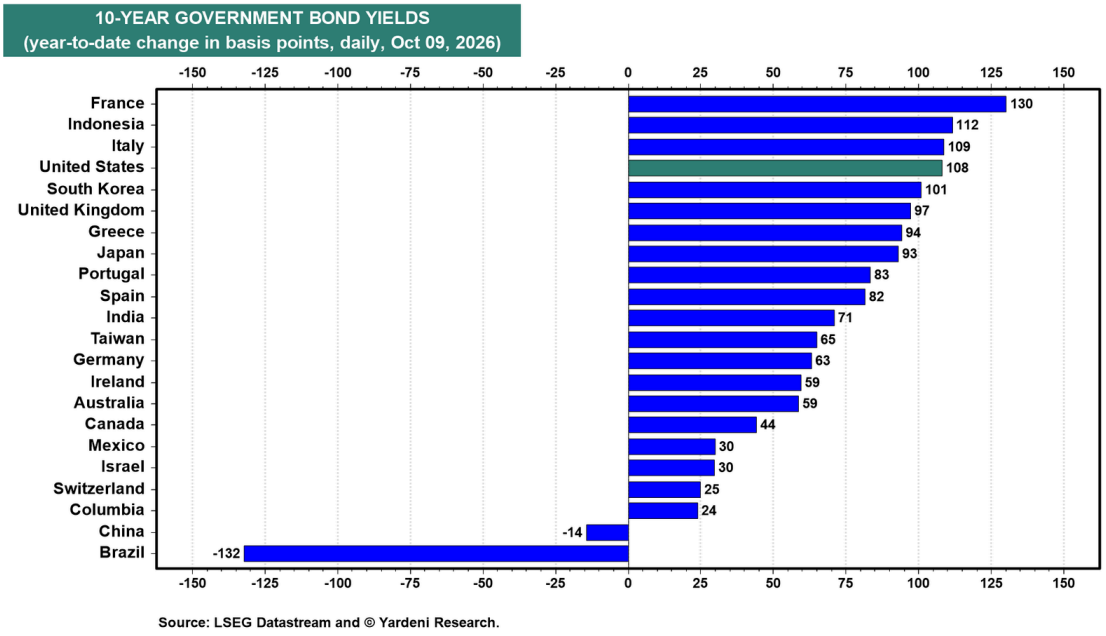
I. Bonds

The global bond market remains jittery, though yields may be starting to stabilize after their rout in recent months (chart). Nevertheless, geopolitical turbulence may still buffet them.



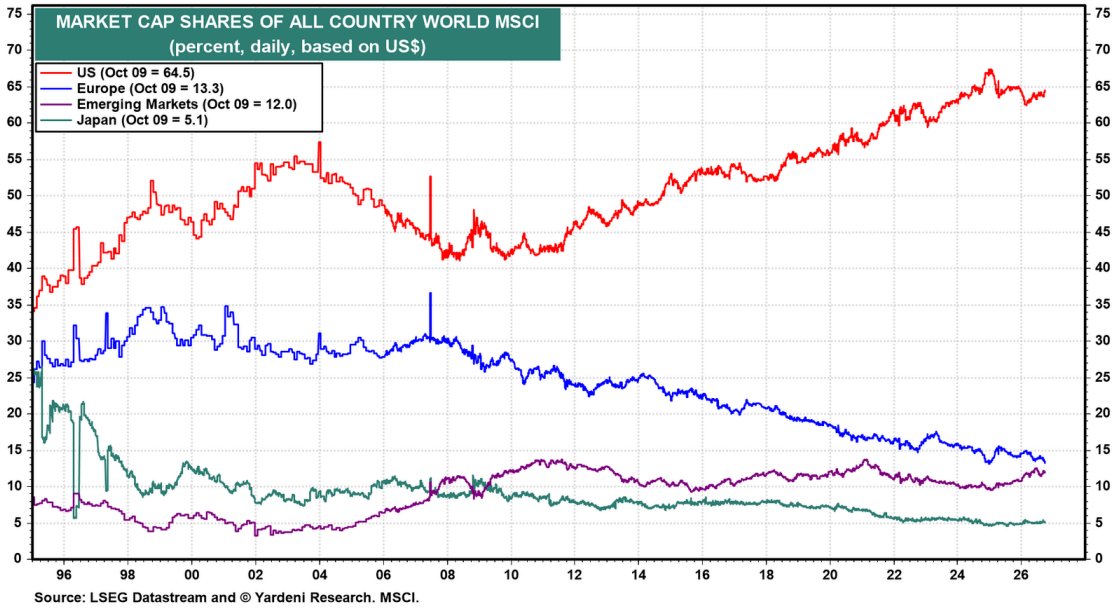
Yields have risen for an assortment of reasons, from better-than-expected economic growth (US) to fiscal instability concerns (France). Brazil and China are the only major countries showing declines in bond yields so far this year (chart).

Bond yields have also risen on higher-for-longer oil prices as the war in the Middle East has continued to disrupt oil supplies. The Houthis have said airliners should avoid Saudi airspace, declaring it a zone of military operations after the Saudi-backed Yemeni government launched an offensive against the group earlier this month. On Saturday, a third attack this week on Riyadh's international airport left 12 people dead and dozens wounded. President Donald Trump said the US will "look at" joining Saudi strikes on Houthis.

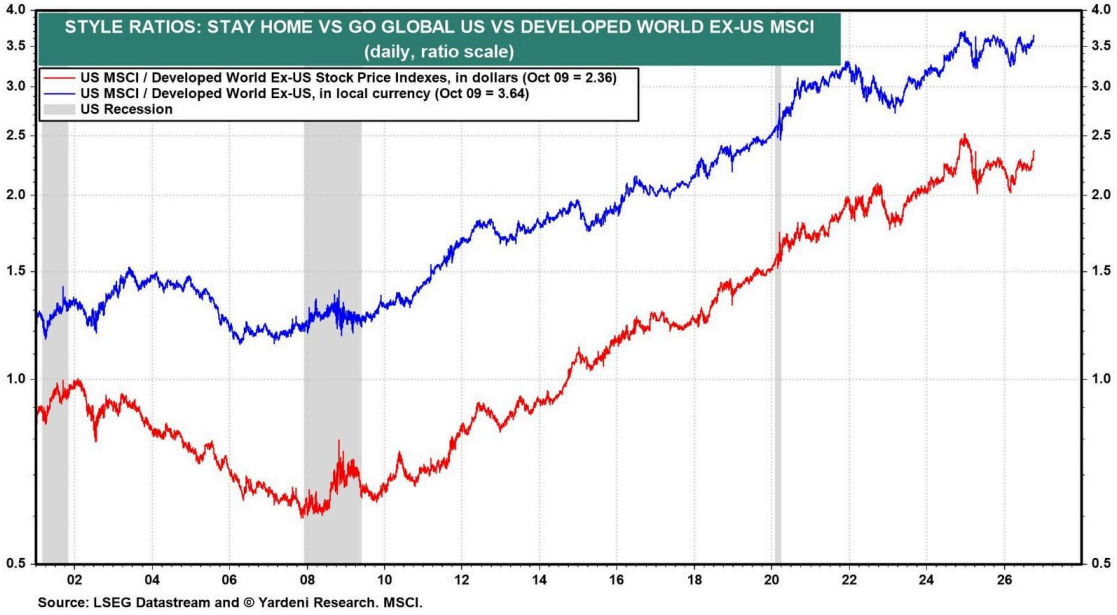


II. Stay Home vs Go Global

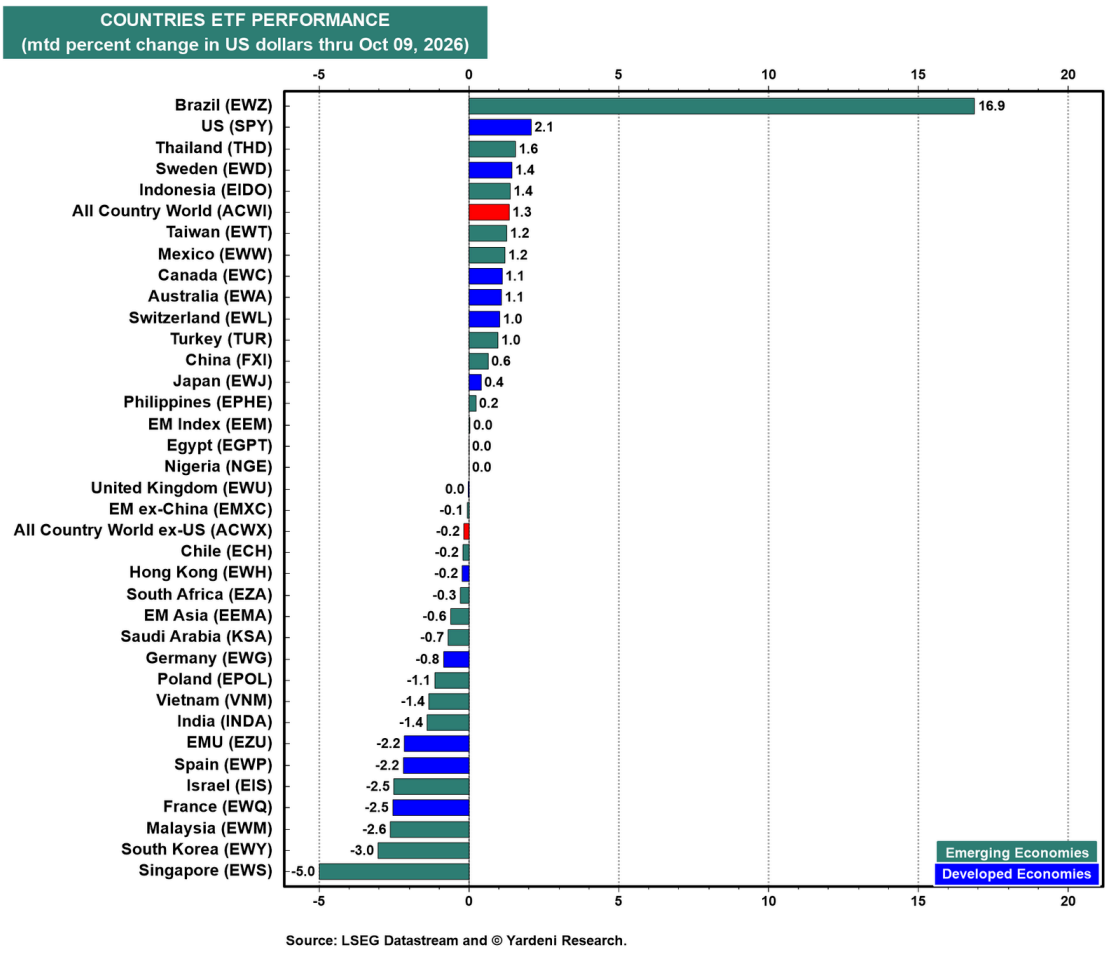
So far this year, the US has held onto its 65% market-cap share of the All Country World (ACW) MSCI (chart). Japan's share has also held steady at 5%. However, Europe's share has slipped (13.3%) to the advantage of the share of the Emerging Markets (12.0%).



Nevertheless, last week, the US outperformed the Developed World ex-US as the French debt crisis weighed on European stock prices (chart).

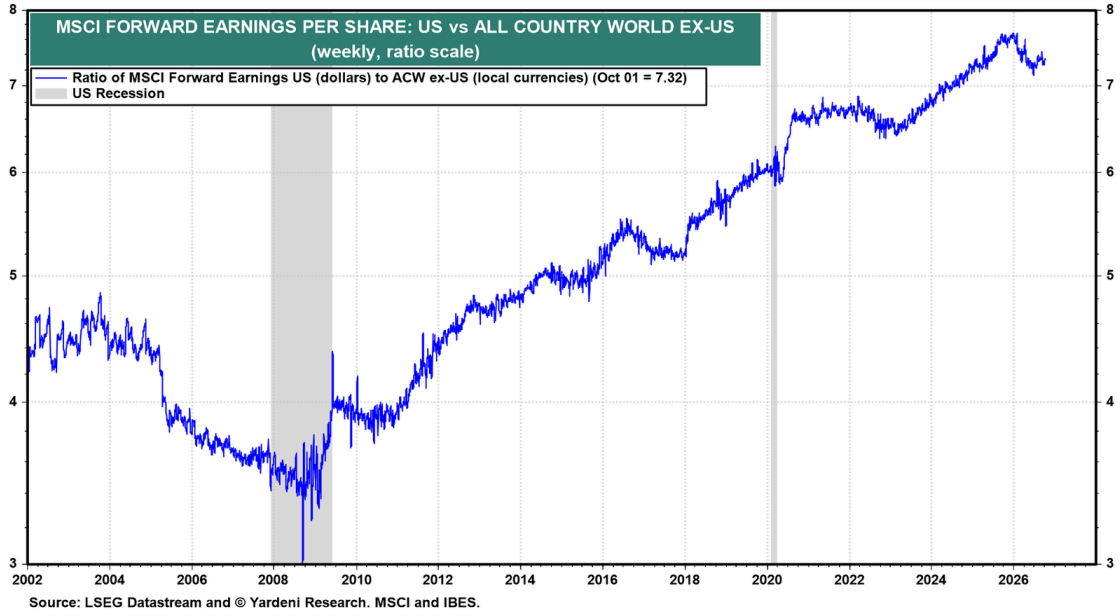


So far this month, the US MSCI is the second-best-performing stock price index we track (chart). Brazil was the best-performing in response to first-round presidential election results showing conservative Flavio Bolsonaro, son of former President Jair Bolsonaro, exceeding expectations in the race against leftist President Luiz Inacio Lula da Silva.

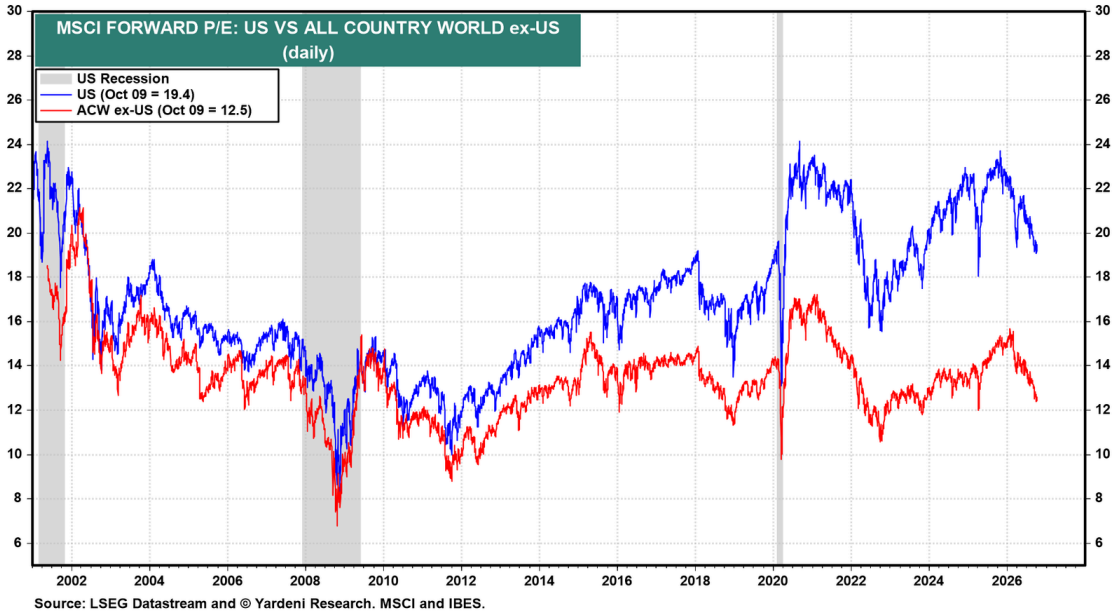


III. Earnings & Valuation

The ratio of forward earnings for the US MSCI to the ACW ex US MSCI fell through the summer this year (chart). It stopped doing so in recent weeks.

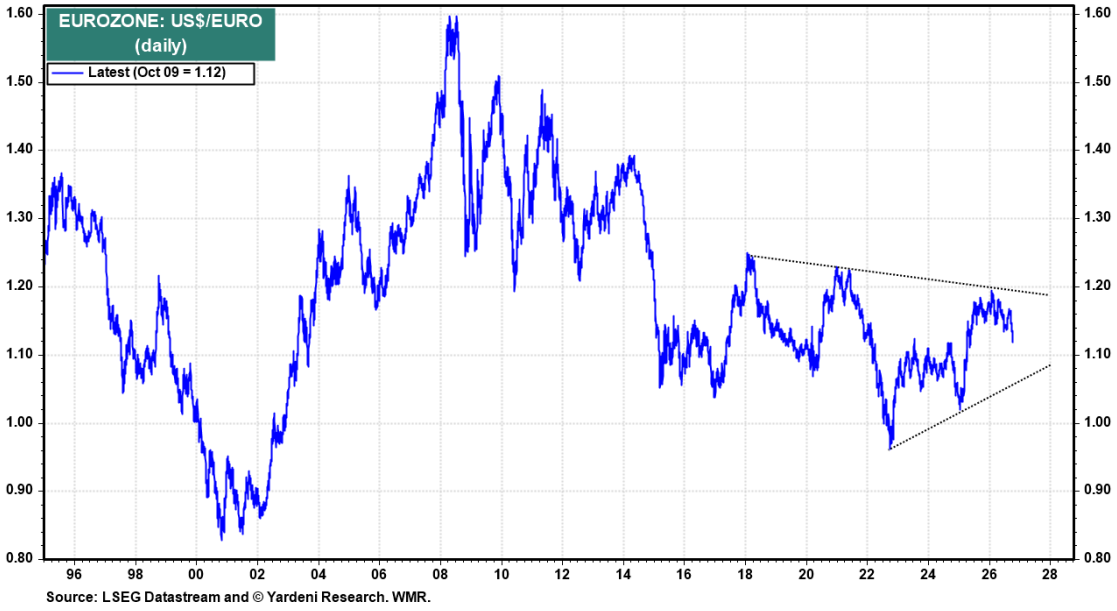


Forward P/Es have been falling worldwide since the start of the year, as forward earnings have outpaced stock price gains (chart).



IV. Currencies

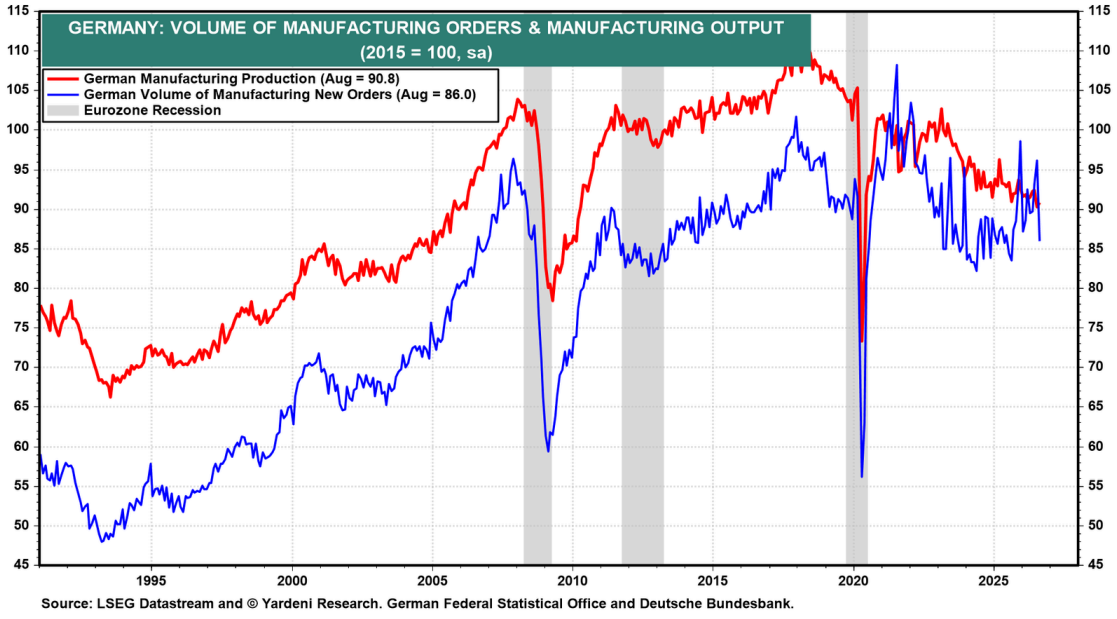
The French debt crisis is also weighing on the euro, which is down from $1.20 at the start of this year to $1.12 (chart).



As a result, the DXY dollar index has rebounded to its best reading since early 2025 (chart). We remain constructive on the US dollar.

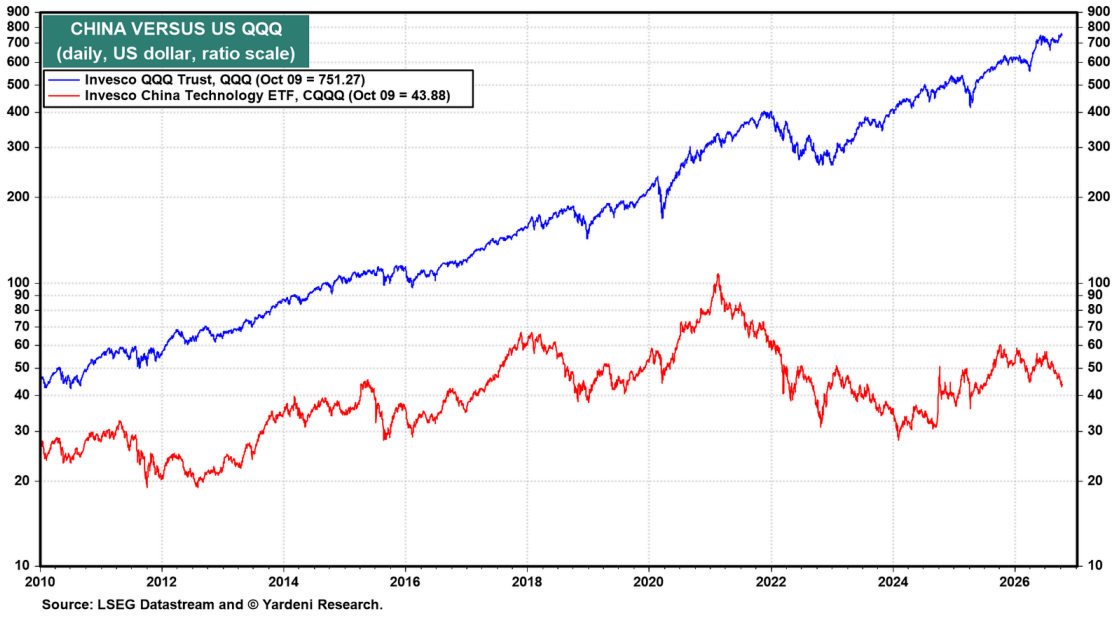
V. Notable

(1) German factory orders fell sharply in August, plunging 10.6% m/m (chart). The primary driver behind the contraction was a steep pullback in large-scale contracts. The headline drop was almost entirely due to a 61.5% slump in large-scale orders for aircraft, ships, trains, and military vehicles, which suffered a major correction following an exceptional surge in July. When these volatile large-scale contracts are excluded, new manufacturing orders actually slipped by just 0.1%, indicating that core industrial demand was essentially treading water rather than experiencing a broad-based collapse.



(2) CQQQ continues to underperform QQQ (chart). While China's long-term industrial policy goals are ambitious, cyclical demand inside China

has struggled to generate the same explosive revenue growth seen in Western tech sectors. Furthermore, China's tech industries carry a persistent regulatory discount. Chinese technology sectors have historically been vulnerable to abrupt policy shifts, anti-monopoly crackdowns, data security reviews, and ongoing geopolitical tensions (including US–China trade restrictions and semiconductor export controls), which compress valuations and deter long-term institutional capital inflows.



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Subject: US MARKETS CALL: Is The 10-Year Bond Yield Back To The Old Normal?
Date: 15 hours ago

US MARKETS CALL: Is The 10-Year Bond Yield Back To The Old Normal?

YARDENI QUICKTAKES

US MARKETS CALL: Is The 10-Year Bond Yield Back To The Old Normal?

By Ed Yardeni & Toby Hearst • 10 Oct 2026

View in browser

Photo by Justin Simmonds / Unsplash

I. Bonds

The Economist just rang the all-clear siren for the bond market (chart). It is safe to buy bonds now that the front cover of the October 10 magazine is titled "Will bonds blow up?" This and other national magazines' cover stories often have been great contrary indicators. It's the "front-page

curse” (once magazines put in the time it takes them to produce a cover article on a financial market trend, investors have moved on).

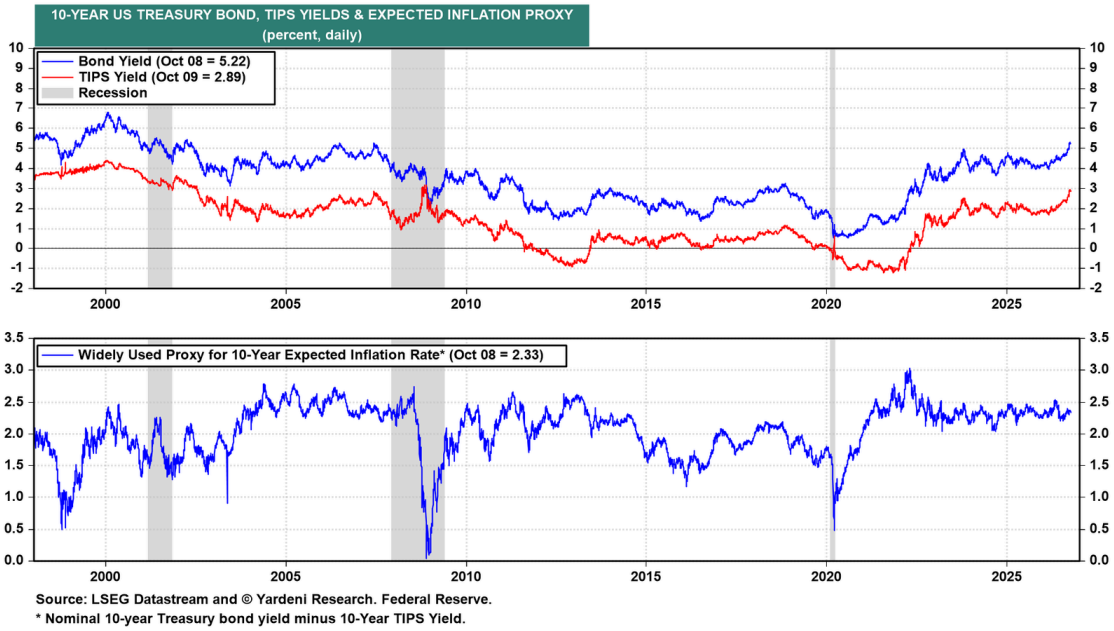
We think the 10-year Treasury bond yield is back to normal this year, i.e., in the 4.00% to 5.00% range (chart). We've disputed the widespread notion that "interest rates are likely to remain higher for longer." That implies that they should come back down at some point. We've argued that "interest rates are likely to *stay normal* for longer." They were abnormally low between the Great Financial Crisis and the Great Virus Crisis, when central banks rigged the fixed-income markets with zero and near-zero interest-rate policies and quantitative easing.

As bond yields rose in recent months to the top end of what we consider the normal range, we expected buyers to be attracted by higher yields, so we expected the range to be maintained (that’s what happened in in 2023, when the yield spiked to 5.00% in late October). This time, the yield

spiked above the normal range, to around 5.25% (as it did during the 4.00%-5.00% normal-range period from 2002-07).

In recent conversations with several of our institutional accounts, many expressed interest in buying bonds at levels above 5.00%.

A slice-and-dice analysis of the 10-year yield shows that its rise this year has been almost entirely attributable to the rise in the comparable TIPS yield (chart). The spread, which is a proxy for inflation expectations over the next 10 years, has been range-bound roughly between 2.00% and 2.50% since 2023.

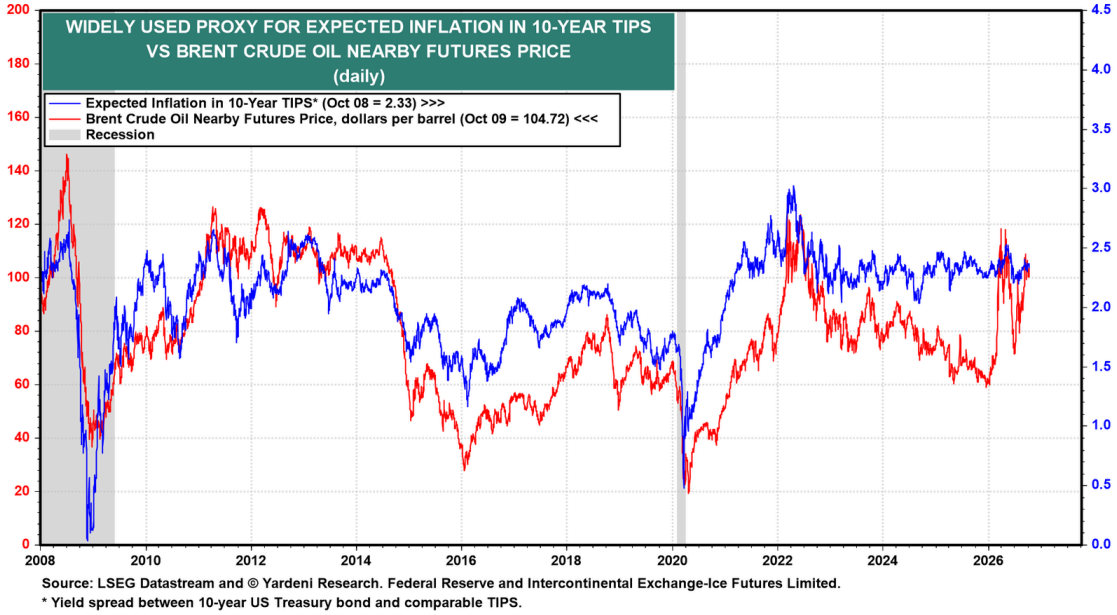


Since 2023, the TIPS yield has closely tracked the Weekly Economic Index, which tracks the growth rate in real GDP on a y/y basis (chart). This

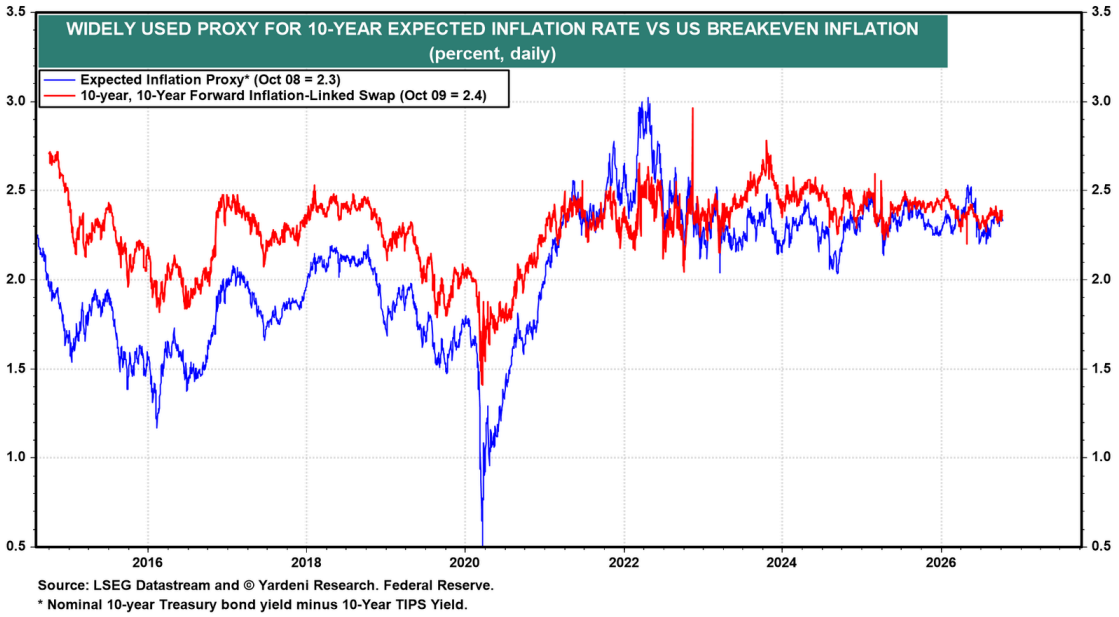
supports our view that the increase in US nominal yields has been driven by better-than-expected economic growth.

In the past, the expected inflation spread was highly correlated with the price of a barrel of crude oil (chart). That correlation has been weaker since 2023. Bond investors seem to believe that inflation will remain low in the 2.00%-2.50% y/y range over the rest of the Roaring 2020s into the Roaring 2030s. We agree.

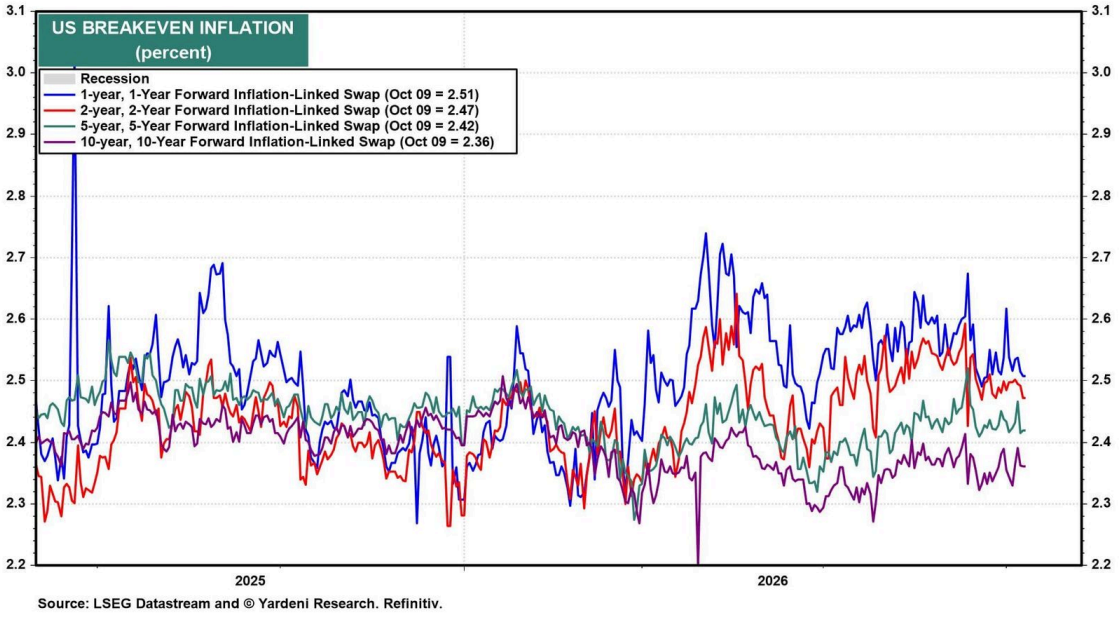
Nevertheless, we can't rule out the possibility that another spike in oil prices, driven by a re-escalation of the war in the Middle East, won't push nominal bond yields higher. If so, that will likely be another buying opportunity.



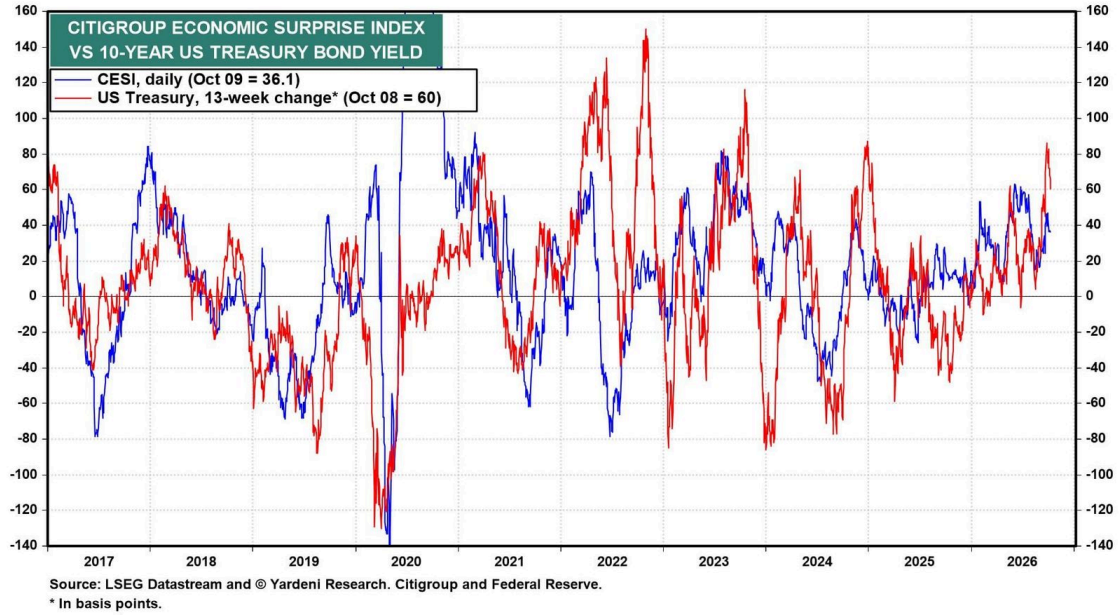
By the way, the expected inflation spread closely tracks the 10-year forward inflation-linked swap (chart).



All major forward inflation-linked swaps are roughly in the 2.00%- 2.50% range (chart).

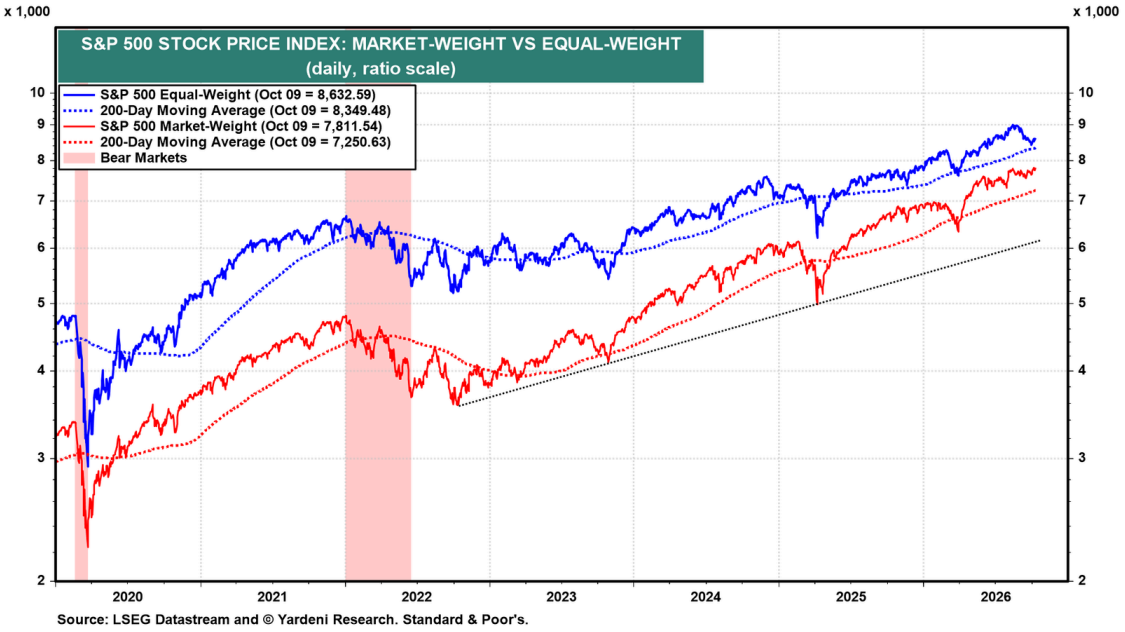


Meanwhile, the link between the 13-week change in the 10-year nominal yield and the Citigroup Economic Surprise Index suggests that the former's recent climb should abate (chart).

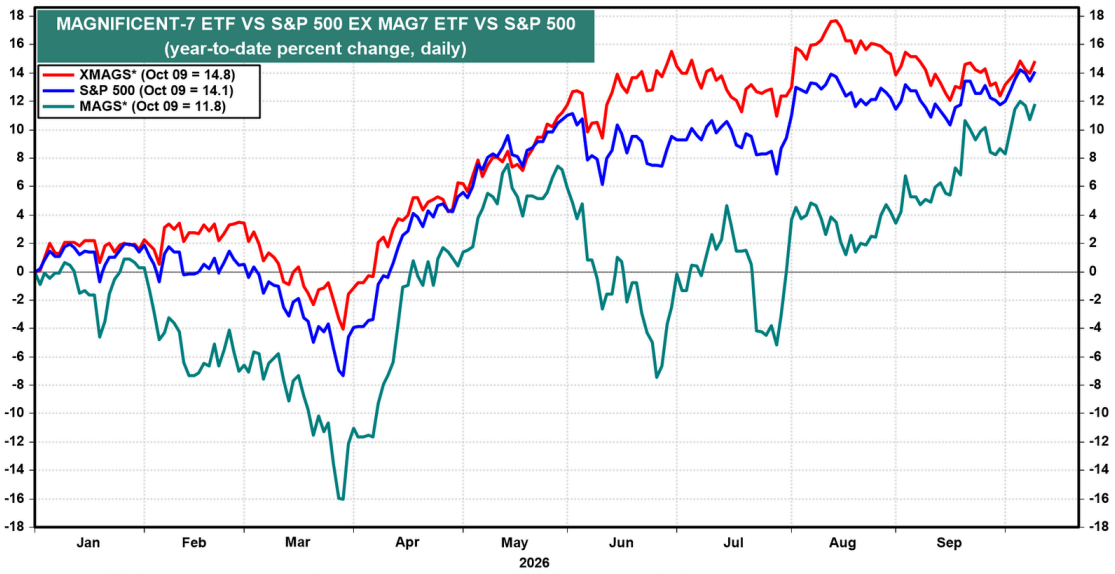


II. Stocks

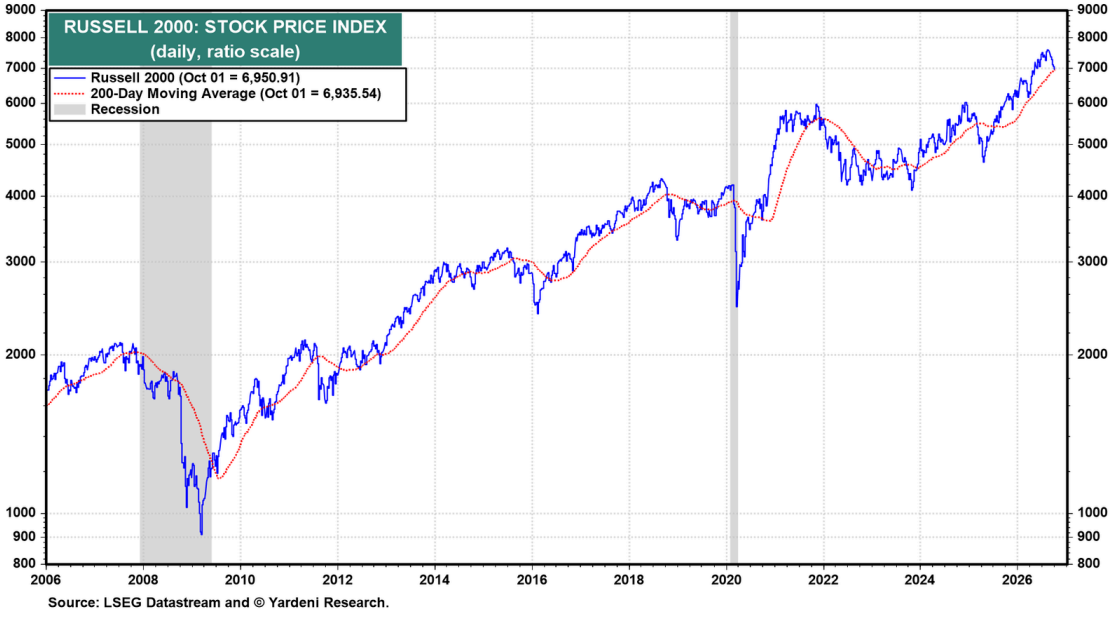
The steep rise in bond yields since mid-August has weighed on the S&P 500 equal-weight stock price index, while the market-weight index rose to record highs last week (chart). The former may be finding support at its 200-day moving average.



The gap between XMAGS's outperformance and MAGS's underperformance has narrowed since mid-August as bond yields jumped (chart). Over the past couple of weeks, they both did well.

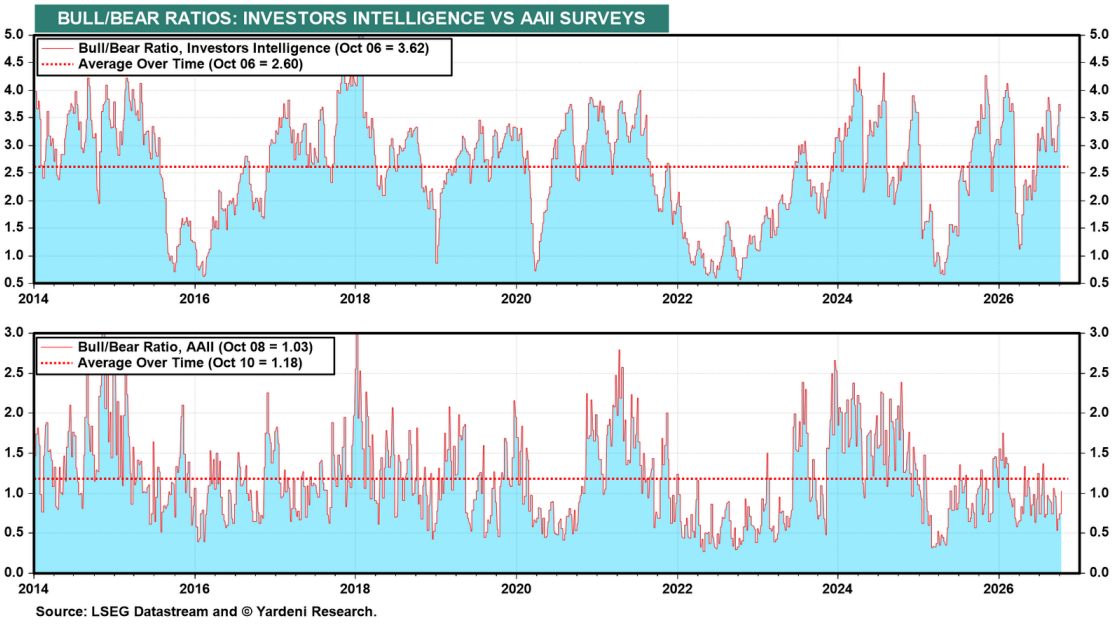


The Russell 2000 also dipped as bond yields soared since the late summer (chart). The index is back at its 200-day moving average, which should hold if US bond yields continue to stabilize.



III. Sentiment

Investor sentiment turned more bullish last week according to the two bull-bear ratios we track. (chart). Like the front-page curse, they also tend to be contrary indicators. However, they aren't bullish enough to be bearish for stocks.



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Subject: Marvell 營收翻十倍？SpaceX 下場做電信商？Google帶核電飛？ - KP思考筆記(第63期)
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KP@FOMOSOC, BENITA HSU, AND ポニョ オタク
OCT 10

READ IN APP

大家好，我是 KP，歡迎來到第 63 期的《週末思考筆記》。

大概兩個月前，我開了一個新欄目，叫「財報解剖」。當時有跟大家講，這是一個新嘗試，不保證更新頻率，先跑跑看。

兩個月下來，總共出了五篇，節奏慢慢穩了，我和團隊也越寫越順。**所以今天想正式跟大家宣佈：這個欄目會一直做下去。**

也就是說，FOMO 研究院現在有三條線：

- **深入分析：** 一個產業或公司主題，深挖到底，週中發出；

- **思考筆記：** 相對短一點，一週的市場觀察與隨筆，週末發出；
- **財報解剖：** 比較硬核，數字會多一點，在一家公司出財報之後發出。

講到這裡，我想多聊幾句我對這三條線的想法。

深入分析做了超過一年了。這一年多，我們一起走過了很多產業、很多公司，有些可能你之前完全沒聽過，有些你可能聽過但沒認真看過。

我很喜歡這個過程，把一個陌生的東西拆開，搞懂它，再跟你講清楚。

> 但我最近開始有一個感覺：廣度鋪開了，是時候多一點往深度走。

財報解剖做的事情，其實跟深入分析不太一樣。

深入分析是帶你認識一個你可能不知道的公司或產業；**財報解剖是挑一家大家都聽過的公司，但我們不只看一次，我們持續看、反覆看、每個季度再看一次。**

我想做的事情是建立一個「追蹤名單」。你作為讀者，可以每個季度回來，看看這家公司跟上一季比怎麼樣了、跟我們上次拆解的時候比怎麼樣了。

老實講，我目前還沒有把同一家公司寫過兩次財報解剖，所以這個「追蹤」的感覺到底會變成什麼樣子，我自己也很期待，要跑起來才知道。

但方向是這樣的：**一個覆蓋名單，慢慢成形，大家可以一起追蹤那些公司的進展。**

完整名單我還沒完全定下來，之後會慢慢跟大家揭曉。

我也不保證每一次財報我都會寫一篇完整的長文，有些季度可能真的沒什麼大變化，但我希望至少會有一個短筆記，跟你講一聲：「嘿，我看了，大概是這樣。」

這樣你不會斷線，我也能把脈搏持續摸著。

好，講完願景，講一下實際的節奏調整。

下週有兩家公司出財報（我不講名字，讓大家猜一下，但你查一下財報日曆大概猜得到），我和團隊會集中火力準備，財報出來之後也會第一時間拆解。

所以下週，會有兩篇財報解剖，取代原本的深入分析。

往後也是這樣：如果某一周有比較多我們要覆蓋的財報，財報解剖可能會頂掉那週的深入分析。每次有這個情況，我會提前跟大家講一聲。

我想強調的是，這不是說深入分析不重要，它依然是主菜。

> 只是跑了一年多，我們把地圖畫得夠大了，接下來我想把一些已經畫上地圖的地方，再鑽深一點。
>
> 與其每週都在開新地圖，不如偶爾回到一個老朋友那裡，看看它這三個月長成什麼樣子了。

做「深入分析」需要極大的腦力激盪，但我後來發現，其實大家的注意力也是有限且寶貴的資源。

如果我們每週都在塞一個全新的、陌生的公司或產業，久了大家也會產生資訊疲勞。「怎麼這週又是一個我沒聽過的名字？」。

所以，偶爾用大家熟悉的公司（財報解剖）來替換一下，不僅能讓我的研究團隊有更充足的時間打磨品質，也能讓大家在閱讀時有更熟悉的錨點，讀起來更輕鬆、更有共鳴，也有一種「喔，原來它變成這樣了」的樂趣。

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這個星期我們有三個主題，但當中有兩個比較長：

- **主題一 | Marvell 說五年後營收會翻十倍？**
- **主題二 | Google 簽約 Constellation Energy，核電準備起飛？**
- **主題三 | SpaceX 準備下場做電信商？**

不妨考慮先讚後看，我們開始!

主題一｜Marvell 說五年後營收會翻十倍？

Marvell 的投資人日上，管理層把 2031 年的營收目標畫在了 700 億到 900 億美元之間，中位數 800 億。這是現在 82 億營收的整整十倍。

華爾街之前怎麼估的？大約 470 億。

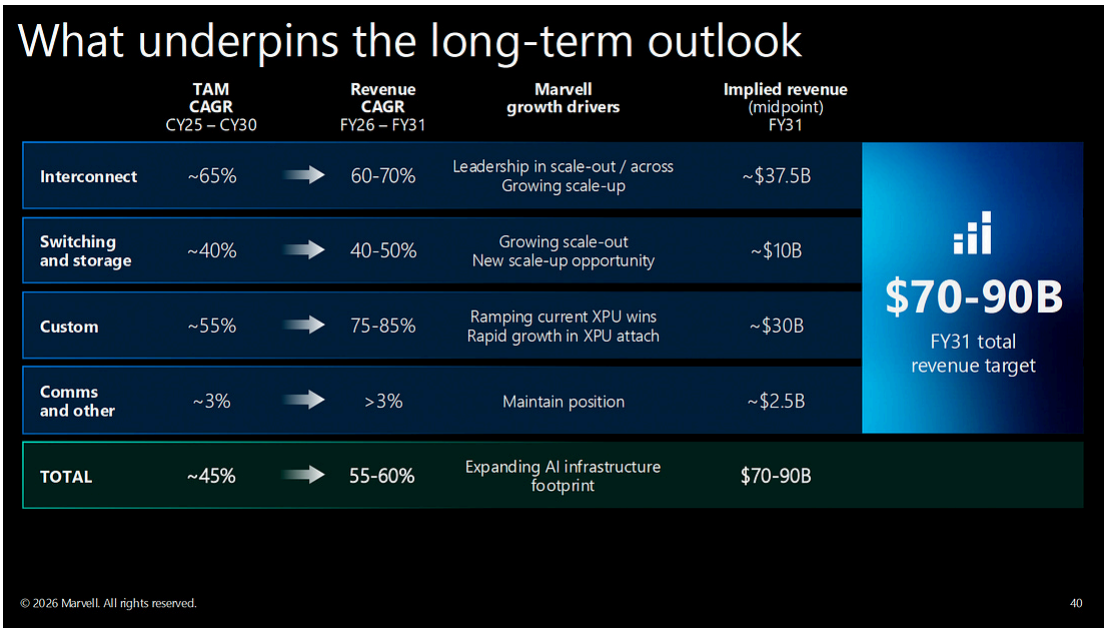
「800 億」這個大餅當然很吸引，但背後是由甚麼組成才是重要的。

我想和大家分享三個洞察。

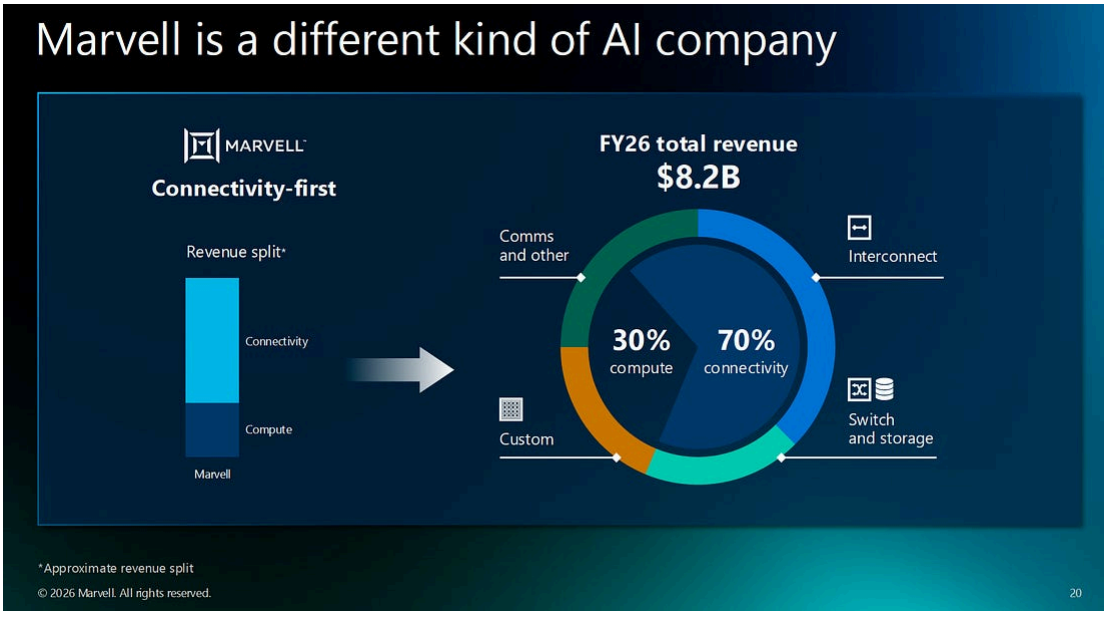
洞察一：光通訊是「本命基本盤」，客製化只是「高槓桿期權」

在 Marvell 規劃的 2031 年 800 億美元營收藍圖中，我們可以把業務拆成兩大核心：

(其實還有交與儲存，通訊等等，但這些比較小，我們暫不討論)



1. **互連與光通訊（Interconnect）：** 預估營收 375 億美元。對比 2030 年該領域 650 億美元的市場規模（TAM），**這意味著 Marvell 預期自己將拿下高達 58% 的市場份額。這是一種「準壟斷」地位。**
2. **客製化晶片（Custom Silicon）：** 預估營收 300 億美元。看起來數字也很大，但對比 2030 年高達 2,350 億美元的客製化晶片市場，Marvell 的份額其實只有 13%。



雖然我在不同文章也說過很多次，但總是很多人仍然視Marvell為一家ASIC公司。

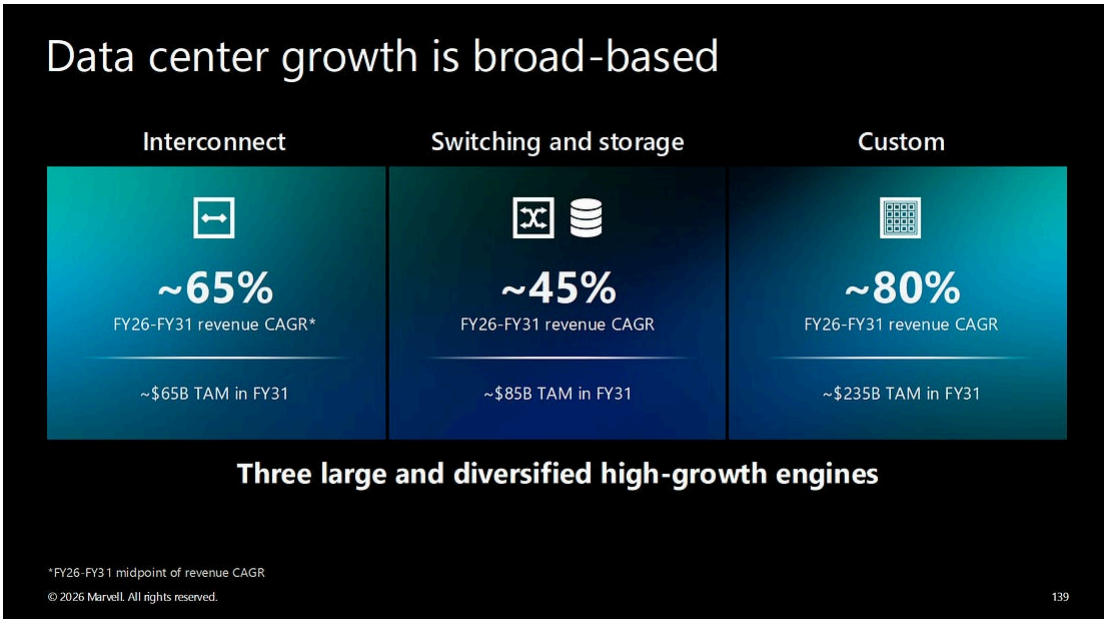
但在客製化晶片市場裡，Marvell 依然只是一個挑戰 Broadcom 霸權的「挑戰者」；但在光通訊與互連領域，它才是領導者。

什麼是互連？在一座大型 AI 資料中心裡面，擺滿了成千上萬顆 AI 晶片。

每一顆晶片就算運算速度再快，如果晶片跟晶片之間傳遞資料太慢，整座機房就只能停下來空等。Marvell 賣的產品，就是讓這些晶片能夠高速互相溝通的關鍵零件。

這些零件包含了許多類型：例如「光訊號處理晶片（DSP）」，負責把電訊號轉換成光訊號，再透過光纖傳送到另一個伺服器機櫃；「交換器晶片」，負責指揮資料流量該往哪裡走；還有專門用來增強銅線訊號品質的晶片。

根據 Marvell 內部的預估，**在伺服器機櫃內部、晶片與晶片之間互相傳輸的資料，就佔了整座 AI 資料中心總流量的 85% 以上。**



如果我們進一步把這兩組數據的「成長率」放在一起比對，會發現一個更值得玩味的細節：

- **互連產品：** Marvell 預估自家的營收年成長率約為 60% 到 70%，而整個市場的成長率大約也是 65%。**這意味著，Marvell 的預測前提並不是「我要去搶下更多市場」，而是「我要在未來五年內，穩穩守住這過半的市占率」。**
- **客製化晶片：** Marvell 預估自家的營收年成長率高達 75% 到 85%，遠遠高於整個市場 55% 的平均成長速度。**這意味著，Marvell 的預測前提是「我要發動猛烈進攻，從競爭對手的手中搶下更多的市占率」。**

一守一攻之間，面臨的風險完全是兩回事。

在防守的這一邊（互連產品），公司最害怕的是「技術世代交替」。互連產品是賣給所有客戶的標準化產品，不依賴單一買家。

但是，只要傳輸速度每提升一個世代（例如從 800G 到 1.6T，再到 3.2T），客戶都會重新評估並選擇供應商。

每當技術進入世代交替的時候，原本穩固的市占率就有可能被競爭對手趁虛而入。

在進攻的這一邊（客製化晶片），潛在客戶池本來就只有那幾家大型雲端服務商。只要其中一家客戶決定把下一代產品的訂單轉給別人，或是決定把產品延後一代推出，營收就會瞬間掉一大塊。

> **所以，Marvell 喊出的 800 億美元營收目標，是由兩個完全不同的賭注拼湊而成的。**
>
> **一個要求它在五年內絕對不能失守陣地，另一個則要求它在五年內必須持續搶勝。這兩個條件必須同時成立，總目標才有可能順利兌現。**

這一點在短期的業績調升中展現得淋漓盡致。

Marvell 將 2028 財年的營收目標從 180 億美元調升至 200 億美元，這多出來的 20 億美元，完全是由光通訊（Scale-out/Scale-up optics）和交換器晶片拉動的，跟客製化晶片沒有關係。

正如執行長 Matt Murphy 所說：「是Interconnect在引領這場競賽。」

> **你買 Marvell，買的其實是 AI 時代「數據傳輸」的物理剛需，客製化晶片更多是附贈的超額期權。**

還有一個現在幾乎是零、但可能改變遊戲的東西：scale-up 光學。

今天這塊的收入基本是零。管理層說放量是 2028 年的事。每顆 XPU 的光學元件價值是數千美元，而且鏈路的兩端都需要光學。

Celestial AI 的光子互連技術，瞄準的是記憶體共享：HBM 距離處理器大約 1.5 毫米，所以今天加記憶體基本等於加處理器。

洞察二：毛利率的障眼法，營業利益率才是真功夫

在展開之前，我們先把兩個名詞講清楚。因為接下來這段分析，全部建立在這個區別上。

毛利率 (Gross Margin)，是營收減去「直接製造成本」之後的比例。

所謂直接製造成本，就是晶圓、封裝、測試這些跟「做出這顆晶片」直接相關的費用。它不包含研發、銷售、行政這些間接開銷。

營業利益率 (Operating Margin)，是把所有營運費用都扣完之後，真正剩下的利潤比例。

除了直接製造成本，它還要扣掉研發費用、銷售費用、行政費用，這才是衡量一家公司「到底賺不賺錢」的真實指標。

一句話總結：毛利率告訴你「產品本身的利潤空間有多大」，營業利益率告訴你「公司在扣完所有開銷之後，真正賺到手多少」。

現在，回到 Marvell。

如果你有關注上一季的財報，你可能會記得：第三季的毛利率指引從 58.9% 微幅下滑到 57.5% 到 58.5%。就這麼一點點，股價當天重挫了 10%。

市場的解讀是，毛利率降了，代表產品利潤在縮水，代表生意變差了，賣。

但這個結論並不是完全正確的。因為毛利率下降的原因，不是產品利潤變薄了，是「營收組合變了」。

這就要講到 Marvell 兩種產品的本質差異。

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第一種：標準光通訊產品。毛利率高，但研發費用重。

這類產品是 Marvell 自己設計、自己開發，然後賣給很多客戶。因為是標準化產品，每顆晶片的直接製造成本相對低，所以毛利率很高。

但代價是，Marvell 必須自己養一大批工程師，持續投入下一代產品的研發。這些研發費用是 Marvell 自己承擔的，全部算在營業費用裡。

第二種：客製化晶片。毛利率較低，但營業利益率非常好。

客製化晶片的直接製造成本比較高，因為要針對每個客戶的需求做定制化設計和驗證，所以毛利率看起來比較低。

但這裡有一個關鍵的結構差異：客製化專案的研發費用，大部分是由客戶承擔的。業界叫 NRE（Non-Recurring Engineering），也就是客戶在專案啟動時就先付一筆研發費。

這意味著，Marvell 做客製化晶片的時候，最貴的那塊「研發」，已經被客戶買單了。

一旦進入量產，規模效應出來，每一顆晶片的邊際成本很低，不需要再額外投入大量研發。

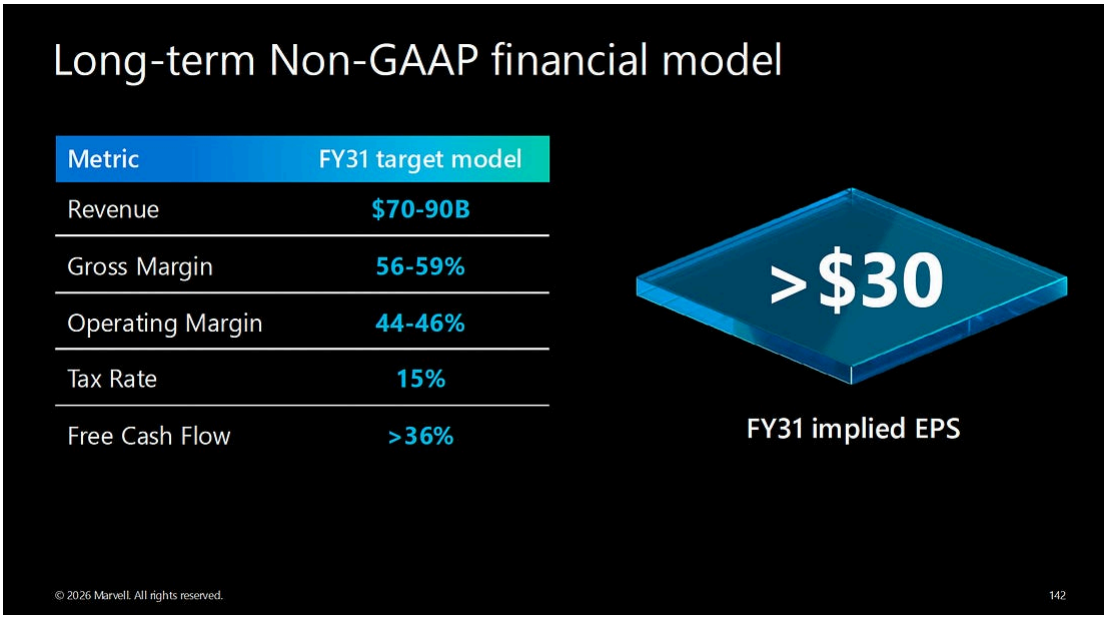
所以結果是：毛利率比較低，但扣完所有費用之後，真正賺到手裡的利潤比例反而更高。

現在你明白上次財報為什麼跌了。

當客製化晶片的營收佔比提高，整體毛利率自然會往下走。因為客製化的毛利率比標準產品低。

但這不代表生意變差了。這代表營收正在從「高毛利但研發自付」的產品，轉向「低毛利但研發由客戶買單」的產品。毛利率表面上掉了，但營業利益率其實在改善。

這次投資人日，管理層直接回答了這個問題。



他們沒有提高毛利率的長期目標。它維持在 56% 到 59%。意思是：毛利率會隨著產品組合波動，這是結構性的。

但他們把營業利益率的長期目標，從過去的 38% 到 40%，一口氣拉到了 44% 到 46%。整整多了六個百分點。

邏輯很簡單：當營收從 82 億成長到 800 億，研發團隊不會變成十倍大，銷售團隊不會變成十倍大，行政費用不會變成十倍大。營業費用的增速，大約只有營收增速的一半。

這就叫營運槓桿。規模越大，每一塊錢營收裡面，真正掉進股東口袋的比例就越高。

管理層甚至給了一個對沖框架：

如果未來客製化晶片賣得比預期好，毛利率會往 56% 那一端靠攏，但營業利益率會被撐住，因為研發費是客戶出的。

如果光通訊賣得比預期好，毛利率會往 59% 那一端走，但研發費用會比較重，會吃掉一些。

不管哪一邊贏，營業利益率都應該守在 44% 到 46%。

因此，最重要的數字，是營業利益率有沒有守住。

而那個「30 in 30」，也就是 2030 年 EPS 超過 30 美元，只有在這個營運槓桿真的兌現的情況下才成立。

> **800 億的營收聽起來很厲害。但如果營業利益率守不住，那 800 億就只是營收很大，結果大部分都拿去養工程師 and 付行政費用了，股東分不到多少。**

洞察三：Google 的 1200 億合約，暫時觀察就好

市場上另一個焦點，是媒體紛紛報導的「Marvell 與 Google 簽署六年、總金額 1,200 億美元的龐大合作訂單」。

這個數字聽起來很嚇人。但如果你去翻今年 8 月 18 日 Marvell 提交給美國證管會（SEC）的原始公告，會發現故事完全不是你想的那樣。

這份公告的真實內容是：Marvell 發放了一批「認股權證」給 Google。這個權利允許 Google 在未來，以每股 206.58 美元的約定價格，購買最多約 5,897 萬股的 Marvell 股票。

但關鍵是，這批認股權證不是一次性給的。

除了第一年按季度發放的一小部分之外，其餘絕大部分被拆分成了 240 批。規則很簡單：Google 每向 Marvell 採購滿 5 億美元的客製化產品，才可以解鎖其中一批。

240 批，乘以每批 5 億美元，算出來的結果正好就是 1,200 億美元。

所以這個數字的真實含義是：如果 Google 想要拿滿所有的認股權證，它必須先累計採購到 1,200 億美元。這是解鎖的「上限」，不是一張已經簽字畫押的採購訂單。

換個角度想，這個合約結構其實透露了一個很有意思的事實：**Marvell 需要拿出自家公司的股票作為商業誘因，才能吸引 Google 在未來持續多下訂單。**

而管理層在投資人日上的說法也非常誠實。他們表示，2031 年的財務預測中，只納入了這筆合作的「一小部分」。而且他們強調：「即便未來沒有拿到任何新的設計案，也能穩穩落在 700 到 900 億美元的預測區間內。」

這句話的意思是：那 1,200 億的天花板，根本不在 2031 年的預測模型裡。它是一個更長期的可能性，不是眼前的保證。

不過，這裡有一個細節必須釐清。

管理層在投資人日上強調的是「沒有任何單一 socket 超過營收的 10%」，而不是「沒有任何單一客戶」。

Socket 是一顆晶片在客戶系統裡佔的那個位子（設計席位）。同一個客戶可以有很多個 socket。比如 Amazon，它的下一代 Trainium、光 DSP、交換器、機房間互連，加起來就是四個不同的 socket。

所以，「單一 socket 不超過 10%」，只代表沒有「單獨一顆晶片」大到能決定全年營收。這並不意味著客戶集中度降低了，Amazon 依然是 Marvell 最大的客戶。

Marvell 真正在做的，是「在同一個大客戶裡，塞進更多不同種類的晶片」。他們現在已經同時向美國四大超大規模數據中心出貨客製化晶片，並且在每個客

戶內部贏下了多個設計席位。

客製化業務已經從「靠單一產品押寶」，走向了「用產品矩陣深度綁定」。這才是它作為期權最真實的狀態。

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真正的信號：十個月，四次上調

不過老實說，現在距離2031年還有5年。在AI時代，5個月也嫌太多。任何遠期預測，往往只是基於「AI 資本支出會無限膨脹」的樂觀外推。

所以2031年的預測數字，聽聽就好。

> **真正能驗證 Marvell 底氣的，不是 2031 年的 800 億，而是過去十個月內，管理階層將 2028 年的營收目標連續調升了四次（從 130 億、150億、180 億，一路拔高到現在的 200 億）。**

這種短期的、高頻率的預期上修，是無法靠吹牛粉飾的。它背後代表的是實打實、正在出貨的光模組與交換器晶片。

Marvell 賣的不是「另一家定制 ASIC 廠商」的故事。

它賣的是：是一個互聯的霸主，加上一張在定制市場裡搶份額的期權，再加上一個因為規模效應而重新定價的利潤率模型。

「Marvell 2031 年要做到 800 億」，這句話的解讀應該是：那 800 億裡面，多少是你已經看見的，多少是你必須相信的？

答案的比例，決定了你願意付多少倍。

> Marvell應該如何解讀，我早在半年前，Marvell還沒爆升的時候和大家深入分析過，裡面的邏輯仍然成立，有興趣的朋友不妨重溫分析。

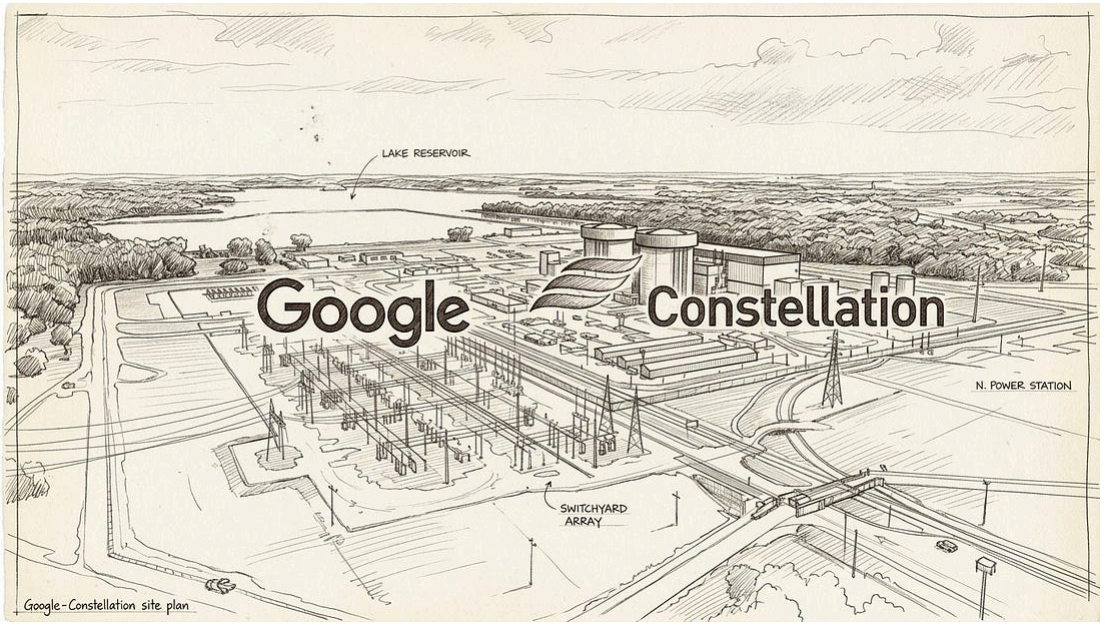
雲端巨頭為何急需ASIC？博通為何估值比NVDA還高？光通訊是Marvell重要支柱？- 深入分析第38期：Broadcom+Marvell

KP@FOMOSOC · MAR 11

想像一下，你正坐在Google或Meta的基礎設施決策會議裡。桌上擺著今年剛被董事會批准的龐大資本支出（CapEx）預算：一個高達千多億美元的鉅額數字。

Read full story

主題二｜Google 簽約 Constellation Energy，核電準備起飛？



當 Google 宣布砸下 43 億美元跟 Constellation Energy 合作時，你可能第一個反應是：核電公司終於要起飛了嗎？

但事實上，Google這次砸下重金，是去讓Constellation Energy升級 11座現有的核電機組，預計在 2028 到 2032 年間「憑空擠出」890 MW 的發電容量。

這在產業裡有個專有名詞，叫做「**核電升級擴容（Nuclear Uprates）**」。

如果把能源產業比喻成房地產，現在市場上最值錢的，早就不是一塊什麼都沒有、需要重新整地、環評、等上 15 年才能蓋好的荒地。

> 現在最值錢的，是一棟：土地早就買好、建照早就拿到、水電管線全部接妥、裡面甚至已經住滿房客，而且頂樓「居然還能合法再加蓋三層樓」的現成大樓。

這多出來的「三層樓」，就是 AI 巨頭們現在不惜代價、用 20 年長約也要搶下來的黃金資產。

時間，是 AI 擴張最大的敵人

為什麼 Google 選擇升級舊廠，而非蓋新廠？**因為蓋新核電廠最大的敵人從來不是技術，而是時間。**

美國喬治亞州的 Vogtle 核電廠新增兩個反應爐，從動工到商轉整整花了十五年，建造成本更是天文數字。

相比之下，我們來算一筆帳：Google 這次幫 CEG 出資 43 億美元換取 890 MW 的新增容量，折算下來，每千瓦（kW）的建置成本大約是 4,800 美元。

這個數字，僅僅是 Vogtle 新建機組單位成本的幾分之一，**而且速度快了近十年。**

「升級擴容」這種特定的增量方式，既便宜，又剛好能趕上 2028 到 2032 年這段最致命的缺電窗口。

缺電有多急迫？負責管理美國多州電網的 PJM 在最新報告中預估，2028 到 2029 年就會出現大約 6.8 吉瓦（GW）的可靠電力缺口，而資料中心正是主要來源。

但在 PJM 最近一輪的容量拍賣中，全新機組加上擴容項目，最終竟然只成功出清了約 525 MW。

面對 6.8 GW 的無底深淵，這點新增電力根本是杯水車薪。**資料中心的擴張速度，不可能乖乖等上十五年。**

過去，核電業面臨最大的難題是前期投資太龐大。

如果你要砸 40 億美元升級設備，董事會一定會問：未來二十年的電價怎麼算？如果五年後電價大跌，這筆錢就白花了。

現在，Google 和 Amazon 站出來了。**他們直接簽下十五到二十年的長期購電合約，把未來的訂單提前包下來。**

就在 Google 宣布這筆交易的前一週，Amazon 才剛跟 Constellation Energy 達成另一筆包含 190 百萬瓦升級項目的長約。兩家巨頭加起來，實質支持了超過 1 吉瓦的核電增量。

這等同於科技巨頭用自身的信用，幫核電業者大幅降低了投資風險。原本在試算表上內部報酬率不夠漂亮、不值得投資的升級專案，瞬間變成了華爾街最愛的穩定獲利資產。

> AI 不僅帶來了電力需求，更徹底改變了核電產業的融資方式。

看懂「升級潛力」這個新指標

如果這個邏輯成立，金融市場對核電公司的估值方式也要跟著改寫。

以前評估一家核電營運商，大家看的是總發電量、電價和燃料成本。**但未來，有一個更重要的指標叫做「升級擴容潛力」。**

市場會開始問：你手上的反應爐還可以再增加多少發電量？每增加一度電需要多少成本？美國核管會核准需要多久？你能不能找到像 Google 這種大客戶簽長約？

假設有兩家公司，各自擁有 10 吉瓦的核電容量。A 公司的機組已經操到技術極限，沒有升級餘地。

B 公司的機組還能再擴容 800 百萬瓦，而且執照能一路延長到 2050 年。在現在的市場眼中，**這兩家公司的經濟價值絕對是天壤之別。**

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反向銷售Gemini

這筆交易另一個有趣的點在於：**Google 同時是客戶，也是技術供應商。**

在簽下購電協議的同時，CEG 也與 Google 簽署了為期 5 年的技術合作，導入 Google Cloud 與 Gemini Enterprise。而 Gemini 會在三個領域使用：

- 加速電力開發、選址與併網規劃。
- 優化核電設備的健康監測與停機維護。
- 保護電網與核電設施的關鍵營運系統。

Google 拿著大筆資金去買電，轉過身來，**又把自己的 AI 與雲端服務賣進了能源產業最核心、最敏感、也最不容出錯的營運系統裡。**

一旦 Gemini 成功在 CEG 的 11 座核電機組中證明了自己能優化停機時間、預測設備故障，這套系統將成為 Google Cloud 攻佔全球公用事業市場最強大的活廣告。

升級不是無限魔法

不過，核電升級擴容是有天花板的。你不可能把一座 1 吉瓦的反應爐，一路改造成 2 吉瓦。

美國核管會將升級分為不同類型，規模最大的擴展功率升級，最多也只能增加約 20% 的發電量，而且需要大規模設備改造。

這意味著，**當市場上那些最容易升級的反應爐都被榨乾之後，下一個需要的百萬瓦電力，最終還是得靠蓋新核電廠、天然氣，或是再生能源搭配儲能來補足。**

所以，在觀察核電產業時，我們可以把公司分成兩種。一種是只靠著核能題材在說故事的公司，另一種是手上真的握有可以變現、且還有升級空間的反應爐的實力派。

這場由 AI 驅動的核能復興確實正在發生。

> 但最直接的贏家，是那些在二、三十年前蓋好、以前大家覺得無聊透頂，現在卻能替 AI 時代擠出關鍵電力的舊核電廠。

我之前寫過兩篇電力的分析，分別分析了公用電力發電商，以及獨立發電商的邏輯。而在獨立發電商中，又再分成不同的種類。有興趣的朋友不妨重溫分析。

核能復興應該買甚麼？獨立發電商如何分析？- 深入分析第42期：CEG，VST，TLN，OKLO (電力系列下集)

KP@FOMOSOC · APR 8

2024 年 9 月，三哩島（Three Mile Island）核電廠，Constellation Energy 的執行長Joe Dominguez站在這個曾經因為 1979 年美國史上最嚴重核災而舉世聞名、甚至帶有幾分禁忌色彩的地方，向全世界宣布了一項重要的決定：

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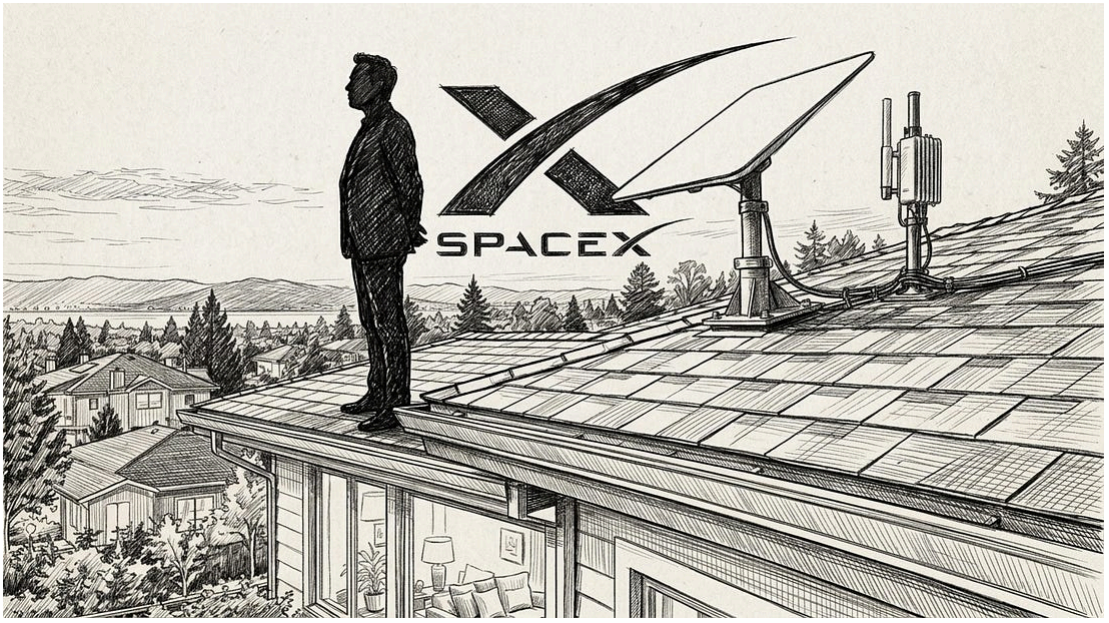
主題三：SpaceX 準備下場做電信商？



美國人這幾十年來用的手機門號服務，基本上都離不開三家大公司：AT&T、Verizon 和 T-Mobile。

這三巨頭花了幾十年的時間來建設行動網路，砸大錢買下大量的無線電頻譜、蓋基地台、鋪設光纖網路，築起了一道外人很難跨越的超高競爭門檻。

不過，兩天前，SpaceX 和 Grain Management 達成協議，準備買下對方手中的一批 800 MHz 低頻頻譜。

這批頻譜最寬的郡大約有 14 MHz 的成對頻段，部分地區甚至只有 5 MHz 左右。整筆交易還需要經過 FCC 核准，目前尚未完成交割。

根據媒體報導，這筆交易大約花了 80 億美元。**消息一出，這三大電信巨頭的股價在盤後全面大跌，跌幅達到 6% 到 7%。**

為什麼只是買下一段看不見的無線電波頻率，就會讓這些傳統電信商的市值瞬間大縮水？SpaceX又在準備甚麼策略？

要搞懂這件事，我們得先了解 SpaceX 過去到底還缺了什麼武器，以及它未來打算怎麼打這場仗。

從「補死角」到「搶客戶」

大家都聽過的星鏈（Starlink），是靠著低軌道衛星來提供網路服務。

一開始，你想用星鏈的網路，還得在自家裝個專門的接收盤。後來 SpaceX 推出了「Direct-to-Cell」技術，讓一般的智慧型手機直接連上衛星，不用再另外花錢買傳統那種笨重的衛星電話。

這聽起來超厲害，但它一開始的商業用途，主要是為了補救傳統基地台訊號打不到的地方。

舉例來說，當你開車經過荒郊野外的公路，附近半個基地台都沒有的時候，手機還是可以靠天上衛星傳求救訊息或是簡訊。

所以，早期 SpaceX 其實是跟電信商 T-Mobile 合作的。T-Mobile 繼續賺客戶的月租費，而 SpaceX 則是幫忙填補網路訊號的死角。

這原本是一個雙贏的安排，大家各取所需。

但問題是，死角批發和自己賣門號，是兩種完全不同的生意。

前者賺的是供應商的錢。用戶每月付給 T-Mobile，SpaceX 只是在基地台打不到的地方把簡訊和少量數據接上衛星。

客戶名單、品牌、月租定價、加價賣方案的權力，通通都在 T-Mobile 手上。衛星用得越多，越像一項被包進電信商資費裡的功能，越不像一門獨立生意。

T-Mobile 的執行長甚至公開說過，衛星覆蓋以後會變成標配，不再是差異化。言下之意就是，配角的批發費，遲早會隨著這件事變便宜。

後者賺的才是電信商的錢。**門號生意的核心在於你擁有客戶、擁有帳單、擁有定價權。**

問題來了，SpaceX 現在已經不甘於只當個配角。

今年 8 月，SpaceX 的總裁 Gwynne Shotwell 很明白地表示，公司有意自己蓋地面網路，把美國無線市場估在大約 6,000 億美元，直接跟大型電信商正面對決。

到了 10 月，SpaceX 更在官方公告裡放話，**Starlink Mobile 的最終目標，就是要成為美國主要的手機電信業者。**

這代表 SpaceX 想做的，早就超出了「讓你在深山裡能收到一格訊號」的範圍。

它是希望有一天，你可以直接打電話退掉 Verizon 的月租方案，改向 Starlink Mobile 繳錢辦手機門號。

> 當「合作夥伴」公開宣示要「搶客戶」的時候，雙方的關係就產生了質變。

電信商的防禦同盟

要理解這場博弈，我們得先聊聊電信業的一個特殊角色：**MVNO（虛擬行動網路電信商）。**

MVNO 就是那些自己沒有實體基地台，靠租用 AT&T、Verizon 或 T-Mobile 的網路來賣門號的公司，比如美國的 Mint Mobile 或 Google Fi。

對新進者來說，當 MVNO 是最省錢、最快的捷徑。

對任何想進入電信市場的玩家而言，最省錢的劇本都是先跟電信商批發地面網路，在城市和室內使用現成的訊號，再搭配自己的技術補強死角。

這樣一來，就能用極輕的資產，迅速推出一張「全美無死角」的資費方案。

但這個模式能順利推進的前提，是「互補大於競爭」。當 SpaceX 只是幫忙補死角時，電信商很樂意賺這筆批發費。

可是，當 SpaceX 公開宣示要搶客戶、甚至準備自己買頻譜蓋地面網路時，三大巨頭聯手警覺：把城市核心網路租給一個準備用低價策略殺價的對手，等於是在割自己的肉。批發賺來的錢，根本彌補不了被挖走的月租費與定價權。

這條原本可能存在的合作路徑，正在迅速變窄。

> 電信商的防禦動作極其迅速。10 月 1 日，AT&T、T-Mobile 和 Verizon 正式簽署合約成立合資公司，將部分頻譜集中起來，用來補足傳統基地台打不到的死角。

合資聲明寫得很溫和，但華爾街普遍將此解讀為對 Starlink Mobile 的戰略防禦。

這場防禦的訊號，在細節中展露無遺。T-Mobile 悄悄將官網首頁上的「with Starlink」字樣撤下，在門面上與 SpaceX 劃清界線。

三大電信商試圖自己掌控頻譜主導權，組建一個不以 Starlink 為核心夥伴的死角平台。

> 而當三大電信商組成同盟時，他們需要一個衛星通訊的同盟，而這個很可能會是 AST SpaceMobile。

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SpaceX 的破局：買下 800 MHz 自己建網

既然租電信商城市網路的難度越來越高，SpaceX 決定自己買頻譜。

我們可以把無線頻譜想像成一條高速公路。馬路越寬，同一時間能塞進去的車子通常就越多；可是，如果這條路根本開不到你家門口，那車子再多也沒用。

通訊頻譜有兩個最關鍵的問題：**第一是訊號能傳多遠，第二是能同時塞多少上網資料。**

(關於頻譜的詳細分析，可以回看我之前講衛星的深入分析，在這裡不再詳述)

SpaceX 之前透過 EchoStar 拿到了一段 2 GHz 的頻譜，這可以當作衛星通訊服務很重要的「容量」來源。但是，頻率比較高的訊號有一個缺點，遇到牆壁之類的障礙物比較容易被擋住。

而這次買下的 800 MHz 頻率比較低、波長比較長，不僅傳播條件好，穿透部分建築物的能力也比較有優勢。

簡單來說，2 GHz 負責提供傳輸大量資料的空間，800 MHz 負責鋪設一張無死角的覆蓋網。

> **這讓 SpaceX 拿到了自己蓋「地面網路」的入場券**，可以把天上的第二代衛星、地面的小型基地台以及這張低頻訊號網結合起來，打造一個完全屬於自己的「Starlink Mobile」。

有趣的是，這批 800 MHz 頻譜的源頭，其實是 T-Mobile 在 2020 年併購 Sprint 時被政府要求剝離的資產。

T-Mobile 自己建立全國 5G 覆蓋網用的是 600 MHz 頻段，800 MHz 對它而言只是重複的冗餘覆蓋。把一個自己根本沒在用的資產賣掉，換回 29 億美元現金加上自己真正需要的 600 MHz 頻譜，這是一筆極度理性的資產置換。

後來這批資產轉到了 Grain 手中，如今又被 SpaceX 以近 80 億美元的高價接手。

> 對 SpaceX 而言，全國性的低頻頻譜幾乎不可能從三大巨頭手中流出，**這段 800 MHz，是馬斯克在美國建立「獨立手機網路」難得能買到的門票。**

自建網的物理現實與時間賽跑

就算拿到了頻譜，SpaceX 接下來面對的依然是極度現實的物理限制。

800 MHz 頻譜確實能改善覆蓋範圍，但它無法憑空創造無限的網路容量。

這次買下的頻寬最厚只有 14 MHz，在某些地區甚至只有 5 MHz。

這是一層極薄的覆蓋毯。如果幾十萬人同時擠在紐約曼哈頓市中心看影片、開視訊會議，這種低頻薄頻寬根本撐不住。

城市尖峰時刻的龐大資料流量，依然需要高頻頻譜與密集的基地台。

因此，SpaceX 的建網策略注定是「衛星為主，微型基站為輔」。

它可能會把小型基站直接掛在用戶家中的 Starlink 接收盤旁邊，或者只在有高密度需求的地方按需部署。這種做法資本支出極低，覆蓋範圍會跟著用戶慢慢長出來。

這意味著 SpaceX 開台第一天，絕對無法提供全國室內無死角的完美體驗。

真正的護城河：不在地面，在火箭

很多人盯著 800 MHz 的薄頻寬，覺得這不足以撼動三大巨頭，卻忽略了 SpaceX 真正的殺手鐧：發射成本與星座規模。

SpaceX 自己造衛星、自己用火箭打上去，FCC 最近還批准了多達 15,000 顆下一代 Direct-to-Cell 衛星的部署。

這種「太空基礎設施」的規模優勢，是 AT&T 或 Verizon 花多少錢都買不到的，也是其他衛星新創難以企及的。

當傳統電信商在地面為了幾 MHz 的頻譜斤斤計較、為了基地台選址和地方政府博弈時，SpaceX 正在頭頂上佈建一張不受地形限制的立體網路。

> 天上用下一代衛星直連手機提供廣覆蓋，地面用剛買下的 800 MHz 鋪一層穿透力強的基礎網。

這套組合拳，才是 SpaceX 敢於直接賣門號的底氣。

定價權的倒數計時

根據公司近期的說法，比較有意義的 5G 級衛星直連服務，大概要到 2028 年上半年才會開始上線。

即使屆時正式開台，初期體驗也不會完美。室外死角會明顯減少，對價格敏感、不在乎極致網速的用戶有吸引力；但城市室內與尖峰時段的容量，仍會明顯落後三大電信商。

SpaceX 開台第一天賣的，會是一個**「覆蓋更廣、價格可能更低的基本服務」**，而不是立刻全面取代現有的高速體驗。

但是，SpaceX 不需要提供完美的 5G 體驗，只要它提供一個「足夠便宜且沒有死角」的替代品，就足以讓那些對價格敏感的消費者心動，進而打破電信巨頭們多年來維持的價格默契。

從投資的角度來看，市場重新定價的焦點，早就超越了「明年會流失多少用戶」。

真正被放上天平的，是未來十年內，這些依賴高資本支出築起護城河的老字號公司，是否還能維持過去那種穩定且豐厚的利潤率。

當破壞者拿到了低頻頻譜這張入場券，並且擁有對手無法複製的太空優勢時，定價權的流失，往往會比用戶的流失來得更快。

關於太空，我在年初的時候寫了兩篇，分為衛星和發射的兩篇，有興趣的朋友可以重溫當中邏輯。

低軌衛星的決戰元年？手機直連是下個戰場？- 深度分析第28期：Starlink，亞馬遜，AST SpaceMobile

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太空，總令人有無數的幻想。

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Will Nvidia use Intel Foveros 3D to replace TSMC’s CoWoS? Passive overflow order Part II

Nvidia 要是真的開始成功地採用 Intel 的先進封裝 Foveros 3D 來取代台積電的 CoWoS，主動式外溢單變成被動式外溢單，大客戶將逐步擴大第二來源 CoWoS like 產能的比重

ANDREW LU ON GLOBAL SEMI/TECHS
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What’s new?

GF Securities 分析師 Jeff Pu 於 10 月 9 日發布最新機構研究報告指出，Nvidia（NVDA）正積極評估於 2028 年下半年，將部分次世代 AI 加速器（對應 Feynman 系列或 Rubin Ultra 延伸架構）之後段先進封裝導入 Intel Foundry 的 Foveros 3D 堆疊技術。值得注意的是，前段 GPU 運算核心 Die 仍將 100% 由台積電（2330.TW）先進製程（2nm / A16）代工製造，但後段 3D 晶片整合與中介層封裝將引入 Intel 作為第二供應商，以解決台積電 CoWoS 產能長期吃緊瓶頸。但長期還是要看 Intel 在EMIB/Foveros 先進封測的良率表現及其成本競爭力。

Positive impacts to Intel and supply chain vendors

Intel（INTC.US）：若順利獲 Nvidia 採用 Foveros，將是 Intel Foundry Services（IFS）在先進封裝領域具指標性的外部客戶突破，驗證其「封裝優先於晶圓代工」商業策略，有助於填補新墨西哥州 Fab 9 及奧勒岡州先進封裝產能，改善代工部門虧損，未來再透過 Foveros 合作量產經驗，擴展延伸到前段晶圓代工訂單。這項合作案如果成真，也將對谷歌跟Intel EMIB 合作案的繼續擴大有幫助。

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測試設備龍頭 Advantest（6857.JP）與 Teradyne（TER.US）：跨 Foundry 封裝架構意味著晶片測試介面與 KGD（Known Good Die）篩選程序複雜度倍增，帶動 SLT（系統級測試）與高階探針卡需求激增。如果台灣先進封裝設備廠能切入 Intel EMIB/Foveros 製造產業鏈，也能受惠。

OSAT 測試與後段外包廠（Amkor AMKR.US）：當先進封裝生態由台積電一站式統包轉向混合代工模式，最後組裝測試與基板重佈線之委外代工比重將同步提升。

Negative impacts to other supply chain vendors

台積電 CoWoS 生態圈本益比溢價承受調整壓力：雖然台積電前段晶圓製造地位依然固若金湯，但過往市場給予台積電「前段製程 + CoWoS 封裝一條龍壟斷」的封閉生態溢價將面臨解構風險。從谷歌EMIB訂單，到Nvidia Foveros 3D 封測統包訂單流失破口一旦擴大，這將讓主動式外溢單變成被動式外溢單，台積電大客戶如谷歌，及Nvidia將逐步擴大第二來源 CoWoS like 產能的比重。

台灣先進封裝設備廠：若 Nvidia 將部分封裝機台資本支出外溢至 Intel 供應鏈，台積電本土供應鏈（弘塑/辛耘濕製程，萬潤，均華/均豪，志聖等 CoWoS 設備大廠）如果無法切入Intel EMIB/Foveros 產業鏈，其訂單能見度將受到多源採購分流影響。

Our advice to investors

評估 Intel Foundry 轉型價值：投資人可能要分拆評估 Intel 的晶圓代工與封裝代工業務。EMIB/Foveros 獲外部頂級客戶認證代表先進封裝毛利率具備修復空間，但 Intel 晶圓製造良率（18A/14A）仍具執行風險。台積電前段 2nm 與 A16 先進製程市占率依然高達 90% 以上，封裝委外反而可使台積電釋放CoWoS產能，專注於毛利率高達 65%+ 的前段晶圓代工，短期結構性影響不大。

Non consensus outside the box thinking

市場普遍將「Nvidia 採用 Foveros」解讀為 Intel 晶圓代工成功扳倒台積電，但本質上是AI 加速器 GPU/TPU/ASIC 由於面積持續擴大，突破光罩極限

（Reticle Limit），先進封裝已不再是晶圓製造的附屬品，而是獨立的微系統架構。Nvidia 此舉絕非放棄台積電，而是想要主導半導體供應鏈建立如同傳統 PCB 一樣的「跨廠標準化封裝匯流排」，實質上是 Nvidia 在剝奪晶圓廠對系統整合的主導權，將議價籌碼完全收回自己手中。但前提還是要看 Intel 在 EMIB/Foveros 先進封測的良率表現及其成本競爭力。

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Start writing

Largan (3008.TW)

FAU to drive strong earnings growth and re-rating; TP up to NT$11,000, Buy

3008.TW	12m Price Target: **NT$11,000.00**	Price: **NT$6,660.00**	Upside: **65.2%**

BUY

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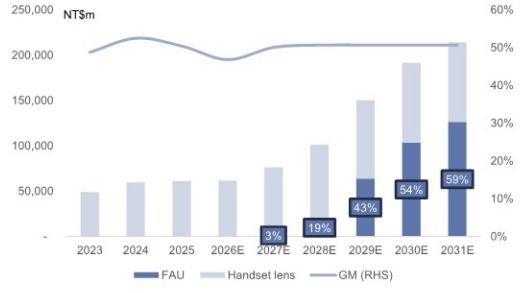
We are positive on Largan and expect its strong alignment technology to continue supporting its expansion from handset lens into FA and FAU for CPO switches. The company's FA has passed customer qualification with strong positive feedback, while customers prefer to purchase its integrated product (FA+FAU), reaffirming our positive view on Largan's business expansion. Management highlights that its strong alignment and automated production provide strong precision, fast production and testing, supporting customer penetration. We model FAU revenue contribution at 3% in 2027E and above 50% in 2030E, driving Largan's earnings growth, margin expansion, and valuation re-rating. Maintain Buy with TP up to NT$11,000 (implies 39.8x 2028E PE).

Exhibit 1: Largan revenues by products and GM trend



Source: Company data, Goldman Sachs Global Investment Research

3-month revenue preview: Largan delivered weak revenue growth in 3Q26 with revenue -11% YoY, which we attribute to a slow smartphone market amid high memory costs, fewer model launches of its major smartphone brand customer, and foldable phones released in 4Q26 (report link). The miss also reflects a slower handset lens pixel mix upgrade than we had previously estimated.

Key Data

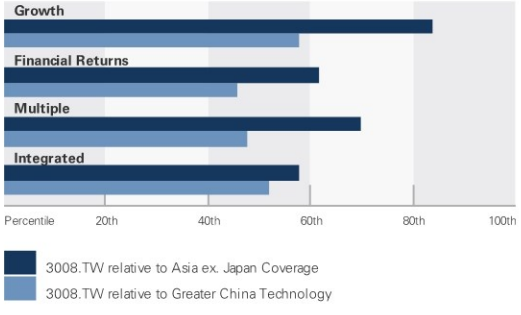
Market cap: NT$896.1bn / $28.1bn
Enterprise value: NT$782.7bn / $24.5bn
3m ADTV: NT$13.2bn / $413.8mn
Taiwan
Greater China Technology
M&A Rank: 3
Leases incl. in net debt & EV?: No

GS Forecast

	12/25	12/26E	12/27E	12/28E
Revenue (NT$ mn) New	**61,147.9**	**61,653.5**	**76,334.7**	**101,306.6**
Revenue (NT$ mn) Old	61,147.9	65,503.4	76,561.8	85,020.5
EBITDA (NT$ mn)	31,290.3	30,251.2	42,329.3	53,616.0
EPS (NT$) New	**159.41**	**162.60**	**207.41**	**278.57**
EPS (NT$) Old	159.41	187.34	217.11	239.29
P/E (X)	14.8	41.0	32.1	23.9
P/B (X)	1.7	4.5	4.2	3.9
Dividend yield (%)	3.3	1.2	1.5	2.1
CROCI (%)	11.2	11.5	14.4	16.2

	9/26	12/26E	3/27E	6/27E
EPS (NT$)	39.62	40.65	49.77	36.05

GS Factor Profile



Source: Company data, Goldman Sachs Research estimates. See disclosures for details.

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BUY

Largan (3008.TW)

Rating since Nov 12, 2024

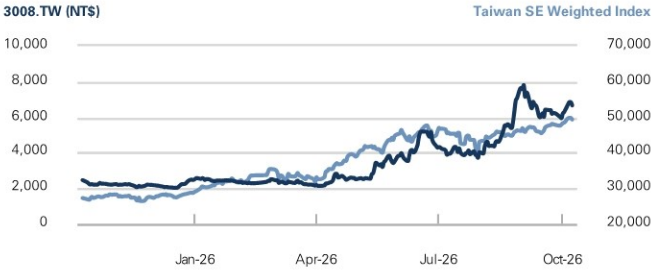
Ratios & Valuation

	12/25	12/26E	12/27E	12/28E
P/E (X)	14.8	41.0	32.1	23.9
P/B (X)	1.7	4.5	4.2	3.9
FCF yield (%)	4.3	1.1	3.7	3.3
EV/EBITDAR (X)	6.4	25.6	17.8	13.8
EV/EBITDA (excl. leases) (X)	6.4	25.6	17.8	13.8
CROCI (%)	11.2	11.5	14.4	16.2
ROE (%)	11.4	11.1	13.4	16.7
Net debt/equity (%)	(64.1)	(60.3)	(66.6)	(68.0)
Net debt/equity (excl. leases) (%)	(64.1)	(60.3)	(66.6)	(68.0)
Interest cover (X)	15,611.8	7,608.9	9,940.7	13,702.5
Days inventory outst, sales	37.1	47.9	40.9	40.4
Receivable days	62.1	62.0	50.0	50.0
Days payable outstanding	21.6	22.0	22.0	22.0
DuPont ROE (%)	11.1	10.7	12.9	15.9
Turnover (X)	0.3	0.3	0.3	0.4
Leverage (X)	1.2	1.2	1.2	1.2
Gross cash invested (ex cash) (NT$)	222,420.9	241,679.6	267,828.7	298,740.5
Average capital employed (NT$)	73,429.5	80,655.1	82,379.7	79,780.5
BVPS (NT$)	1,395.59	1,484.98	1,579.62	1,727.59

Growth & Margins (%)

	12/25	12/26E	12/27E	12/28E
Total revenue growth	2.8	0.8	23.8	32.7
EBITDA growth	3.4	(3.3)	39.9	26.7
EPS growth	(19.7)	2.0	27.6	34.3
DPS growth	(19.0)	2.4	27.0	35.0
EBIT margin	38.5	35.2	39.2	40.7
EBITDA margin	51.2	49.1	55.5	52.9
Net income margin	34.8	35.1	36.3	36.7

Price Performance



	3m	6m	12m
Absolute	68.6%	192.1%	168.0%
Rel. to the Taiwan SE Weighted Index	56.4%	105.9%	47.1%

Source: FactSet. Price as of 8 Oct 2026 close.

Income Statement (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Total revenue	61,147.9	61,653.5	76,334.7	101,306.6
Cost of goods sold	(30,310.8)	(32,786.7)	(38,112.7)	(49,958.8)
SG&A	(1,985.5)	(1,885.5)	(1,832.0)	(1,620.9)
R&D	(5,293.3)	(5,295.8)	(6,488.4)	(8,509.8)
Other operating inc./(exp.)	–	–	–	–
EBITDA	**31,290.3**	**30,251.2**	**42,329.3**	**53,616.0**
Depreciation & amortization	(7,732.0)	(8,565.7)	(12,427.8)	(12,398.9)
EBIT	**23,558.3**	**21,685.5**	**29,901.5**	**41,217.1**
Net interest inc./(exp.)	4,258.6	3,896.3	3,913.0	4,104.2
Income/(loss) from associates	87.0	83.3	–	–
Pre-tax profit	**25,900.2**	**26,446.2**	**33,814.6**	**45,321.3**
Provision for taxes	(4,340.0)	(4,539.3)	(5,883.7)	(7,891.5)
Minority interest	(284.9)	(241.0)	(248.7)	(250.2)
Preferred dividends	–	–	–	–
Net inc. (pre-exceptionals)	**21,275.4**	**21,665.9**	**27,682.2**	**37,179.6**
Post-tax exceptionals	–	–	–	–
Net inc. (post-exceptionals)	**21,275.4**	**21,665.9**	**27,682.2**	**37,179.6**
EPS (basic, pre-except) (NT$)	**159.41**	**162.60**	**207.41**	**278.57**
EPS (diluted, pre-except) (NT$)	**157.21**	**161.03**	**204.55**	**276.22**
EPS (basic, post-except) (NT$)	**159.41**	**162.60**	**207.41**	**278.57**
EPS (diluted, post-except) (NT$)	**157.21**	**161.03**	**204.55**	**276.22**
DPS (NT$)	77.87	79.76	101.32	136.83
Div. payout ratio (%)	48.9	49.1	48.9	49.1

Balance Sheet (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Cash & cash equivalents	115,941.7	115,177.0	136,906.6	152,609.6
Accounts receivable	10,461.7	10,483.6	10,430.0	17,325.2
Inventory	6,713.2	9,455.6	7,669.0	14,778.3
Other current assets	21,168.8	21,168.8	21,168.8	21,168.8
Total current assets	**154,285.5**	**156,285.0**	**176,174.5**	**205,881.9**
Net PP&E	51,472.4	60,978.8	57,943.2	52,866.6
Net intangibles	474.6	351.6	344.5	213.7
Total investments	26.5	26.5	26.5	26.5
Other long-term assets	14,528.1	14,528.1	14,528.1	14,528.1
Total assets	**220,787.0**	**232,169.9**	**249,016.8**	**273,516.7**
Accounts payable	1,727.9	2,224.5	2,369.9	3,652.5
Short-term debt	0.0	0.0	0.0	0.0
Short-term lease liabilities	–	–	–	–
Other current liabilities	28,020.6	28,214.1	31,194.1	35,898.7
Total current liabilities	**29,748.5**	**30,438.5**	**33,564.1**	**39,551.2**
Long-term debt	–	–	–	–
Long-term lease liabilities	–	–	–	–
Other long-term liabilities	170.6	170.6	170.6	170.6
Total long-term liabilities	**170.6**	**170.6**	**170.6**	**170.6**
Total liabilities	**29,919.1**	**30,609.1**	**33,734.6**	**39,721.7**
Preferred shares	–	–	–	–
Total common equity	188,867.2	199,801.1	213,771.1	232,534.0
Minority interest	**2,000.8**	**1,759.8**	**1,511.1**	**1,261.0**
Total liabilities & equity	**220,787.0**	**232,169.9**	**249,016.8**	**273,516.7**
Net debt, adjusted	(122,308.6)	(121,543.8)	(143,273.5)	(158,976.5)

Cash Flow (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Net income	25,900.2	21,665.9	27,682.2	37,179.6
D&A add-back	7,732.0	8,565.7	12,427.8	12,398.9
Minority interest add-back	(284.9)	(241.0)	(248.7)	(250.2)
Net (inc)/dec working capital	(1,208.2)	(2,267.7)	1,985.6	(12,721.8)
Other operating cash flow	(6,058.9)	–	–	–
Cash flow from operations	**26,080.2**	**27,723.0**	**41,846.9**	**36,606.6**
Capital expenditures	(12,340.0)	(17,849.1)	(9,160.2)	(7,091.5)
Acquisitions	(2,040.4)	–	–	–
Divestitures	(2.6)	(100.0)	(225.0)	(100.0)
Others	5,147.2	–	–	–
Cash flow from investing	**(9,235.7)**	**(17,949.1)**	**(9,385.2)**	**(7,191.5)**
Repayment of lease liabilities	–	–	–	–
Dividends paid (common & pref)	(13,013.1)	(10,538.6)	(10,732.1)	(13,712.2)
Inc/(dec) in debt	(203.4)	–	–	–
Other financing cash flows	(1,344.6)	0.0	0.0	0.0
Cash flow from financing	**(14,561.2)**	**(10,538.6)**	**(10,732.1)**	**(13,712.2)**
Total cash flow	**2,283.4**	**(764.7)**	**21,729.7**	**15,703.0**
Free cash flow	13,740.3	9,873.9	32,686.7	29,515.1

Source: Company data, Goldman Sachs Research estimates.

We expect Oct revenues to be at a similar level to Sep, and to sequentially decline in Nov and Dec, resulting in 4Q26 revenue growth of +7% QoQ.

Exhibit 2: Largan monthly revenues preview

	Jul 2026	Aug 2026	Sep 2026	Oct 2026E	Nov 2026E	Dec 2026E	2Q26	3Q26	4Q26E
Rev (NT$m)	4,877	5,012	5,779	5,895	5,600	5,299	13,665	15,651	16,794
YoY	-11%	-16%	-7%	-6%	6%	-6%	17%	-11%	-2%
MoM/QoQ	31%	3%	15%	2%	-5%	-5%	-12%	15%	7%
GS estimates (NT$m)	4,826	6,756	7,285				13,750	18,866	
Act. Vs. GS	1%	-26%	-21%				-1%	-17%	

Source: Company data, Goldman Sachs Global Investment Research

Earnings revision

Our net income estimates are revised by -13% / -4% / +16% in 2026-28E mainly on changes in revenues, GM, and Opex ratio, reflecting: **(1) handset lens:** we continue to expect Largan's lens pixel mix upgrade, but at a slower pace, reflecting the 3Q26 miss and slow smartphone market amid high memory costs, **(2) FAU:** we estimate revenue contribution will begin in 2H27, with 0.6m / 5m in 2027 / 28E, along with product mix upgrade toward higher bandwidth FAU that should support ASP and GM. We expect FAU revenue contribution of 3% in 2027E and exceed 50% in 2030E, **(3) GM:** we lower our GM estimates to reflect the slower pace of handset lens pixel mix upgrade and the company's higher costs on green energy. We expect GM to sustain at ~50% in 2026-28E, supported by continuous handset lens pixel mix upgrade and strong demand of FAU, **(4) Opex ratio:** we continue to expect rising opex, while the opex ratio would improve as strong revenue growth driven by FAU and FAU's higher ASP relative to handset lens enhance operating efficiency.

Exhibit 3: Largan earnings revision

	2026E			2027E			2028E		
NT m	Old	New	Chg	Old	New	Chg	Old	New	Chg
Revenue	65,503	61,654	-6%	76,562	76,335	0%	85,021	101,307	19%
GP	33,681	28,867	-14%	42,012	38,222	-9%	47,968	51,348	7%
OP	25,995	21,685	-17%	32,893	29,902	-9%	37,757	41,217	9%
Net income	24,904	21,666	-13%	28,976	27,682	-4%	31,937	37,180	16%
EPS (Diluted)	185.12	161.03	-13%	214.48	204.55	-5%	237.27	276.22	16%
Margins									
GM	51.4%	46.8%		54.9%	50.1%		56.4%	50.7%	
OPM	39.7%	35.2%		43.0%	39.2%		44.4%	40.7%	
NM	38.0%	35.1%		37.8%	36.3%		37.6%	36.7%	

Source: Goldman Sachs Global Investment Research

Exhibit 4: Largan revenues and GM revision

Revenues (NT$m)	2026E	2027E	2028E	GM	2026E	2027E	2028E
NEW	61,654	76,335	101,307	NEW	46.8%	50.1%	50.7%
FAU	-	2,074	19,191	FAU		46.0%	47.4%
Periscope lens	19,653	23,178	26,702	Periscope lens	38.4%	38.3%	38.2%
20MPx+ handset lens	11,911	26,562	34,714	20MPx+ handset lens	70.5%	70.4%	70.4%
Other handset lens	17,833	13,078	9,657	Other handset lens	57.1%	56.4%	55.6%
Others	12,256	11,443	11,042	Others	22.5%	20.3%	20.3%
OLD	65,503	76,562	85,021	OLD	51.4%	54.9%	56.4%
FAU	63	1,067	4,518	FAU	50.3%	50.4%	50.5%
Periscope lens	18,610	18,570	20,741	Periscope lens	38.9%	38.5%	38.5%
20MPx+ handset lens	16,705	31,766	38,447	20MPx+ handset lens	72.9%	73.0%	75.0%
Other handset lens	17,796	13,029	9,611	Other handset lens	62.7%	62.3%	62.0%
Others	12,330	12,130	11,705	Others	24.8%	24.8%	24.9%
CHG	-6%	0%	19%	CHG (NEW - OLD)	-5%	-5%	-6%
FAU	-100%	94%	325%	FAU	-50%	-4%	-3%
Periscope lens	6%	25%	29%	Periscope lens	-1%	0%	0%
20MPx+ handset lens	-29%	-16%	-10%	20MPx+ handset lens	-2%	-3%	-5%
Other handset lens	0%	0%	0%	Other handset lens	-6%	-6%	-6%
Others	-1%	-6%	-6%	Others	-2%	-5%	-5%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 5: Largan 3Q26 results snapshot

NT m	3Q25	2Q26	3Q26	QoQ	YoY	GS	Act / GS	Cons.	Act/Cons.
Revenue	17,677	13,665	15,651	15%	-11%	18,866	-17%	18,196	-14%
GP	8,352	6,753	6,607	-2%	-21%	9,339	-29%	9,120	-28%
OP	6,265	5,148	4,735	-8%	-24%	7,094	-33%	6,946	-32%
Net income	7,080	4,670	5,182	11%	-27%	6,699	-23%	6,723	-23%
Margins									
GM	47.2%	49.4%	42.2%			49.5%		50.1%	
OPM	35.4%	37.7%	30.3%			37.6%		38.2%	
NM	40.1%	34.2%	33.1%			35.5%		36.9%	

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Compared to Bloomberg consensus, we are 4% / 16% ahead in 2027 / 28E net income, mainly on higher revenues and lower opex ratio, reflecting our positive view on Largan's FAU, which is in strong demand with bandwidth upgrade.

Exhibit 6: GS vs. Bloomberg consensus

	2027E			2028E		
NT m	GS	BB	Diff %	GS	BB	Diff %
Revenue	76,335	72,074	6%	101,307	90,479	12%
GP	38,222	36,929	4%	51,348	45,410	13%
OP	29,902	28,454	5%	41,217	34,680	19%
Net income	27,682	26,732	4%	37,180	31,942	16%
Margins						
GM	50.1%	51.2%		50.7%	50.2%	
OPM	39.2%	39.5%		40.7%	38.3%	
NM	36.3%	37.1%		36.7%	35.3%	

Source: Goldman Sachs Global Investment Research, Bloomberg

Valuation

We change our valuation methodology from 2027E PE to discounted PE to better capture the company's long term opportunities in FAU. Our target PE multiple (29.8x) is derived from (1) peers' avg. ratio of trading PE to forward year fundamentals (NI YoY and OPM), which is at 0.5x, and (2) Largan's forward year fundamentals: 2030-31E NI YoY at

19% on avg. and OPM at 41% on avg. We apply the target PE multiple to Largan's 2030E EPS and discount it back to 2027E via a COE of 10.7%, which is based on beta at 1.1x, risk free at 3.5%, and market risk premium at 6.5%. Our new target price is NT$11,000 (vs. NT$6,263 previously). Our target price implies 39.8x 2028E PE, the first full year of FAU contribution, which is within Largan's historical trading range of 8.0x to 45.1x, reflecting our positive view on Largan's expansion from handset lens to FAU, which broadens its exposure from consumer electronics to AI data center that enjoys strong market growth, and could support a valuation re-rating. We note that Taiwan AI supply chain also enjoys re-rating when expanding from consumer electronics to AI data center. Maintain Buy.

Exhibit 7: Largan discounted PE

NT$m	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Key assumptions									
Revenues (NT$m)	**48,842**	**59,458**	**61,148**	**61,654**	**76,335**	**101,307**	**150,128**	**191,476**	**214,294**
FAU	-	-	-	-	2,074	19,191	63,908	103,530	126,348
Others	48,842	59,458	61,148	61,654	74,261	82,115	86,221	87,945	87,945
YoY	**2%**	**22%**	**3%**	**1%**	**24%**	**33%**	**48%**	**28%**	**12%**
FAU						825%	233%	62%	22%
Others	2%	22%	3%	1%	20%	11%	5%	2%	0%
FAU revenues contribution					*3%*	*19%*	*43%*	*54%*	*59%*
FAU shipment (k units)					592	4,789	14,368	21,551	25,862
YoY						*710%*	*200%*	*50%*	*20%*
FAU ASP (US$)					117	134	148	160	163
YoY						*14%*	*11%*	*8%*	*2%*
P&L									
Revenue	**48,842**	**59,458**	**61,148**	**61,654**	**76,335**	**101,307**	**150,128**	**191,476**	**214,294**
Gross profit	**23,808**	**31,209**	**30,837**	**28,867**	**38,222**	**51,348**	**76,093**	**97,050**	**108,616**
Gross margin	48.7%	52.5%	50.4%	46.8%	50.1%	50.7%	50.7%	50.7%	50.7%
OPEX	5,986	7,177	7,279	7,181	8,320	10,131	15,013	19,148	21,429
Opex ratio	12.3%	12.1%	11.9%	11.6%	10.9%	10.0%	10.0%	10.0%	10.0%
Operating profit	**17,821**	**24,033**	**23,558**	**21,685**	**29,902**	**41,217**	**61,081**	**77,903**	**87,186**
OPM	36.5%	40.4%	38.5%	35.2%	39.2%	40.7%	40.7%	40.7%	40.7%
Non-op	4,280	8,142	2,342	4,761	3,913	4,104	4,104	4,104	4,104
Pre-tax profit	**22,102**	**32,174**	**25,900**	**26,446**	**33,815**	**45,321**	**65,185**	**82,007**	**91,291**
Tax	4,200	5,963	4,340	4,539	5,884	7,891	11,350	14,279	15,896
Tax rate	19.0%	18.5%	16.8%	17.2%	17.4%	17.4%	17.4%	17.4%	17.4%
Minority	-	296	285	241	249	250	250	250	250
Net profit	**17,902**	**25,915**	**21,275**	**21,666**	**27,682**	**37,180**	**53,584**	**67,478**	**75,145**
EPS (NTD, diluted)	**132.62**	**191.44**	**157.21**	**161.03**	**204.55**	**276.22**	**398.10**	**501.32**	**558.28**
YoY	*-21%*	*44%*	*-18%*	*2%*	*27%*	*35%*	*44%*	*26%*	*11%*
TP implied P/E					53.8	39.8	27.6	21.9	19.7

2030E target P/E	29.8
Target multiple x EPS	14,939
Discounted back to 2027; TP (NTD)	**11,000**
Implied 2027 P/E	53.8

COE assumption	
Beta	1.1
Risk free	3.5%
Market risk premium	6.5%
COE	**10.7%**

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 8: Largan peer comparison

Company	Ticker	Rating	PE	NI YoY		OPM		Ratio
			2030E	2030E	2031E	2030E	2031E	
Largan	3008.TW	Buy	13.3	26%	11%	40.7%	40.7%	0.2
Largan (TP implied)	3008.TW	Buy	29.8	26%	11%	40.7%	40.7%	0.5
Company	**Ticker**	**Rating**	**PE**	**NI YoY**		**OPM**		**Ratio**
			2027E	**2027E**	**2028E**	**2027E**	**2028E**	
Peers								
AAC	2018.HK	Neutral	12.4	17%	12%	9.7%	10.0%	0.5
T&S	300570.SZ	NC	47.1	96%	62%	24.2%	25.4%	0.5
VPEC	2455.TW	Buy	48.9	67%	55%	35.2%	37.3%	0.5
Sumitomo	5802.JT	Buy	17.2	18%	25%	9.8%	11.2%	0.5
Average			**31.4**	**50%**	**38%**	**19.7%**	**21.0%**	**0.5**

NC: Not Covered; data from Bloomberg consensus. Price as of Oct 9, 2026.

Source: Goldman Sachs Global Investment Research, Bloomberg

Exhibit 9: Taiwan AI server supply chain enjoys strong re-rating when expanding from consumer electronics to AI data center

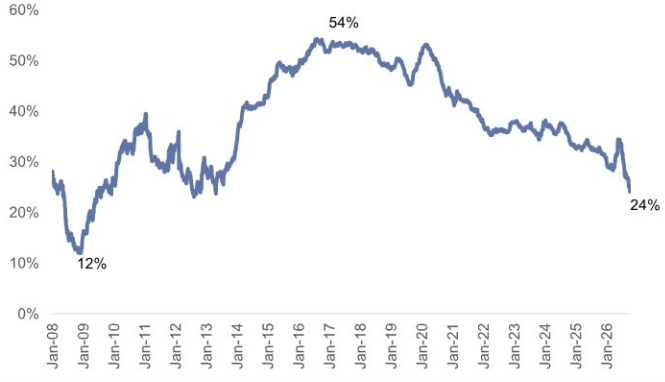
Company	Ticker	2020-2025 PE		
		Trough	Peak	Diff
Largan	3008.TW	10.6	28.0	163%
Taiwan AI server supply chain				
Landmark	3081.TWO	28.9	144.1	398%
VPEC	2455.TW	18.1	74.1	310%
Auras	3324.TWO	8.8	54.6	523%
Fositek	6805.TW	11.3	49.9	343%
AVC	3017.TW	4.7	32.8	604%
Inventec	2356.TW	9.9	31.0	213%
King Slide	2059.TW	9.0	30.2	236%
Gigabyte	2376.TW	3.8	29.6	685%
Mitac	3706.TW	2.5	28.7	1044%
Chenbro	8210.TW	6.7	25.6	282%
Wiwynn	6669.TW	6.1	23.4	287%
Wistron	3231.TW	6.0	21.8	262%
Quanta	2382.TW	6.8	20.4	201%
Hon Hai	2317.TW	7.7	17.5	129%
Avg.		**9.3**	**41.7**	**394%**

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 10: Largan 12M forward PE ratio

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 11: Largan QFII holdings



Source: TEJ

Financial tables

Exhibit 12: Largan P&L

NT$m	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2023	2024	2025	2026E	2027E	2028E
Revenues	15,544	13,665	15,651	16,794	18,095	13,322	19,655	25,263	48,842	59,458	61,148	61,654	76,335	101,307
GP	7,680	6,753	6,607	7,827	9,110	6,375	9,995	12,743	23,808	31,209	30,837	28,867	38,222	51,348
OP income	5,812	5,148	4,735	5,990	7,137	4,923	7,852	9,989	17,821	24,033	23,558	21,685	29,902	41,217
Net income	6,123	4,670	5,182	5,690	6,643	4,812	7,230	8,996	17,902	25,915	21,275	21,666	27,682	37,180
EPS, diluted (NT$)	46.11	35.61	39.52	39.79	49.09	35.56	53.43	66.48	132.62	191.44	157.21	161.03	204.55	276.22
YoY														
Revenues	7%	17%	-11%	-2%	16%	-3%	26%	50%	2%	22%	3%	1%	24%	33%
GP	-4%	8%	-21%	-5%	19%	-6%	51%	63%	-9%	31%	-1%	-6%	32%	34%
OP income	-5%	6%	-24%	-5%	23%	-4%	66%	67%	-13%	35%	-2%	-8%	38%	38%
Net income	-5%	352%	-27%	-15%	8%	3%	40%	58%	-21%	45%	-18%	2%	28%	34%
EPS, diluted (NT$)	-3%	362%	-25%	-19%	6%	0%	35%	67%	-21%	44%	-18%	2%	27%	35%
QoQ														
Revenues	-10%	-12%	15%	7%	8%	-26%	48%	29%						
GP	-7%	-12%	-2%	18%	16%	-30%	57%	27%						
OP income	-8%	-11%	-8%	27%	19%	-31%	60%	27%						
Net income	-9%	-24%	11%	10%	17%	-28%	50%	24%						
EPS, diluted (NT$)	-6%	-23%	11%	1%	23%	-28%	50%	24%						
Margins														
GM	49.4%	49.4%	42.2%	46.6%	50.3%	47.9%	50.9%	50.4%	48.7%	52.5%	50.4%	46.8%	50.1%	50.7%
OPM	37.4%	37.7%	30.3%	35.7%	39.4%	37.0%	40.0%	39.5%	36.5%	40.4%	38.5%	35.2%	39.2%	40.7%
NM	39.4%	34.2%	33.1%	33.9%	36.7%	36.1%	36.8%	35.6%	36.7%	43.6%	34.8%	35.1%	36.3%	36.7%
Ratios														
Opex ratio	12%	12%	12%	11%	11%	11%	11%	11%	12%	12%	12%	12%	11%	10%
Tax rate	15%	23%	14%	17%	17%	17%	17%	17%	19%	19%	17%	17%	17%	17%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 13: Largan balance sheet

NT$ mn	2023	2024	2025	2026E	2027E	2028E
Balance Sheet						
Cash and equivalents	107,490	113,658	115,942	115,177	136,907	152,610
Accounts receivable	10,091	10,360	10,462	10,484	10,430	17,325
Inventory	4,591	5,733	6,713	9,456	7,669	14,778
Other current assets	12,149	16,012	21,169	21,169	21,169	21,169
Current assets	**134,321**	**145,764**	**154,285**	**156,285**	**176,174**	**205,882**
Net PP&E/Fixed assets	41,135	46,936	51,472	60,979	57,943	52,867
Net intangibles	239	505	475	352	345	214
Other long-term assets	19,443	23,322	14,555	14,555	14,555	14,555
Non-current assets	**60,817**	**70,763**	**66,502**	**75,885**	**72,842**	**67,635**
Total assets	**195,138**	**216,527**	**220,787**	**232,170**	**249,017**	**273,517**
Accounts payable	1,731	1,855	1,728	2,224	2,370	3,653
Short-term debt	-	203	-	-	-	-
Other current liabilities	27,787	28,520	28,021	28,214	31,194	35,899
Current liabilities	**29,517**	**30,578**	**29,749**	**30,439**	**33,564**	**39,551**
Long-term debt	-	-	-	-	-	-
Other long-term liabilities	110	561	171	171	171	171
Non-current liabilities	**110**	**561**	**171**	**171**	**171**	**171**
Total liabilities	**29,627**	**31,139**	**29,919**	**30,609**	**33,735**	**39,722**
Common stock	1,335	1,335	1,335	1,335	1,335	1,335
Retained earnings	160,871	175,969	185,818	196,752	210,722	229,485
Other common equity	3,305	6,215	1,714	1,714	1,714	1,714
Total common equity	**165,510**	**183,519**	**188,867**	**199,801**	**213,771**	**232,534**
Minority interest	-	1,869	2,001	1,760	1,511	1,261
Total equity	**165,510**	**185,388**	**190,868**	**201,561**	**215,282**	**233,795**
BVPS	1,226.07	1,369.51	1,410.38	1,498.06	1,590.78	1,736.96
Cash conversion cycle						
Account receivable days	68	63	62	62	50	50
Inventory days	71	67	75	90	82	82
Net payable days	25	23	22	22	22	22
Cash conversion cycle	114	106	116	130	110	110
Ratios						
ROE (common equity)	11%	15%	11%	11%	13%	17%
ROA	9%	13%	10%	10%	12%	14%
Net debt to total equity	-67%	-64%	-64%	-60%	-67%	-68%
Total liabilities to total assets	15%	14%	14%	13%	14%	15%
Dupont analysis						
Asset turnover	0.3	0.3	0.3	0.3	0.3	0.4
Leverage (assets to equity)	1.2	1.2	1.2	1.2	1.2	1.2
Net margin	36.7%	43.6%	34.8%	35.1%	36.3%	36.7%
ROE (total equity)	11%	15%	11%	11%	13%	17%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 14: Largan cash flow

NT$ mn	2023	2024	2025	2026E	2027E	2028E
Cash flow statement						
Net income	22,102	32,174	25,900	21,666	27,682	37,180
Minority interest add-back	-	(296)	(285)	(241)	(249)	(250)
Depreciation and amortization add-back	5,422	6,230	7,732	8,566	12,428	12,399
(Increase)/decrease in working capital	(1,400)	(1,287)	(1,208)	(2,268)	1,986	(12,722)
Other operating cash flow items	(7,925)	(5,243)	(6,059)	-	-	-
Cash flow from operating	**18,199**	**31,579**	**26,080**	**27,723**	**41,847**	**36,607**
Capital expenditure	(8,237)	(11,133)	(12,340)	(17,849)	(9,160)	(7,091)
Other investment cash flow items	(2,915)	(4,933)	3,104	(100)	(225)	(100)
Cash flow from investing	**(11,151)**	**(16,066)**	**(9,236)**	**(17,949)**	**(9,385)**	**(7,191)**
Dividends paid	(11,412)	(9,009)	(13,013)	(10,539)	(10,732)	(13,712)
Change in common stock	-	-	-	-	-	-
Increase/(decrease) in short-term debt	(19)	203	(203)	-	-	-
Increase/(decrease) in long-term debt	-	-	-	-	-	-
Other financing cash flow items	1,680	(1,958)	(241)	-	-	-
Cash flow from financing	**(9,750)**	**(10,764)**	**(13,458)**	**(10,539)**	**(10,732)**	**(13,712)**
Net change in cash	(2,682)	6,168	2,283	(765)	21,730	15,703
FCF	9,962	20,446	13,740	9,874	32,687	29,515
Ratio						
Capex to revenue	17%	19%	20%	29%	12%	7%

Source: Company data, Goldman Sachs Global Investment Research

Price Target Risks and Methodology - Largan

Valuation: We are Buy rated on Largan and have a 12-month target price of NT$11,000, which is based on discounted PE. Our target P/E multiple of 29.8x is derived from (1) peers' avg. ratio of trading PE to forward year fundamentals (NI YoY and OPM), which is at 0.5x, and (2) Largan's forward year fundamentals: 2030-31E NI YoY at 19% on avg. and OPM at 41% on avg. We apply the target PE multiple to Largan's 2030E EPS and discount it back to 2027E via a COE of 10.7%, which is based on beta at 1.1x, risk free at 3.5%, and market risk premium at 6.5%. Our target price of NT$11,000 implies 39.8x 2028E PE, the first full year of FAU contribution, which is within Largan's historical trading range of 8.0x to 45.1x, reflecting our positive view on Largan's expansion from handset lens to FAU, which broadens its exposure from consumer electronics to AI data center that enjoys strong market growth, and could support a valuation re-rating. We note that Taiwan AI supply chain also enjoys re-rating when expanding from consumer electronics to AI data center. Maintain Buy.

Key risks: (1) slower-than-expected smartphone market growth, (2) fiercer competition in handset lenses, (3) slower-than-expected smartphone camera lens specification upgrades or Largan's handset lens pixel mix upgrade, (4) slower-than-expected FAU ramp up, and (5) fiercer competition in FAU.

Disclosure Appendix

Reg AC

We, Verena Jeng and Allen Chang, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

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Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Rating and pricing information

AAC (Neutral, HK$37.36), Sumitomo Electric Industries (Buy, ¥2,361) and VPEC (Buy, NT$558.00)

The rating(s) for Largan is/are relative to the other companies in its/their coverage universe: AAC, ACM Research, AMEC, ASMPT, AVC, AccoTest, Anji Micro, Asus, Auras, BOE, BYDE, Biren, CR Micro, CXMT, Cambricon, Chenbro, China Mobile (HK), China Telecom, China Tower Corp., China Unicom, Chinasoft Intl, Compal, Desay SV, E Ink, E-Town Semis, EHang, Empyrean, Eoptolink, FOCI, Fositek, Foxconn Industrial Internet, Gigabyte, Gigadevice, Glodon Co., HTC Corp., Hikvision, Hon Hai, Horizon Robotics, Hua Hong, Huace Navigation, Huaqin Co.(A), Huaqin Co.(H), Hwatsing, Iluvatar, InnoScience, Innolight (A), Innolight (H), Inspur, Insta360, Inventec, JCET, Kematek, King Slide, Kingdee, Kingsoft Office, LandMark, Largan, Lenovo, Lingyi, Luxshare (A), Luxshare (H), Maxscend, Meitu, MetaX, Mitac, Montage (A), Montage (H), NAURA, NSIG, Nexchip, OmniVision, Pegatron, Pony AI Inc. (ADR), Pony AI Inc. (H), Quanta, RoboTechnik, Ruijie Networks, SG Micro, SICC, SMIC (A), SMIC (H), SZS, Sangfor, SenseTime, Shengyi Tech, Shennan Circuits, StarPower, Sunny Optical, TFC Optical, Thundersoft, Tongyu Communication, Transsion, UMT, UNIS, VPEC, Vanchip, VeriSilicon, Victory Giant, WNC, WUS, WeRide, Wistron, Wiwynn, YJ Semitech, YOFC, Yonyou, ZTE (A), ZTE (H), iFlytek

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

There are no company-specific disclosures for: Largan (NT$6,660.00)

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	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	59%	43%

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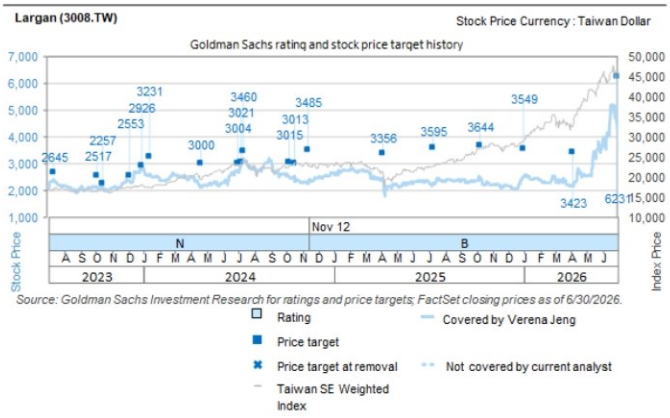
Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Price target and rating history chart(s)



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

Largan (3008.TW)

Date of report	Target price (NT$)	Closing price (NT$)
13-Jul-26	6,263.00	4,345.00
30-Jun-26	6,231.00	4,290.00
07-Apr-26	3,423.00	2,170.00
01-Jan-26	3,549.00	2,495.00
09-Oct-25	3,644.00	2,470.00
11-Jul-25	3,595.00	2,285.00
06-Apr-25	3,356.00	2,450.00
12-Nov-24	3,485.00	2,255.00
18-Oct-24	3,013.00	2,325.00
07-Oct-24	3,015.00	2,525.00
11-Jul-24	3,460.00	3,200.00
08-Jul-24	3,021.00	3,030.00
01-Jul-24	3,004.00	2,700.00
21-Apr-24	3,000.00	2,120.00
12-Jan-24	3,231.00	2,600.00
29-Dec-23	2,926.00	2,870.00
06-Dec-23	2,553.00	2,320.00
15-Oct-23	2,257.00	2,175.00

Price targets shown in table(s) are unadjusted for corporate actions.

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From: 陸行之 Andrew Lu on global semis and techs <andrewhclu@substack.com>
Subject: TSMC outsources silicon interposer production to GFS; Active overflow order Part I
Date: 9 hours ago

TSMC outsources silicon interposer production to GFS; Active overflow order Part I

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TSMC outsources silicon interposer production to GFS; Active overflow order Part I

因矽中介層產能不足，台積電首次合作外包給競爭者GFS，這應該算台積電的主動外溢訂單

ANDREW LU ON GLOBAL SEMI/TECHS
OCT 9 • PAID

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What is new?

半導體晶圓代工領域誕生了一項具備歷史分水嶺意義的競合大案，全球晶圓代工龍頭台積電（TSMC）與長期競爭對手美商格芯（GlobalFoundries, GFS）正式簽署一項為期五年、價值 20 億美元，2028上半年開始量產的製造供應協議。根據合約，格芯將在其位於美國紐約州馬爾他（Malta, NY）的旗艦 Fab 8 晶圓廠，專門為台積電的 CoWoS 先進封裝生態系生產最關鍵的「矽中介層（Silicon Interposers）」。Silicon interposer 做完RDL層後與邏輯晶片與記憶體進行接合，對高頻寬、訊號完整性與封裝良率至關重要；它不是最先進 GPU 邏輯 die，也不代表 GlobalFoundries 取得 Nvidia／AMD 的先進節點代工。

格芯因此成為全美國境內第一座具備量產能力、專為先進封裝供貨的本土矽中介層供應基地，徹底解決了台積電亞利桑那鳳凰城 Fab 21 在美生產 4nm/3nm/2nm 裸晶後，因美國缺乏先進矽中介層製造而被迫將晶圓空運回台灣封裝或將矽中介層空運到美國的荒謬地緣斷鏈難題。

GlobalFoundries reaches agreement to establish U.S.-based supply of silicon interposers for advanced AI packaging

October 8, 2026

Multi-year $2 billion agreement will expand capacity for advanced packaging ecosystem and support growing demand for AI and high-performance computing technologies

MALTA, N.Y., Oct. 8, 2026 – GlobalFoundries (Nasdaq: GFS) (GF) today announced a manufacturing agreement with TSMC to establish a U.S.-based supply of silicon interposers for TSMC's CoWoS® advanced packaging ecosystem. Under the agreement, GF will provide manufacturing service to TSMC, and add fabrication capacity at its Malta, New York facility, supporting growing demand for high-performance computing and AI systems. The agreement provides a foundation to expand capacity over time as customer demand grows.

GF expects to expand capacity at its Malta facility and establish the first U.S.-based source of silicon interposers supporting advanced packaging technologies, including embedded deep trench capacitor components. The additional manufacturing capacity will provide greater scale and flexibility for delivering advanced packaging solutions across multiple product generations.

"Advanced packaging is becoming increasingly critical to delivering the performance, power efficiency and scale required for next-generation AI systems," said Ed Kaste, senior vice president of CMOS Business at GF. "By providing manufacturing service using GF's trusted U.S. manufacturing footprint, we are creating a secure, scalable source of essential advanced-packaging elements that will help customers accelerate innovation and strengthen the semiconductor supply chain."

Volume production is expected to begin ramping at GF's Malta site during the first half of 2028. The agreement has an initial term of five years and provides a framework for future capacity expansion.

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Positively affect what supply chain vendors and why?

美商格芯（GlobalFoundries, GFS）：我們估計台積電 20 億美元保證採購訂單約佔 Globalfoundries 2028-2032 累積營收增量的3-4%，明顯改善紐約 Fab 8 的成熟與特殊製程產能利用率，且無需承擔極紫外光（EUV）微縮風險即可跨入高毛利的 AI 先進封裝關鍵環節。但要是台積電在2030年用玻璃中介層替代矽中介層轉進 CoPoS，這訂單上修的機會將被移除。

台積電：以極低資本開支成功化解美國白宮與國防部對「晶片在美製造、封裝在台完成」的強烈地緣政治施壓，建立完全在美閉環的 CoWoS 替代供應鏈（台積電4/3/2nm 邏輯晶片+GFS的Silicon Interposer+Amkor的 WoS 封測），同時將台積電美國的產能完全留給最高附加價值的晶粒堆疊（Chip-on-Wafer）。

美系 AI 晶片巨頭（Nvidia、AMD、谷歌）：美國國防與聯邦政府機構採購的敏感 AI 晶片，自此可符合「全美在地製造與先進封裝（100% On-shore

Packaging）」標準，合規阻礙完全消除。

EDA、封裝協同設計、良率分析與量測： 跨廠製造需要更嚴格 PDK、版圖、電性、翹曲與缺陷資料交換。

矽中介層，HBM、ABF substrate、先進測試與封裝設備： Interposer 供給擴張可帶動可組裝的 GPU/ASIC-HBM 套數，但必須與其餘環節同步。

Negatively affect what supply chain vendors and why?

台灣二線矽中介層代工廠（聯電 2303.TW、世界先進 5347.TW）：台積電過去在台灣將部分 CoWoS-S 矽中介層外包給聯電等本土夥伴；隨著美系客戶指定美國在地化生產，聯電爭取海外 CoWoS 衍生中介層訂單的空間遭到分流，美國第二來源降低稀缺溢價並提高客戶議價權。

Our outside-the-box, non-consensus thinking

台積電把關鍵的 CoWoS 中介層交給技術競爭對手，是否引狼入室？我們的非共識觀點指出，矽中介層本質上是高耗面積、低電晶體密度的低毛利率材料。矽中介層通常使用 65nm 甚至 90nm 老舊製程製造，上面幾乎沒有主動電晶體，純粹是用深矽穿孔（TSV）進行高密度金屬布線 RDL，但它卻要霸佔珍貴的 12 吋潔淨室機台空間。台積電將這種「吃產能、賺工錢」的低毛利環節甩給急需產能利用率的格芯，自己則全力聚焦在單片要價 3 萬美元的 3nm/2nm 核心運算晶粒與頂層高難度鍵合。台積電不是出讓技術，而是進行高超的微笑曲線資本利潤套利。

而且這筆交易的真正戰略價值不僅僅是美國製造，而是把先進封裝拆成可外包的模組。若 TSMC 能標準化 interposer PDK、測試與良率資料，未來可像前段 foundry 的多廠調度一樣管理封裝供應鏈；這會提高總體產能韌性，但也把競爭焦點轉到跨廠協同、良率 ownership 與責任界面。長期勝者可能不是單一設備商，而是能把 design-to-package 數位線索串起來的 EDA／manufacturing data 平台。

Our advice to investors

我們不該把協議解讀為 TSMC 技術外流或 GlobalFoundries 進入最先進邏輯製程。應繼續追蹤四項指標：1. 1H28 是否能實際量產、2. GFS 的 silicon interposer 良率與產能、3. TSMC 內部 CoWoS 瓶頸釋放幅度、4. 是否追加訂單。US2bn若平均分五年為US400mn／年，但估值模型應按 2028 起的爬坡曲線認列。

這項世紀大和解是台積電進行地緣政治壓力卸載的操作。台積電不僅沒有因此喪失封裝主導權，反而用外包中介層的代價，將原本對立的格芯轉化為自身的利益同盟，並在美國本土牢牢築起 CoWoS 生態圈。我們維持對台積電長期的正面評價並同時看好格芯（GFS）的結構性轉變。

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